

Group Assignment Trimester 1 25/26

2025-09-16

Task

Description

You are a consultant for Pacific Insights Analytics (PIA), a boutique data analytics firm specializing in workforce performance. PIA has been hired by Huaxia Communications, a major Chinese company operating several large call centers across the country. Huaxia is facing rising operational challenges. Staff attrition is driving up recruitment and training costs and retaining top performers is becoming increasingly difficult. To address these issues, Huaxia has provided you with historical data from one of its call centers. This data comes from multiple administrative sources, including worker surveys, HR records, and call center productivity logs. Because the data was collected from different systems, it contains inconsistencies, formatting problems, and other quality issues. Your task is to clean, explore, and analyze the data in order to generate clear, evidence-based recommendations that can help Huaxia's management improve retention and boost productivity. Important: You should not run any regression models. The goal is to understand the data and identify patterns that can inform management decisions through EDA and hypothesis testing.

Declaration of AI Usage

In the preparation of this project, I made use of AI tools, specifically OpenAI's ChatGPT, as a support resource. The tool was employed to enhance efficiency in tasks such as structuring ideas, clarifying concepts, refining wording, and checking technical accuracy.

At all times, the use of AI served as a complement to my own knowledge, analysis, and critical thinking. The outputs generated were carefully reviewed, adapted, and integrated by us to ensure originality, accuracy, and academic integrity. The final work reflects my own understanding, reasoning, and skills, with AI functioning only as a leverage of these.

Setup

Packages

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```

library(haven)
library(dplyr)
library(ggplot2)
library(ggcorrplot)
library(ISOweek)
library(stargazer)

##
## Please cite as:
##
## Hlavac, Marek (2022). stargazer: Well-Formatted Regression and Summary Statistics Tables.
## R package version 5.2.3. https://CRAN.R-project.org/package=stargazer
library(fmsb)

```

Loading the data

```

data_perf <- read_dta("../data/performance_labeled.dta")
data_attitude <- read_dta("../data/attitude_labeled.dta")
data_endperiod <- read_dta("../data/endperiod_outcomes_labeled.dta")
data_sum_vol <- read_dta("../data/summary_volunteer_labeled.dta")
data_wage_new <- read_dta("../data/wage_new_labeled.dta")

# convert to tibbles
performance <- as_tibble(data_perf)
attitude <- as_tibble(data_attitude)
endperiod <- as_tibble(data_endperiod)
volunteer <- as_tibble(data_sum_vol)
wage <- as_tibble(data_wage_new)

# remove raw data
rm(data_perf)
rm(data_attitude)
rm(data_endperiod)
rm(data_sum_vol)
rm(data_wage_new)

```

Cleaning data

Handy functions for cleaning

Here come some handy functions to reduce duplicate code for cleaning the data:

```

# convert to numeric, if its a number (not NA)
as_numeric <- function (x) {
  x <- as.character(x)
  # only convert if it looks numeric
  if (grepl("^[-+]?[0-9]*\\.?[0-9]+([eE][-+]?[0-9]+)?$", x)) {
    as.numeric(x)
  } else {
    NA_real_
  }
}

```

```

# convert to integer, if its a number (not NA)
as_integer <- function(x) {
  # only convert if it looks numeric
  x <- as.character(x)
  if (grepl("^[~+]?[0-9]*\\.?[0-9]+([eE][~+]?[0-9]+)?$", x)) {
    as.integer(x)
  } else {
    NA_real_
  }
}

# Extract a numeric values correctly from a string value, considering multiple cases
extract_numeric <- function(x) {
  # remove leading/trailing spaces
  x <- trimws(x)

  # normalize dash types (en-dash, em-dash) to minus
  x <- gsub("-", "-", x)

  # allow scientific notation: replace commas in mantissa
  if (grepl("[eE]", x)) {
    # if comma before e → decimal comma
    x <- sub(",", ".", x)
    # remove thousand separators before e
    x <- gsub("\\.(?=[\\d]{3}[eE])", "", x, perl = TRUE)
    x <- gsub(",(?=[\\d]{3}[eE])", "", x, perl = TRUE)
    return(as.numeric(x))
  }

  # remove spaces
  x <- gsub(" ", "", x)

  # case 1: comma used as decimal
  if (grepl(",", x) & !grepl("\\.", x)) {
    x <- gsub(",", ".", x)
  }

  # case 2: both comma and dot appear
  # assume comma is thousands separator, remove it
  if (grepl(",.*\\.", x)) {
    x <- gsub(",", "", x)
  }

  as_numeric(x)
}

# find all non numeric characters in a string and return them
find_non_numeric_chars <- function(x) {
  non_numeric <- gsub("[0-9]", "", x) # remove digits, minus, dot, comma
  unique(non_numeric[non_numeric != ""])
}

# find all rows with non numeric characters in a string and return them
find_non_numeric_rows <- function(x) {
  non_numeric <- !grepl("^[~+]?[0-9]+([0-9]+)?$", x) # remove digits, minus, dot, comma

```

```

  non_numeric <- non_numeric[non_numeric != ""]
  x[non_numeric]
}

```

Clean volunteer

```
glimpse(volunteer)
```

```

## Rows: 140
## Columns: 9
## $ personid <chr> "33350", "40034", "31292", "21654", "36908", "13980", "44794~
## $ age <dbl> 26, 19, 27, 29, 23, 1, 23, 25, 21, 23, 23, 30, 22, 22, 23, 2~
## $ tenure <dbl> 22, 9, 24, 37, 15, 51, 3, 42, 3, 34, 24, 25, 3, 66, 9, 70, 4~
## $ children <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", "0", ~
## $ bedroom <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", ~
## $ commute <chr> "120", "300", "135", "80", "60", " 120", "40 min", "60", "1~
## $ gender <chr> "m", "woman", "F", "f", "m", "woman", "male", "woman", "MALE~
## $ married <chr> "SINGLE", "N", "NO", "N", "NO", " single", " N", "NO", "sin~
## $ high_educ <chr> "Y", "no", "YES", "YES", "N", "NO", "n", "YES", "no", "Y", "~

```

```
stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = FALSE)
```

```

##
## =====
## Statistic N    Mean St. Dev.  Min  Median   Max
## -----
## age      140 23.536  4.352    1    23    35
## tenure   140 22.971 24.296  2.000 13.000 150.000
## -----

```

```

# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases

```

```
find_non_numeric_chars(volunteer$commute)
```

```

## [1] " "      " min"    " MIN"    " "      " minute" " minutes" ". h"
## [8] " MIN"

```

```
find_non_numeric_rows(volunteer$commute)
```

```

## [1] " 120"      "40 min"      " 50 MIN"      " 170"      " 210"
## [6] "40 minute" "240 minutes" "1.2 h"        " 180"      " 90 MIN"
## [11] "40 minute" "30 minutes"  " 180"        "120 MIN"   " 120"
## [16] "50 min"    "100 min"     " 120"        "60 minutes" "1.2 h"
## [21] "180 minutes" " 150"       "2.3 h"

```

```
# here we only need to take care of the decimal dots (will be added to the extraction function)
```

We can see a lot of columns with numerical values within strings and additional chars. Furthermore, categorical variables like married and gender contain different labels for the same categories.

```

# convert to integer to remove leading zeros an than to str
volunteer$personid <- as.character(as.integer(volunteer$personid))
# Clean Numbers with extra characters
# Remove non-numeric characters, except dots and commas and convert to numeric
volunteer$commute <- sapply(volunteer$commute, extract_numeric)

```

```
## Warning in FUN(X[[i]], ...): NAs introduced by coercion
```

```

## Warning in FUN(X[[i]], ...): NAs introduced by coercion
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## Warning in FUN(X[[i]], ...): NAs introduced by coercion

volunteer$commute <- sapply(volunteer$commute, as_numeric)

# convert "no" strings in the children column to 0 children
volunteer$children <- ifelse(volunteer$children %in% c("No", ""), "0", volunteer$children)
# convert to boolean by first converting to integer
volunteer$children <- as.logical(sapply(volunteer$children, as_integer))
# Convert Tenure to integer, as float dont make sense for months
volunteer$tenure <- sapply(volunteer$tenure, as_integer)

# convert bedroom to boolean by first converting 0s to False, 1 to true
volunteer$bedroom<- ifelse(
  volunteer$bedroom == "0",
  0,
  1
)
volunteer$bedroom <- sapply(volunteer$bedroom, as_integer)
# normalize gender data by
# lower all charcters
volunteer$gender <- tolower(volunteer$gender)
# trim whitespaces
volunteer$gender <- trimws(volunteer$gender)
# cast all possible options for male and female
volunteer$gender <- ifelse(volunteer$gender %in% c("m", "male", "man"), "male", volunteer$gender)
volunteer$gender <- ifelse(volunteer$gender %in% c("woman", "female", "f", "w","fem"), "female", volunteer$gender)

# using a similar approach to the gender we convert the married status now
volunteer$married <- tolower(volunteer$married)
# trim whitespaces
volunteer$married <- trimws(volunteer$married)
# cast all possible options for married/not married
volunteer$married <- ifelse(volunteer$married %in% c("married", "yes", "y", "true"), "TRUE", volunteer$married)
volunteer$married <- ifelse(volunteer$married %in% c("no", "not married", "n", "na", "single","false"), "FALSE", volunteer$married)

# using a similar approach to the married column we convert the high_educ status now
volunteer$high_educ <- tolower(volunteer$high_educ)
# trim whitespaces
volunteer$high_educ <- trimws(volunteer$high_educ)
# cast all possible options for high_educ/not high_educ
volunteer$high_educ <- ifelse(volunteer$high_educ %in% c("yes", "y"), "TRUE", volunteer$high_educ)
volunteer$high_educ <- ifelse(volunteer$high_educ %in% c("no", "n","false"), "FALSE", volunteer$high_educ)

# convert to boolean
volunteer$high_educ <- as.logical(volunteer$high_educ)

# create numerical variable cols for gender and married
volunteer$gender_num <- sapply(ifelse(volunteer$gender=="male",0,1), as_integer)
volunteer$married_num <- sapply(ifelse(volunteer$married==FALSE,0,1), as_integer)
volunteer$high_educ_num <- sapply(ifelse(volunteer$high_educ==FALSE,0,1), as_integer)

```

```
volunteer$children_num <-sapply(ifelse(volunteer$children==FALSE,0,1), as_integer)
```

```
# remove duplicates on same personid
volunteer <- volunteer %>%
  distinct(personid, .keep_all = TRUE)
```

```
# Now we have much better insights using stargazer
stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N    Mean    St. Dev. Min Median Max
## -----
## age            135 23.452    4.315    1    23    35
## tenure         135 23.385    24.608    2    13   150
## children       135  0.148    0.357    0     0     1
## bedroom        135  0.978    0.148    0     1     1
## commute        122 104.721   68.544    1    85   300
## high_educ       135  0.415    0.495    0     0     1
## gender_num     135  0.496    0.502    0     0     1
## married_num    135  0.200    0.401    0     0     1
## high_educ_num  135  0.415    0.495    0     0     1
## children_num   135  0.148    0.357    0     0     1
## -----
```

Clean attitude

```
glimpse(attitude)
```

```
## Rows: 2,379
## Columns: 5
## $ personid   <dbl> 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, ~
## $ year_week  <chr> "202249", "202250", "202251", "202252", "202253", "202302", ~
## $ exhaustion <dbl> 9, 8, 8, 6, 12, 12, 12, 12, 10, 12, 11, 12, 12, 12, 14, 12, ~
## $ negative   <dbl> 20, 21, 20, 17, 19, 18, 20, 16, 18, 24, 18, 19, 20, 19, 18, ~
## $ positive   <dbl> 20, 25, 24, 22, 19, 19, 19, 22, 20, 23, 23, 22, 22, 24, 23, ~
```

```
stargazer(as.data.frame(attitude), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean      St. Dev.   Min  Median  Max
## -----
## personid      2,379 33,005.050 10,331.780 4,122 36,288 45,442
## exhaustion    2,379   8.612      7.794    0.000  7.000  36.000
## negative      2,379  16.651      6.848    8.000 16.000  40.000
## positive      2,379  24.215      6.668    8.000 24.000  40.000
## -----
```

```
# checking unique values to decide int vs dbl
unique(attitude$exhausting)
```

```
## Warning: Unknown or uninitialised column: `exhausting`.
```

```
## NULL
```

```
unique(attitude$positive)
```

```
## [1] 20.00000 25.00000 24.00000 22.00000 19.00000 23.00000 21.00000 18.00000
## [9] 17.00000 8.00000 16.00000 32.00000 34.00000 30.00000 29.00000 31.00000
## [17] 24.24000 28.00000 22.56000 27.00000 26.00000 28.13000 33.00000 35.00000
## [25] 9.00000 12.00000 14.00000 13.00000 15.00000 27.19000 11.00000 10.00000
## [33] 38.00000 26.28000 36.00000 23.79000 23.48000 37.00000 39.00000 40.00000
## [41] 21.91000 22.13000 23.46000 27.18000 26.88000 27.11000 26.04000 25.04065
## [49] 27.05000
```

```
unique(attitude$negative)
```

```
## [1] 20.00000 21.00000 17.00000 19.00000 18.00000 16.00000 24.00000 28.00000
## [9] 22.00000 26.00000 25.00000 27.00000 10.00000 15.00000 14.00000 13.00000
## [17] 17.55000 11.00000 12.00000 18.53000 9.00000 17.87716 8.00000 14.85000
## [25] 15.47000 23.00000 37.00000 16.22000 30.00000 17.98000 18.19000 32.00000
## [33] 29.00000 40.00000 18.77000 31.00000 19.75000 17.57000 14.89000 15.04000
## [41] 34.00000 39.00000 36.00000 35.00000 15.55000 15.94000 16.75610 15.10000
```

We see that we do not have missing values, and the data types for all columns already fit. We only need to convert personid to a string.

```
# convert to integer to remove leading zeros and then to str
attitude$personid <- as.character(as.integer(attitude$personid))
# remove duplicates on same personid and year_week
attitude <- attitude %>%
  distinct(personid, year_week, .keep_all = TRUE)
```

Clean endperiod

```
glimpse(endperiod)
```

```
## Rows: 135
## Columns: 4
## $ personid      <dbl> 4122, 6278, 7720, 8834, 8854, 10098, 10356, 12426, 1297~
## $ promote_switch <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, 1, 0~
## $ quitjob       <dbl> 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, 0~
## $ costofcommute <chr> "18", "12", "9", "0", "4", "12", "0", "0", "0", "0", "6.00"~
```

```
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean      St. Dev.   Min  Median  Max
## -----
## personid      135 32,716.780 10,835.700 4,122 37,292 45,442
## promote_switch 135 0.163      0.371      0      0      1
## quitjob       135 0.296      0.458      0      0      1
## -----
```

```
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
```

```
find_non_numeric_chars(endperiod$costofcommute)
```

```
## [1] "."      "." / " ."      " "      "." "NA"      "CNY ."
## [7] " "      ","      "."      "¥."      " "
```

```
find_non_numeric_rows(endperiod$costofcommute)
```

```
## [1] "6.00 "      "-"      "25.00 / "    "11.8181819915771"
## [5] " 14.00"     "NA"      "CNY 3.00"    " 6"
## [9] "2,00"       "17.7272720336914" " 0"          "4.54545450210571"
## [13] "2,00"       "0.00 "    "¥10.00"      " 10"
## [17] "0,00"
```

here we only need to take care of the decimal dots (will be added to the extraction function), any ot

No missing values, promote_switch and quitjob are logical values, though we also will create a numerical version (0-1) for them to compute statistics. costofcommute need to be converted to doubles.

```
# convert to integer to remove leading zeros an than to str
endperiod$personid <- as.character(as.integer(endperiod$personid))
# convert to integers
endperiod$promote_switch_num <- sapply(endperiod$promote_switch, as_integer)
endperiod$quitjob_num <- sapply(endperiod$quitjob, as_integer)
# convert to logical (too have numerical and categorical variables)
endperiod$promote_switch <- as.logical(endperiod$promote_switch_num)
endperiod$quitjob <- as.logical(endperiod$quitjob_num)
# extract numbers from string. Both, dots and commas are used as decimal separators
endperiod$costofcommute<- sapply(endperiod$costofcommute, extract_numeric)
# remove duplicates on same personid
endperiod <- endperiod %>%
  distinct(personid, .keep_all = TRUE)
```

```
glimpse(endperiod)
```

```
## Rows: 135
## Columns: 6
## $ personid      <chr> "4122", "6278", "7720", "8834", "8854", "10098", "1~
## $ promote_switch <lgl> FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FALSE, FA~
## $ quitjob        <lgl> FALSE, TRUE, FALSE, FALSE, FALSE, TRUE, FALSE, FALS~
## $ costofcommute  <dbl> 18.00000, 12.00000, 9.00000, 0.00000, 4.00000, 12.0~
## $ promote_switch_num <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 1, 0, 0, 0, ~
## $ quitjob_num    <int> 0, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 1, ~
```

```
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N Mean St. Dev. Min Median Max
## -----
## promote_switch 135 0.163 0.371    0    0    1
## quitjob        135 0.296 0.458    0    0    1
## costofcommute  127 7.308 7.195  0.000 6.000 55.000
## promote_switch_num 135 0.163 0.371    0    0    1
## quitjob_num    135 0.296 0.458    0    0    1
## -----
```

Cost of commute has two missing values, which needs to be considered during analysis.

Clean performance

```
glimpse(performance)
```

```
## Rows: 9,870
## Columns: 12
## $ personid      <dbl> 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122, 4122~
## $ year_week     <chr> "202201", "202202", "202203", "202204", "202205", "2~
## $ perform1      <chr> "-1.14160859584808", "0.414592355489731", "1.0139772~
## $ phonecall     <chr> "-1.13219404220581", "0.312170952558517", "1.0971519~
## $ phonecallraw  <chr> "223", "499", "649", "709", "403", "760", "268", "73~
## $ homethatweek  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ logphonecall  <chr> "5.40717172622681", "6.21260595321655", "6.475432872~
## $ logcallpersec <chr> "-5.10147857666016", "-5.15978050231934", "-5.142978~
## $ logcalllength <chr> "10.5086498260498", "11.372386932373", "11.618411064~
## $ logcall_dayworked <chr> "9.41003799438477", "9.58062744140625", "9.826651573~
## $ logdaysworked <chr> "1.0986123085022", "1.7917594909668", "1.79175949096~
## $ date          <chr> "2022-01-03", "2022-01-10", "2022-01-17", "2022-01-2~
```

```
stargazer(as.data.frame(performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean      St. Dev.   Min  Median  Max
## -----
## personid      9,870 32,493.740 10,623.780 4,122 36,908 45,442
## homethatweek  9,870  0.196      0.397      0      0      1
## -----
```

```
unique(performance$homethatweek)
```

```
## [1] 0 1
```

```
# check character columns with numerical values to extract for non numeric chars to
# be aware for special cases
```

```
find_non_numeric_chars(performance$perform1)
```

```
## [1] "." "-." ".e"
```

```
#find_non_numeric_rows(performance$perform1)
```

```
find_non_numeric_chars(performance$phonecall)
```

```
## [1] "." "-."
```

```
find_non_numeric_chars(performance$phonecallraw)
```

```
## [1] "NA" " " " " "
```

```
find_non_numeric_chars(performance$homethatweek)
```

```
## character(0)
```

```
find_non_numeric_chars(performance$logphonecall)
```

```
## [1] "." "NA" ","
```

```
find_non_numeric_chars(performance$logcallpersec)
```

```
## [1] "." "NA" ","
```

```
find_non_numeric_chars(performance$logcalllength)
```

```
## [1] "." "NA" ","
```

```
find_non_numeric_chars(performance$logcall_dayworked)
```

```
## [1] "." "NA" ", "
```

```
find_non_numeric_chars(performance$logdaysworked)
```

```
## [1] "." ", "
```

here we need to take care of the decimal dots, dashes to negative(negative) and scientific notation w

Personid needs to be converted to a string, as well as all the call and performance measurements need to be converted to doubles. Homethatweek needs to be converted to a logical and numerical

```
# convert to integer to remove leading zeros an than to str
performance$personid <- as.character(as.integer(performance$personid))
# trim year_week and remove all not numerical characters
performance$year_week <- trimws(performance$year_week)
performance$year_week<- gsub("[^0-9.]", "", performance$year_week)
# convert to float/doubles
performance$perform1 <- sapply(performance$perform1, as_numeric)
performance$phonecall <- sapply(performance$phonecall, as_numeric)
# convert to integer
performance$phonecallraw <- sapply(performance$phonecallraw, as_integer)
# convert to boolean
performance$homethatweek_num <- sapply(performance$homethatweek, as_integer)
performance$homethatweek <- sapply(performance$homethatweek, as_integer)
# convert to float/doubles
performance$logphonecall <- sapply(performance$logphonecall, as_numeric)
performance$logcallpersec <- sapply(performance$logcallpersec, as_numeric)
performance$logcalllength <- sapply(performance$logcalllength, as_numeric)
performance$logcall_dayworked <- sapply(performance$logcall_dayworked, as_numeric)
performance$logdaysworked <- sapply(performance$logdaysworked, as_numeric)
# convert to Date
performance$date <- as.Date(performance$date)

# remove duplicates on same personid, year_week
performance <- performance %>%
  distinct(personid, year_week, .keep_all = TRUE)
```

```
glimpse(performance)
```

```
## Rows: 9,870
## Columns: 13
## $ personid      <chr> "4122", "4122", "4122", "4122", "4122", "4122", "412~
## $ year_week     <chr> "202201", "202202", "202203", "202204", "202205", "2~
## $ perform1      <dbl> -1.1416086, 0.4145924, 1.0139773, 1.4812315, -0.1150~
## $ phonecall     <dbl> -1.13219404, 0.31217095, 1.09715199, 1.41114438, -0.~
## $ phonecallraw  <dbl> 223, 499, 649, 709, 403, 760, 268, 73, 301, 629, 529~
## $ homethatweek  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ logphonecall  <dbl> 5.407172, 6.212606, 6.475433, 6.563856, 5.998937, 6.~
## $ logcallpersec <dbl> -5.101479, -5.159781, -5.142978, -5.161007, -5.15217~
## $ logcalllength <dbl> 10.508650, 11.372387, 11.618411, 11.724862, 11.15110~
## $ logcall_dayworked <dbl> 9.410038, 9.580627, 9.826652, 9.778953, 9.359349, 9.~
## $ logdaysworked <dbl> 1.098612, 1.791759, 1.791759, 1.945910, 1.791759, 1.~
## $ date          <date> 2022-01-03, 2022-01-10, 2022-01-17, 2022-01-24, 202~
## $ homethatweek_num <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
```



```
# only trimming needed
```

We have no numerical values yet. basewage, can be converted to integer, gross_wage and bonustotal can be converted to double

```
# convert to integer to remove leading zeros an than to str
```

```
wage$personid <- as.character(as.integer(wage$personid))
wage$basewage <- as.integer(wage$basewage)
```

```
## Warning: NAs introduced by coercion
```

```
wage$grosswage <- sapply(wage$grosswage, as_numeric)
wage$wage_month <- trimws(wage$wage_month)
wage$wage_month <- gsub("[^0-9.]", "", wage$wage_month)
wage <- rename(wage, year_month=wage_month)
wage$bonustotal <- sapply(wage$bonustotal, as_numeric)
# remove duplicates on same personid, yearmonth
wage <- wage %>%
  distinct(personid, year_month, .keep_all = TRUE)
```

```
glimpse(wage)
```

```
## Rows: 3,007
## Columns: 5
## $ basewage    <int> 1650, 1650, 1650, 1650, 1650, 1650, 1800, 1800, 1800, 1800,~
## $ grosswage   <dbl> 3650.76, 4775.52, 4358.00, 4801.00, 4045.12, 5497.76, 3123.~
## $ personid    <chr> "4122", "4122", "4122", "4122", "4122", "4122", "4122", "41~
## $ year_month  <chr> "202201", "202202", "202203", "202204", "202205", "202206",~
## $ bonustotal  <dbl> 2000.76, 3125.52, 2708.00, 3151.00, 2395.12, 3847.76, 1323.~
```

```
stargazer(as.data.frame(wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.   Min     Median     Max
## -----
## basewage      2,987 1,666.271  221.174    650     1,600     2,850
## grosswage     2,978 3,135.544 1,066.355  48.850  2,979.950 14,553.000
## bonustotal    2,993 1,502.263  926.276    0.000   1,325.170 12,853.000
## -----
```

All columns again contain missing data. Grosswage is not the sum of basewage+bonustotal all the time:
Cases:

- NA in basewage and grosswage or basewage and bonus total or bonustotal and grosswage: remove row
- NA in basewage but grosswage and bonustotal are given: $\text{basewage} = \text{grosswage} - \text{bonustotal}$
- NA in bonustotal but grosswage and basewage are given: $\text{bonustotal} = \text{grosswage} - \text{basewage}$
- $\text{Grosswage} \neq \text{basewage} + \text{bonustotal}$: recalculate
- wage yearmonth sometimes misses the month

```
# case 2
```

```
wage <- wage %>% filter(!is.na(basewage) & !is.na(grosswage) | !is.na(bonustotal) & !is.na(grosswage) |
```

```
# case 3
```

```
wage <- wage %>% mutate(basewage = ifelse(is.na(basewage), grosswage-bonustotal, basewage))
```

```

# case 4
wage <- wage %>% mutate(bonustotal = ifelse(is.na(bonustotal), grosswage-basewage, bonustotal))

# case 5
wage <- wage %>% mutate(grosswage = ifelse(grosswage != bonustotal+basewage, bonustotal+basewage, grosswage))

# case 6
wage <- wage %>% filter(nchar(year_month)==6)

```

Merging data together

Merging

```

# First we merge all volunteer related data together on personid (unique for a person)
volunteer_endperiod <- merge(volunteer, endperiod, by="personid", all=TRUE)
# Second we merge all weekly based timeseries data together per personid
attitude_performance <- merge(attitude, performance, by=c("personid", "year_week"), all=TRUE)
# As wage only has a yearmonth column, we cannot directly merge it with an other table containing times
# The goal now is to create one big dataframe with all volunteer information, their performance, attitude
attitude_performance <- attitude_performance %>%
  mutate(
    week_date = ISOweek2date(
      paste0(substr(year_week,1,4), "-W", substr(year_week,5,6), "-1")
    ),
    year_month = format(week_date, "%Y%m") # gives e.g. "202401", "202412"
  )
# merge on year_month
attitude_performance_wage <- merge(attitude_performance, wage, by=c("personid", "year_month"), all.x=TRUE)
# merge on personid
volunteer_endperiod_attitude_performance_wage <- merge(attitude_performance_wage, volunteer_endperiod, by="personid", all=TRUE)

# lets try to get a dataframe with ALL data and no NAs at all
distinct_all <- na.omit(volunteer_endperiod_attitude_performance_wage)

```

Analyzing merged dataframes

```
stargazer(as.data.frame(volunteer_endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N   Mean   St. Dev.   Min   Median   Max
## -----
## age            135 23.452   4.315     1     23     35
## tenure         135 23.385   24.608     2     13    150
## children       135  0.148   0.357     0      0      1
## bedroom        135  0.978   0.148     0      1      1
## commute        122 104.721  68.544     1     85    300
## high_educ      135  0.415   0.495     0      0      1
## gender_num     135  0.496   0.502     0      0      1
## married_num    135  0.200   0.401     0      0      1
## high_educ_num  135  0.415   0.495     0      0      1
## children_num   135  0.148   0.357     0      0      1

```

```
## promote_switch      135  0.163  0.371    0    0    1
## quitjob             135  0.296  0.458    0    0    1
## costofcommute       127  7.308  7.195   0.000 6.000 55.000
## promote_switch_num  135  0.163  0.371    0    0    1
## quitjob_num         135  0.296  0.458    0    0    1
## -----
```

looks good

```
stargazer(as.data.frame(attitude_performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.  Min    Median  Max
## -----
## exhaustion      2,379  8.612    7.794    0.000  7.000  36.000
## negative        2,379 16.651    6.848    8.000 16.000 40.000
## positive        2,379 24.215    6.668    8.000 24.000 40.000
## perform1        9,827 -0.015    0.988   -3.031  0.049  4.163
## phonecall       9,716 -0.006    0.963   -3.113  0.070  5.828
## phonecallraw    9,561 440.258 142.558    1     445   1,264
## homethatweek    9,870  0.196    0.397    0      0      1
## logphonecall    9,574  6.009    0.481    0.000  6.098  7.142
## logcallpersec   9,576 -5.170    0.160   -5.951 -5.169 -1.099
## logcalllength   9,576 11.175    0.521    2.485 11.273 12.116
## logcall_dayworked 9,576  9.474    0.457    2.485  9.552 10.359
## logdaysworked  9,855  1.685    0.282    0.000  1.792  1.946
## homethatweek_num 9,870  0.196    0.397    0      0      1
## -----
```

when using this dataframe we have to be cautious as attitude has way less columns than attitude n(att

```
stargazer(as.data.frame(attitude_performance_wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.  Min    Median  Max
## -----
## exhaustion      2,379  8.612    7.794    0.000  7.000  36.000
## negative        2,379 16.651    6.848    8.000 16.000 40.000
## positive        2,379 24.215    6.668    8.000 24.000 40.000
## perform1        9,827 -0.015    0.988   -3.031  0.049  4.163
## phonecall       9,716 -0.006    0.963   -3.113  0.070  5.828
## phonecallraw    9,561 440.258 142.558    1     445   1,264
## homethatweek    9,870  0.196    0.397    0      0      1
## logphonecall    9,574  6.009    0.481    0.000  6.098  7.142
## logcallpersec   9,576 -5.170    0.160   -5.951 -5.169 -1.099
## logcalllength   9,576 11.175    0.521    2.485 11.273 12.116
## logcall_dayworked 9,576  9.474    0.457    2.485  9.552 10.359
## logdaysworked  9,855  1.685    0.282    0.000  1.792  1.946
## homethatweek_num 9,870  0.196    0.397    0      0      1
## basewage        10,023 1,601.750 176.618  527.350 1,600.000 2,450.000
## grosswage       9,919  3,097.672 990.788  1,140.280 2,877.000 14,553.000
## bonustotal      10,023 1,494.926 912.498    0.000  1,284.590 12,853.000
## -----
```

```
stargazer(as.data.frame(volunteer_endperiod_attitude_performance_wage), type = "text", median = TRUE, h
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## exhaustion      2,379    8.612    7.794      0.000     7.000     36.000
## negative        2,379   16.651    6.848      8.000    16.000    40.000
## positive        2,379   24.215    6.668      8.000    24.000    40.000
## perform1        9,827   -0.015    0.988     -3.031     0.049     4.163
## phonecall       9,716   -0.006    0.963     -3.113     0.070     5.828
## phonecallraw    9,561  440.258  142.558      1       445     1,264
## homethatweek    9,870    0.196    0.397      0         0         1
## logphonecall    9,574    6.009    0.481      0.000     6.098     7.142
## logcallpersec   9,576   -5.170    0.160     -5.951    -5.169    -1.099
## logcalllength   9,576   11.175    0.521      2.485    11.273    12.116
## logcall_dayworked 9,576    9.474    0.457      2.485     9.552    10.359
## logdaysworked  9,855    1.685    0.282      0.000     1.792     1.946
## homethatweek_num 9,870    0.196    0.397      0         0         1
## basewage       10,023  1,601.750 176.618   527.350  1,600.000 2,450.000
## grosswage       9,919  3,097.672 990.788   1,140.280 2,877.000 14,553.000
## bonustotal      10,023  1,494.926 912.498    0.000    1,284.590 12,853.000
## age            10,079   23.485    4.191      1         23         35
## tenure         10,079   23.893    24.773      2         18        150
## children       10,079    0.141    0.348      0         0         1
## bedroom        10,079    0.974    0.160      0         1         1
## commute        9,118  103.519   67.620      1         80        300
## high_educ      10,079    0.400    0.490      0         0         1
## gender_num     10,079    0.498    0.500      0         0         1
## married_num    10,079    0.181    0.385      0         0         1
## high_educ_num  10,079    0.400    0.490      0         0         1
## children_num   10,079    0.141    0.348      0         0         1
## promote_switch  10,079    0.182    0.386      0         0         1
## quitjob        10,079    0.238    0.426      0         0         1
## costofcommute   9,488    7.213    7.231      0.000     6.000    55.000
## promote_switch_num 10,079    0.182    0.386      0         0         1
## quitjob_num    10,079    0.238    0.426      0         0         1
## -----
```

```
stargazer(as.data.frame(distinct_all), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## exhaustion      1,785    8.983    8.105      0.000     8.000     36.000
## negative        1,785   16.699    7.088      8.000    16.000    40.000
## positive        1,785   23.718    6.730      8.000    24.000    40.000
## perform1        1,785    0.070    0.866     -2.906     0.096     2.772
## phonecall       1,785    0.030    0.814     -3.098     0.046     2.266
## phonecallraw    1,785  439.319  126.850      2       442     835
## homethatweek    1,785    0.508    0.500      0         1         1
## logphonecall    1,785    6.032    0.367      0.693     6.091     6.727
## logcallpersec   1,785   -5.158    0.129     -5.541    -5.157    -4.075
## logcalllength   1,785   11.191    0.375      5.561    11.242    11.952
## logcall_dayworked 1,785    9.465    0.354      3.951     9.506    10.104
## logdaysworked  1,785    1.726    0.182      0.693     1.792     1.946
```

```

## homethatweek_num    1,785    0.508    0.500    0    1    1
## basewage            1,785 1,683.305 158.587    1,300    1,650    2,150
## grosswage           1,785 3,328.000 1,046.364 1,740.000 3,095.000 14,553.000
## bonustotal          1,785 1,644.695 969.980    40.000    1,408.500 12,853.000
## age                 1,785    23.519    3.254    18    23    34
## tenure              1,785    23.685    26.683    2    13    150
## children            1,785    0.076    0.264    0    0    1
## bedroom             1,785    0.941    0.235    0    1    1
## commute             1,785 104.871    67.073    20    80    300
## high_educ           1,785    0.345    0.475    0    0    1
## gender_num          1,785    0.459    0.498    0    0    1
## married_num         1,785    0.111    0.314    0    0    1
## high_educ_num       1,785    0.345    0.475    0    0    1
## children_num        1,785    0.076    0.264    0    0    1
## promote_switch      1,785    0.182    0.386    0    0    1
## quitjob             1,785    0.000    0.000    0    0    0
## costofcommute       1,785    6.749    6.160    0.000    5.000    30.000
## promote_switch_num  1,785    0.182    0.386    0    0    1
## quitjob_num         1,785    0.000    0.000    0    0    0
## -----

```

When joining everything together we can see that we do not have all data for each person and its work weeks, what was to be expected. When removing all NAs we can see that the numbers slightly change and for examples all quitters are gone. Therefore we have to use the `volunteer_endperiod_attitude_performance_wage` with caution and eliminate NAs for each calculation individually. Though this dataframe is handy to use for the further work.

`volunteer_endperiod`: Cross sectional data of all employee related information.

`attitude_performance`: Panel data on how an employee performed in a certain week of the year and how he/she felt.

`attitude_performance_wage`: Panel data on how an employee performed in a certain week of the year, how he/she felt in that week and their wage during that month. Caution: contains two different time series.

`volunteer_endperiod_attitude_performance_wage`: Like `attitude_performance_wage`, but with `volunteer_endperiod` data.

General Analysis

Population: All employees of Huaxia

Our sample: Depending on the table we use and what we collect the unit of interest might differ:

Attitude, performance (time series): weekly observations of a person

volunteer (cross sectional): the person itself

wage (time series): the wage of a person in a month

endperiod (cross sectional): behavior of a person(?)

`volunteer_endperiod_attitude_performance_wage`(panel data): repeated observations of different employees across multiple time periods, combining attitudes performance, volunteer, endperiod, and wages

Get first basic insights of the tables before eliminating null values and merging data together. For a better overview we will look at each table alone, not on the composition table.

Statistics

Performance

```
stargazer(as.data.frame(performance), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.   Min    Median   Max
## -----
## perform1      9,827 -0.015    0.988   -3.031  0.049   4.163
## phonecall     9,716 -0.006    0.963   -3.113  0.070   5.828
## phonecallraw  9,561 440.258 142.558    1     445   1,264
## homethatweek  9,870  0.196    0.397    0      0      1
## logphonecall  9,574  6.009    0.481    0.000  6.098   7.142
## logcallpersec 9,576 -5.170    0.160   -5.951 -5.169  -1.099
## logcalllength 9,576 11.175    0.521    2.485 11.273 12.116
## logcall_dayworked 9,576  9.474    0.457    2.485  9.552 10.359
## logdaysworked 9,855  1.685    0.282    0.000  1.792  1.946
## homethatweek_num 9,870  0.196    0.397    0      0      1
## -----
```

```
# perform plot
stats <- performance %>% summarise(
  min = min(perform1, na.rm = TRUE),
  max = max(perform1, na.rm = TRUE),
  mean = mean(perform1, na.rm = TRUE),
  median = median(perform1, na.rm = TRUE),
  sd = sd(perform1, na.rm = TRUE)
)

p<-ggplot(performance, aes(y = perform1)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Performance Score", title = "Distribution of performances per week of employees") +
  theme_minimal() +
  theme(axis.title.x = element_blank(),
        axis.text.x = element_blank(),
        axis.ticks.x = element_blank())

# add annotations
p + annotate("text", x = 0.27, y = 4,
            label = paste0("SD: ", round(stats$sd,2)),
            hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 3.6,
            label = paste0("Mean: ", round(stats$mean,2)),
            text = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 3.2,
            label = paste0("Median: ", round(stats$median,2)),
            hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 2.8,
```

```

label = paste0("Min: ", round(stats$min,2)),
hjust = 0, color = "#0d0c3f") +
annotate("text", x = 0.27, y = 2.4,
label = paste0("Max: ", round(stats$max,2)),
hjust = 0, color = "#0d0c3f")

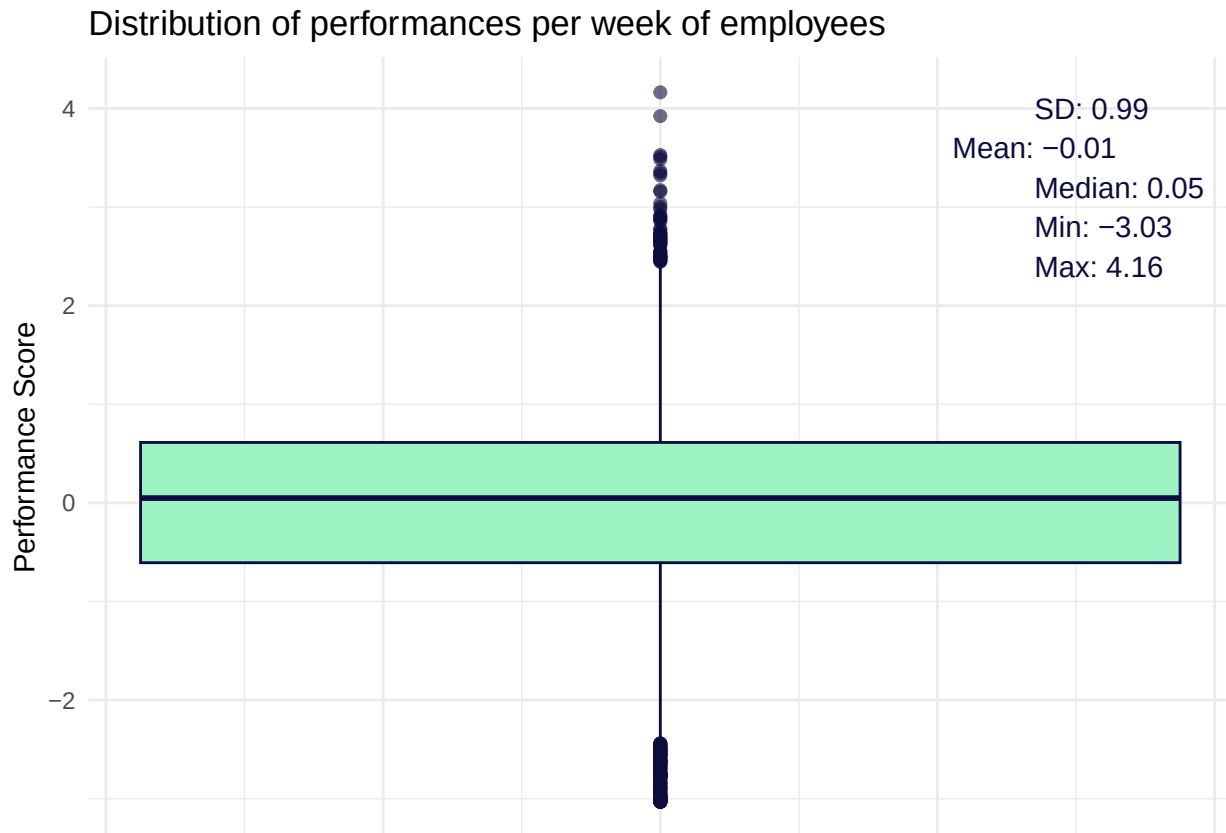
```

```

## Warning in annotate("text", x = 0.27, y = 3.6, label = paste0("Mean: ", :
## Ignoring unknown parameters: `text`

## Warning: Removed 43 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



```

ggsave("performance_distribution.png", width = 7, height = 5, dpi = 300)

```

```

## Warning: Removed 43 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```

A lot of missing values, 296 at max, which will fall out from the data set when working with it.
Working from home rate is at 19,64% on average, probably as a result of a Work-from-home-policy.

Volunteer

```

stargazer(as.data.frame(volunteer), type = "text", median = TRUE, header = TRUE)

```

```

##
## =====
## Statistic      N   Mean   St. Dev. Min Median Max
## -----
## age            135 23.452   4.315    1    23   35

```

```
## tenure      135 23.385  24.608  2   13   150
## children    135  0.148  0.357  0    0    1
## bedroom     135  0.978  0.148  0    1    1
## commute     122 104.721  68.544  1   85   300
## high_educ   135  0.415  0.495  0    0    1
## gender_num  135  0.496  0.502  0    0    1
## married_num 135  0.200  0.401  0    0    1
## high_educ_num 135  0.415  0.495  0    0    1
## children_num 135  0.148  0.357  0    0    1
## -----
```

No missing values. The average worker is 23,5 years old with a mean commute time of 100 minutes per week while there are greater differences, and is for almost 23 months within the company. 15% have children, 20% are married, more than 40% have a higher education. Both, men and women work at the company.

Endperiod

```
stargazer(as.data.frame(endperiod), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N  Mean  St. Dev.  Min  Median  Max
## -----
## promote_switch 135 0.163  0.371    0    0      1
## quitjob        135 0.296  0.458    0    0      1
## costofcommute  127 7.308  7.195   0.000 6.000  55.000
## promote_switch_num 135 0.163  0.371    0    0      1
## quitjob_num    135 0.296  0.458    0    0      1
## -----
```

On average, an employee had cost of commute of 7.5 but this number may heavily differ to some employees, 16% got a promotion, while almost 30% quit their job at some point.

Wage

```
stargazer(as.data.frame(wage), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N    Mean    St. Dev.    Min    Median    Max
## -----
## basewage      2,997 1,665.835  222.317   527.350   1,600.000 2,850.000
## grosswage     2,968 3,170.622 1,023.961   53.100   2,986.500 14,553.000
## bonustotal    2,997 1,502.992  929.753  -1,596.900 1,325.170 12,853.000
## -----
```

```
stats <- wage %>% summarise(
  min = min(grosswage, na.rm = TRUE),
  max = max(grosswage, na.rm = TRUE),
  mean = mean(grosswage, na.rm = TRUE),
  median = median(grosswage, na.rm = TRUE),
  sd = sd(grosswage, na.rm = TRUE)
)
```

```
p <- ggplot(wage, aes(y = grosswage)) +
```

```

geom_boxplot(
  outlier.alpha = 0.6,
  outlier.size = 1.8,
  fill = "#9df3c4",
  color = "#0d0c3f"
) +
labs(y = "€ per month", title = "Distribution of Gross Wages") +
theme_minimal()

# add annotations
p + annotate("text", x = 0.27, y = 14000,
  label = paste0("SD: ", round(stats$sd,0), "€"),
  hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 13000,
    label = paste0("Mean: ", round(stats$mean,0), "€"),
    hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 0.27, y = 12000,
    label = paste0("Median: ", round(stats$median,0), "€"),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 11000,
    label = paste0("Min: ", round(stats$min,0), "€"),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 0.27, y = 10000,
    label = paste0("Max: ", round(stats$max,0), "€"),
    hjust = 0, color = "#0d0c3f")

## Warning: Removed 29 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



```
ggsave("gross_wage_distribution.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 29 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

On average an employee had a base wage of around 1650€ which can slightly differ. A heavily uneven distribution can be seen for the bonuses for employees, while some refer no bonus at all others receive almost 13.000€ as bonus, on average it is 900€ per employee.

Attitude

```
stargazer(as.data.frame(attitude), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic    N    Mean  St. Dev.  Min  Median  Max
## -----
## exhaustion  2,379  8.612   7.794    0.000  7.000   36.000
## negative    2,379  16.651   6.848    8.000  16.000  40.000
## positive    2,379  24.215   6.668    8.000  24.000  40.000
## -----
```

Exhaustion scores are mostly low and clustered around 7–9 but some people reach much higher (up to 36), negative scores center near 16 with moderate variation across the group, and positive scores center near 24 with a similar moderate spread, meaning most participants stay within roughly one standard deviation (about ± 7 points) of these averages but a few reach the extreme minimum or maximum values

All together

```
stargazer(as.data.frame(volunteer_endperiod_attitude_performance_wage), type = "text", median = TRUE, h
```

```
##
## =====
## Statistic          N      Mean    St. Dev.    Min      Median      Max
## -----
## exhaustion         2,379     8.612     7.794     0.000     7.000     36.000
## negative           2,379    16.651     6.848     8.000    16.000     40.000
## positive           2,379    24.215     6.668     8.000    24.000     40.000
## perform1           9,827    -0.015     0.988    -3.031     0.049     4.163
## phonecall          9,716    -0.006     0.963    -3.113     0.070     5.828
## phonecallraw       9,561   440.258   142.558      1      445     1,264
## homethatweek       9,870     0.196     0.397      0        0        1
## logphonecall       9,574     6.009     0.481     0.000     6.098     7.142
## logcallpersec      9,576    -5.170     0.160    -5.951    -5.169    -1.099
## logcalllength      9,576    11.175     0.521     2.485    11.273    12.116
## logcall_dayworked  9,576     9.474     0.457     2.485     9.552    10.359
## logdaysworked     9,855     1.685     0.282     0.000     1.792     1.946
## homethatweek_num   9,870     0.196     0.397      0        0        1
## basewage           10,023  1,601.750  176.618   527.350  1,600.000  2,450.000
## grosswage           9,919  3,097.672  990.788  1,140.280  2,877.000  14,553.000
## bonustotal          10,023  1,494.926  912.498     0.000   1,284.590  12,853.000
## age                 10,079    23.485     4.191      1       23       35
## tenure              10,079    23.893    24.773      2       18      150
## children            10,079     0.141     0.348      0        0        1
## bedroom            10,079     0.974     0.160      0        1        1
## commute             9,118   103.519    67.620      1       80      300
## high_educ           10,079     0.400     0.490      0        0        1
## gender_num          10,079     0.498     0.500      0        0        1
## married_num         10,079     0.181     0.385      0        0        1
## high_educ_num       10,079     0.400     0.490      0        0        1
## children_num        10,079     0.141     0.348      0        0        1
## promote_switch      10,079     0.182     0.386      0        0        1
## quitjob             10,079     0.238     0.426      0        0        1
## costofcommute       9,488     7.213     7.231     0.000     6.000    55.000
## promote_switch_num  10,079     0.182     0.386      0        0        1
## quitjob_num         10,079     0.238     0.426      0        0        1
## -----
##
```

```
df <- volunteer_endperiod_attitude_performance_wage
df <- dplyr::select(df, where(is.numeric))

# create a summary df as preparation for the radar plot rendering
create_summary <- function(df, drop_all_na = TRUE, keep_group_index = TRUE) {
  num <- df %>%
    select(where(is.numeric)) %>%
    mutate(across(everything(), ~ ifelse(is.finite(.x), .x, NA_real_)))

  if (drop_all_na) {
    keep <- vapply(num, function(x) any(!is.na(x)), logical(1))
    num <- num[, keep, drop = FALSE]
  }
  stopifnot(ncol(num) > 0)
```

```

summary <- num %>%
  summarise(
    across(everything(),
      list(mean = ~mean(.x, na.rm = TRUE),
           min  = ~min(.x,  na.rm = TRUE),
           max  = ~max(.x,  na.rm = TRUE)),
      .names = "{.col}_{.fn}")
  )
summary

}

# Expects a dataframe with *_min, *_max, *_mean columns
render_radar_plot <- function (df, pad = 1e-6) {
  # separate min, max, mean
  max_vals <- df %>% select(ends_with("_max")) %>% rename_with(~ sub("_max$", "", .x))
  min_vals <- df %>% select(ends_with("_min")) %>% rename_with(~ sub("_min$", "", .x))
  mean_vals <- df %>% select(ends_with("_mean")) %>% rename_with(~ sub("_mean$", "", .x))

  # avoid zero-range axes (min == max) that break fmsb scaling
  rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
  z <- which(is.na(rng) | rng == 0)
  if (length(z)) {
    max_vals[1, z] <- as.numeric(max_vals[1, z]) + pad
    min_vals[1, z] <- as.numeric(min_vals[1, z]) - pad
  }

  web <- rbind(max_vals, min_vals, mean_vals)
  web[] <- lapply(web, as.numeric)

  radarchart(web, na.itp = TRUE)
}

radar_by_group <- function(df, group_col = "group_index", stat = "mean") {
  num <- df %>%
    mutate(across(where(is.numeric), ~ ifelse(is.finite(.x), .x, NA_real_)))

  # 1) aggregate per group
  fun <- switch(stat,
    mean = ~mean(.x, na.rm = TRUE),
    median = ~median(.x, na.rm = TRUE),
    stop("stat must be 'mean' or 'median'"))
  grp <- num %>%
    group_by(.data[[group_col]]) %>%
    summarise(across(where(is.numeric), fun), .groups = "drop")

  # 2) build fmsb input: first two rows = per-variable max/min over all groups
  vals <- grp %>% select(-all_of(group_col))
  rownames(vals) <- grp[[group_col]]
  max_vals <- summarise_all(vals, max, na.rm = TRUE)
  min_vals <- summarise_all(vals, min, na.rm = TRUE)

  # pad zero-range axes

```

```

rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
z <- which(is.na(rng) | rng == 0)
if (length(z)) {
  max_vals[1, z] <- as.numeric(max_vals[1, z]) + 1e-6
  min_vals[1, z] <- as.numeric(min_vals[1, z]) - 1e-6
}

web <- rbind(max_vals, min_vals, vals)

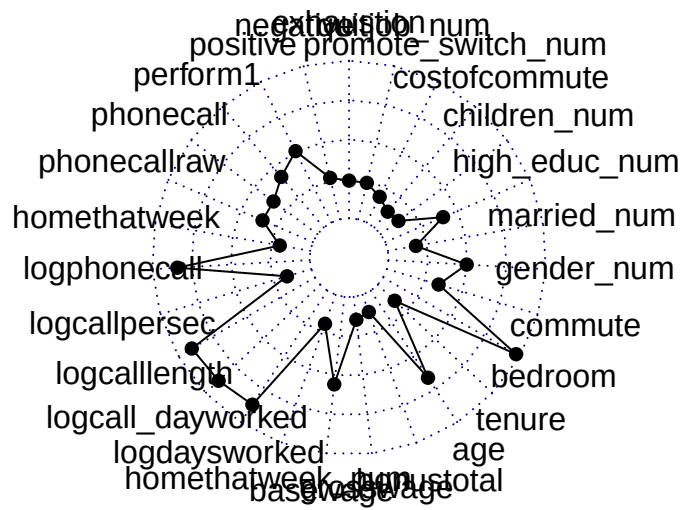
# 3) colors and legend
n <- nrow(web) - 2
pal <- brewer.pal(max(3, min(8, n)), "Set2")[seq_len(n)]
fill <- adjustcolor(pal, alpha.f = 0.25)

radarchart(
  web,
  pcol = pal,
  pfcpl = fill,
  plwd = 2,
  plty = 1,
  cglcol = "grey80",
  cglty = 1,
  cglwd = 0.8,
  axislabcol = "grey30",
  vlce = 0.8
)

legend("topright",
  legend = rownames(web)[-c(1,2)],
  bty = "n", pch = 15, pt.cex = 1.3,
  col = fill, text.col = "grey20")
}

summary <- create_summary(df)
render_radar_plot(summary)

```

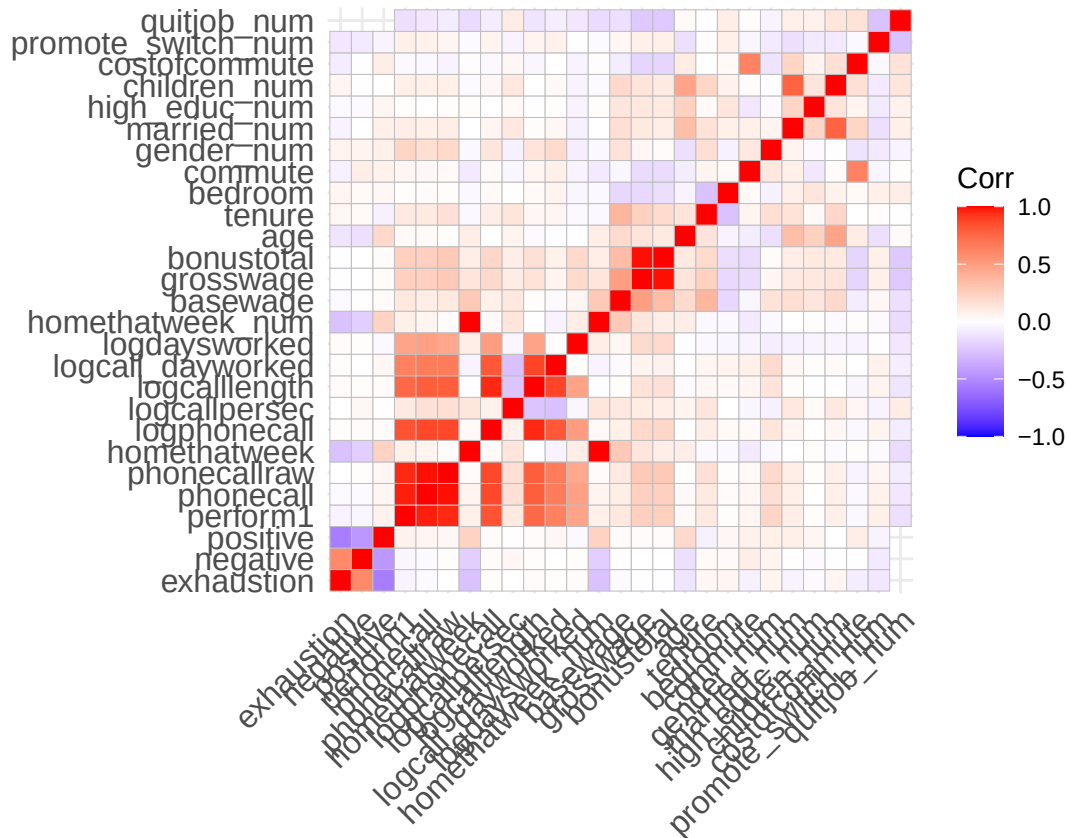


Basic correlation analysis

```
nums <- dplyr::select(volunteer_endperiod_attitude_performance_wage, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



Here we can see that the performance metric “perform1” has a correlation to phonecall, phonecallraw, logphonecall, logcalllength, which gives us a hint on how this metrics was calculated. Furthermore we can see, a negative correlation between “negative” and “positive” from the attitude dataset, which makes sense as these are opposites. Because of the timeseries data of the performance data set, the light correlation can be explained as the attitude may change from day to day, or not (what we can not infer from that). There is also a slight correlation between exhaustion and negative attitude, which seems logical. What we cannot see are significant correlations between quit_job and any variable and other than the cluster itself between perform1 and any other variable. Quit_job has moderate negative correlations to bonuses/grosswage and promotions.

We cannot see any clear indicators on how to fight churn rate and/or what motivates people to perform better.

Detailed exploration analysis

Function for radar plots

```
# A function to render web plots for our data with some modificatino parameters
# The function first selects the desired column and than summarises the data based on what groups to co
radar_by_group <- function(df, include_group_indices = c(), group_col = "group_index", stat = "mean", ti
  # open device to save as png
  if(save==TRUE){
    png(paste0(title, ".png"), width = 2400, height = 2000, res = 300)
  }
  # check for parameters
  stopifnot(group_col %in% names(df))
  # select, based on parameters, columns
  final_selection = c(group_col, "gender_num", "married_num", "bedroom", "children_num", "high_educ_num"
  if (show_full_attitude==TRUE) {
    selection = c("negative", "positive")
    final_selection <- c(selection, final_selection)
  }
  if (show_full_performance==TRUE) {
    selection = c("logdaysworked", "logcall_dayworked", "logcalllength", "logcallpersec", "logphonecall", "pl
    final_selection <- c(selection, final_selection)
  }
  if (show_full_wage==TRUE) {
    selection = c("bonustotal", "basewage")
    final_selection <- c(selection, final_selection)
  }
  if (show_commute==TRUE) {
    selection = c("commute", "costofcommute")
    final_selection <- c(selection, final_selection)
  }
  # chek if group col is in any of the selected field, if so, remove it
  if (group_col %in% sub("_num$", "", final_selection)){
    final_selection <- final_selection[final_selection!=paste0(group_col, "_num")]
  }
  print(final_selection)
  df <- df %>% select(final_selection)
  # only used numeric columns (+ group column) for computation and only real numbers
  num <- df %>%
    select(all_of(group_col), where(is.numeric)) %>%
    mutate(across(where(is.numeric), ~ ifelse(is.finite(.x), .x, NA_real_)))

  # remove "_num" to shorten names of var
  num <- num %>% rename_with(~ sub("_num$", "", .x))
  # select the min max values based on the 90%/10% quantile of the overall data
  max_vals <- summarise(num, across(where(is.numeric), ~ quantile(.x, 0.9, na.rm = TRUE)))
  min_vals <- summarise(num, across(where(is.numeric), ~ quantile(.x, 0.1, na.rm = TRUE)))
  # aggregate per group (mean or median)
  agg_fun <- switch(stat,
    mean = ~mean(.x, na.rm = TRUE),
    median = ~median(.x, na.rm = TRUE),
    stop("stat must be 'mean' or 'median'"))
  vals <- num %>%
    group_by(.data[[group_col]]) %>%
```

```

    summarise(across(where(is.numeric), agg_fun), .groups = "drop")

# get the correct label
if (length(include_group_indices)>0){
  print(vals)
  print(group_col)
  print(include_group_indices)
  labels <- vals %>% filter(.data[[group_col]] %in% include_group_indices)
  print(labels)
  labels <- as.character(labels[[group_col]])
  print(labels)
}

# again check for not numeric values and drop them if needed
keep <- vapply(vals, function(x) any(!is.na(x)), logical(1))
vals <- vals[, keep, drop = FALSE]
# check for enough values to display
if (ncol(vals) < 3) stop("Need 3 valid numeric variables after cleaning.")
# fill remaining NA cells with column means to avoid NA polygons
cm <- vapply(vals, function(x) mean(x, na.rm = TRUE), numeric(1))
for (j in seq_along(vals)) vals[[j]][is.na(vals[[j]])] <- cm[j]
# per-variable max/min, pad zero ranges
rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
z <- which(is.na(rng) | rng == 0)
pad <- 1e-6
rng <- as.numeric(max_vals[1, ] - min_vals[1, ])
z <- which(is.na(rng) | rng == 0)
if (length(z)) {
  mx <- as.numeric(max_vals[1, ])
  mn <- as.numeric(min_vals[1, ])
  mx[z] <- mx[z] + pad
  mn[z] <- mn[z] - pad
  max_vals[1, ] <- as.list(mx)
  min_vals[1, ] <- as.list(mn)
}

# ensure base data.frames to simplify row assignment
max_vals <- as.data.frame(max_vals)
min_vals <- as.data.frame(min_vals)
# sub select values from desired groups
if (length(include_group_indices)>0) {
  vals <- vals %>% filter(.data[[group_col]] %in% include_group_indices)
}

# remove group index to not be displayed as category in the web chart
vals <- vals %>% select(-all_of(group_col))
if (group_col %in% colnames(max_vals)) {
  max_vals <- max_vals %>% select(-all_of(group_col))
}
if (group_col %in% colnames(min_vals)) {
  min_vals <- min_vals %>% select(-all_of(group_col))
}

# cap values if they exceed the bounds
vals <- vals %>%
  mutate(across(
    everything(),

```

```

    ~ pmax(
      pmin(.x, max_vals[[cur_column()]] [1]), # upper bound for this var
      min_vals[[cur_column()]] [1]          # lower bound for this var
    )
  ))
# put everything together
web <- rbind(max_vals, min_vals, vals)
web[] <- lapply(web, function(x) as.numeric(x)) # ensure numeric matrix

# # make it look nice
n <- nrow(web) - 2
cols <- grDevices::hcl(h = seq(15, 375, length.out = n + 1)[1:n],
  c = 80, l = 60)
fills <- grDevices::adjustcolor(cols, alpha.f = 0.25)
cols <- c("#2CB1A6", # teal-mint
  "#1B1E3F", # deep navy
  "#94A3B8", # cool gray
  "#A7F3D0", # light mint
  "#F4A261") # muted coral
fills <- adjustcolor(cols, alpha.f = 0.25)
# plot it
radarchart(web,
  pcol = cols[seq_len(nrow(web)-2)],
  pfcol = fills[seq_len(nrow(web)-2)],
  plwd = 2, plty = 1,
  cglcol = "grey80", cglwd = 0.8, axislabcol = "grey30",
  vlcex = 1.2, na.itp = TRUE)
title(main = title)
# add a little legend
if(!is.null(group_col)){
  legend("topright", legend = labels,
    bty = "n", pch = 15, pt.cex = 1.3, col = cols[seq_len(nrow(web)-2)], text.col = "grey20")
}
# close device
if(save==TRUE){
  dev.off()
}
}

```

Quitter vs no-quitters

Lets compare quitter vs non-quitter to better understand both groups:

```

df <- volunteer_endperiod_attitude_performance_wage
quitter_df <- df %>% filter(quitjob == TRUE)
non_quitter_df <- df %>% filter(quitjob == FALSE)

# general impression
stargazer(as.data.frame(quitter_df), type = "text", median = TRUE, header = TRUE)

##
## =====
## Statistic          N      Mean    St. Dev.    Min      Median      Max
## -----

```

```
## perform1      2,383 -0.239  1.020  -3.031  -0.209  3.367
## phonecall     2,316 -0.182  1.036  -3.113  -0.145  5.828
## phonecallraw  2,269 419.852 150.717    1      419   1,264
## homethatweek  2,401  0.088  0.283    0      0      1
## logphonecall  2,267  5.939  0.563    0.000   6.038   7.142
## logcallpersec 2,264 -5.141  0.192  -5.721  -5.148  -1.099
## logcalllength 2,266 11.075  0.608    2.485  11.198  12.002
## logcall_dayworked 2,266  9.414  0.517    2.485   9.506  10.210
## logdaysworked 2,398  1.636  0.327    0.000   1.609   1.946
## homethatweek_num 2,401  0.088  0.283    0      0      1
## basewage      2,383 1,560.699 185.905  527.350 1,550.000 2,450.000
## grosswage     2,360 2,687.155 793.984 1,140.280 2,523.000 10,830.070
## bonustotal    2,383 1,125.618 725.063    0.000   971.690  9,330.070
## age           2,401 23.676  3.088    19     23     32
## tenure        2,401 23.439 28.649     2     10    150
## children       2,401  0.221  0.415     0      0      1
## bedroom        2,401  1.000  0.000     1      1      1
## commute        2,118 104.710 76.718     1    120    300
## high_educ      2,401  0.461  0.499     0      0      1
## gender_num     2,401  0.450  0.498     0      0      1
## married_num    2,401  0.239  0.427     0      0      1
## high_educ_num  2,401  0.461  0.499     0      0      1
## children_num   2,401  0.221  0.415     0      0      1
## promote_switch 2,401  0.000  0.000     0      0      0
## quitjob        2,401  1.000  0.000     1      1      1
## costofcommute  2,332  9.136 10.395    0.000   6.000   55.000
## promote_switch_num 2,401  0.000  0.000     0      0      0
## quitjob_num    2,401  1.000  0.000     1      1      1
## -----
```

```
stargazer(as.data.frame(non_quitter_df), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## exhaustion     2,379   8.612    7.794     0.000     7.000    36.000
## negative       2,379  16.651    6.848     8.000    16.000    40.000
## positive       2,379  24.215    6.668     8.000    24.000    40.000
## perform1       7,444   0.057    0.966    -3.031     0.119     4.163
## phonecall      7,400   0.049    0.932    -3.113     0.127     3.758
## phonecallraw   7,292  446.607  139.324     1      452    1,141
## homethatweek   7,469   0.231    0.422     0      0      1
## logphonecall   7,307   6.031    0.451     0.000     6.114     7.040
## logcallpersec  7,312  -5.179    0.147    -5.951    -5.175    -3.296
## logcalllength  7,310  11.205    0.487     3.989    11.296    12.116
## logcall_dayworked 7,310   9.492    0.435     2.639     9.567    10.359
## logdaysworked  7,457   1.701    0.263     0.000     1.792     1.946
## homethatweek_num 7,469   0.231    0.422     0      0      1
## basewage       7,640 1,614.554 171.635   650.000 1,600.000 2,300.000
## grosswage      7,559 3,225.839 1,011.125 1,264.430 3,010.000 14,553.000
## bonustotal     7,640 1,610.117 934.160    0.000   1,383.000 12,853.000
## age            7,678  23.426    4.479     1      23      35
## tenure         7,678  24.035   23.430     2      19     150
## children       7,678   0.116    0.320     0      0      1
```

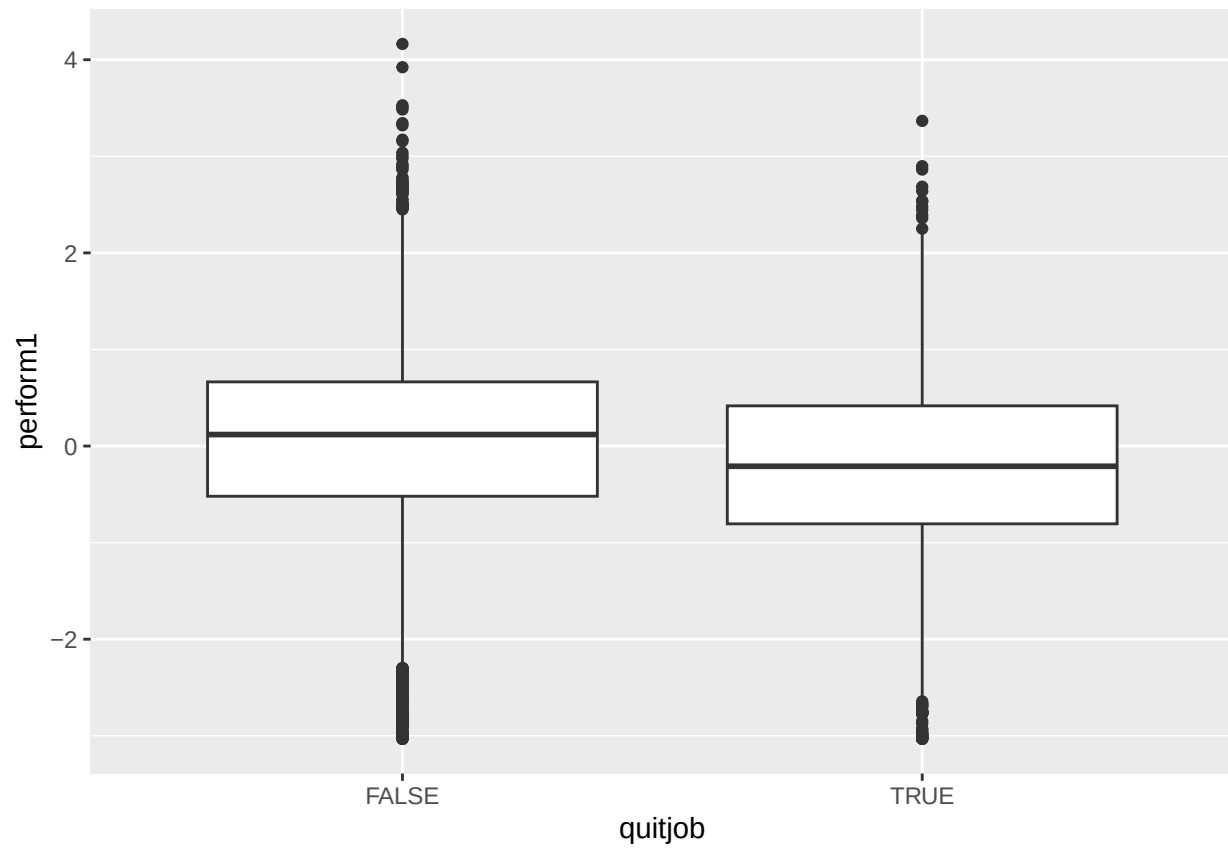
```
## bedroom          7,678    0.966    0.182      0        1        1
## commute          7,000   103.158   64.618      2       80       300
## high_educ        7,678    0.380    0.486      0        0        1
## gender_num       7,678    0.513    0.500      0        1        1
## married_num      7,678    0.163    0.369      0        0        1
## high_educ_num    7,678    0.380    0.486      0        0        1
## children_num     7,678    0.116    0.320      0        0        1
## promote_switch   7,678    0.239    0.427      0        0        1
## quitjob          7,678    0.000    0.000      0        0        0
## costofcommute    7,156    6.587    5.703    0.000    6.000    30.000
## promote_switch_num 7,678    0.239    0.427      0        0        1
## quitjob_num      7,678    0.000    0.000      0        0        0
## -----
```

```
stargazer(as.data.frame(df), type = "text", median = TRUE, header = TRUE)
```

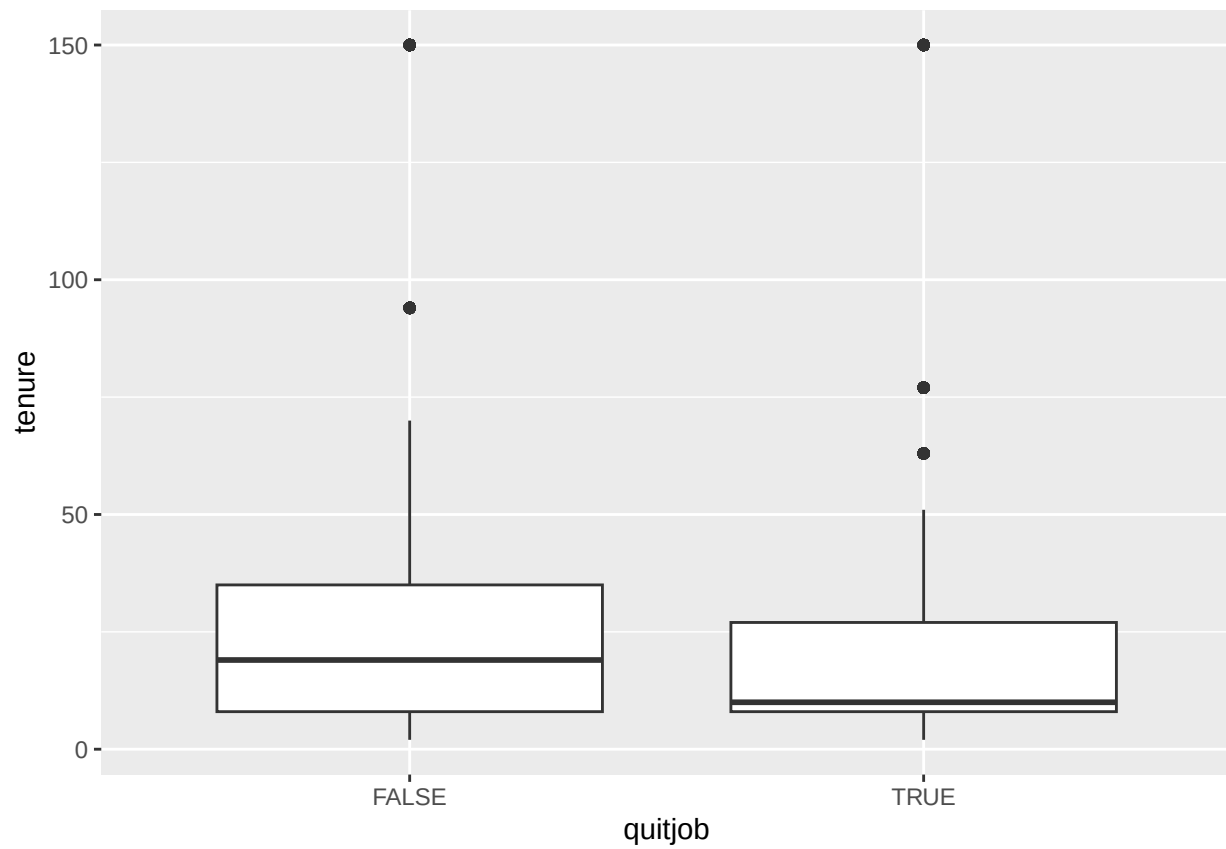
```
##
## =====
## Statistic          N      Mean    St. Dev.    Min      Median      Max
## -----
## exhaustion         2,379     8.612     7.794     0.000     7.000    36.000
## negative           2,379    16.651     6.848     8.000    16.000    40.000
## positive           2,379    24.215     6.668     8.000    24.000    40.000
## perform1           9,827    -0.015     0.988    -3.031     0.049     4.163
## phonecall          9,716    -0.006     0.963    -3.113     0.070     5.828
## phonecallraw       9,561   440.258   142.558      1      445    1,264
## homethatweek       9,870     0.196     0.397      0        0        1
## logphonecall       9,574     6.009     0.481     0.000     6.098     7.142
## logcallpersec      9,576    -5.170     0.160    -5.951    -5.169    -1.099
## logcalllength      9,576    11.175     0.521     2.485    11.273    12.116
## logcall_dayworked  9,576     9.474     0.457     2.485     9.552    10.359
## logdaysworked     9,855     1.685     0.282     0.000     1.792     1.946
## homethatweek_num   9,870     0.196     0.397      0        0        1
## basewage           10,023  1,601.750  176.618   527.350  1,600.000  2,450.000
## grosswage          9,919   3,097.672  990.788  1,140.280  2,877.000  14,553.000
## bonustotal         10,023  1,494.926  912.498     0.000   1,284.590  12,853.000
## age                10,079    23.485     4.191      1        23        35
## tenure             10,079    23.893    24.773      2        18       150
## children            10,079     0.141     0.348      0        0        1
## bedroom            10,079     0.974     0.160      0        1        1
## commute            9,118   103.519    67.620      1       80       300
## high_educ          10,079     0.400     0.490      0        0        1
## gender_num         10,079     0.498     0.500      0        0        1
## married_num        10,079     0.181     0.385      0        0        1
## high_educ_num      10,079     0.400     0.490      0        0        1
## children_num       10,079     0.141     0.348      0        0        1
## promote_switch     10,079     0.182     0.386      0        0        1
## quitjob            10,079     0.238     0.426      0        0        1
## costofcommute      9,488     7.213     7.231    0.000    6.000    55.000
## promote_switch_num 10,079     0.182     0.386      0        0        1
## quitjob_num        10,079     0.238     0.426      0        0        1
## -----
```

```
# plot per numeric variable
ggplot(df, aes(x=quitjob, y=perform1)) + geom_boxplot()
```

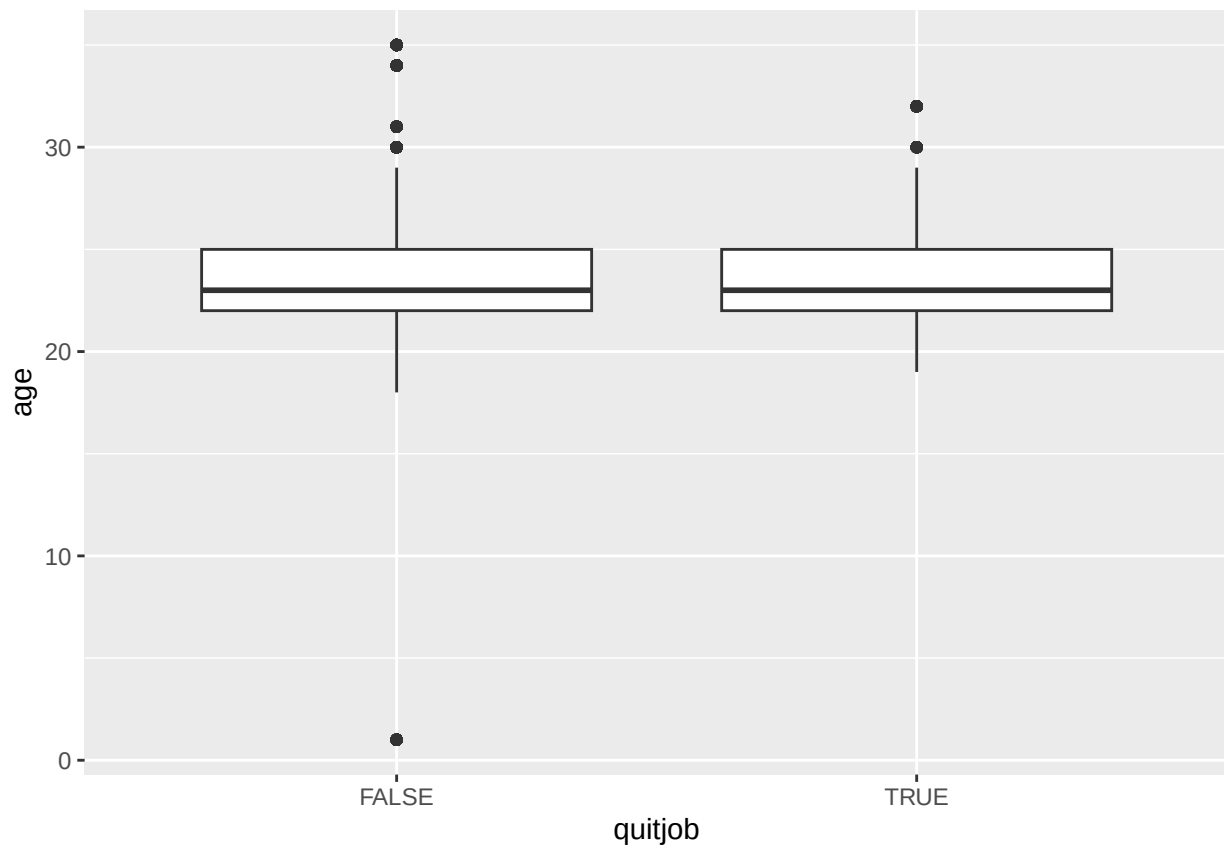
```
## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```
ggplot(df, aes(x=quitjob, y=tenure)) + geom_boxplot()
```

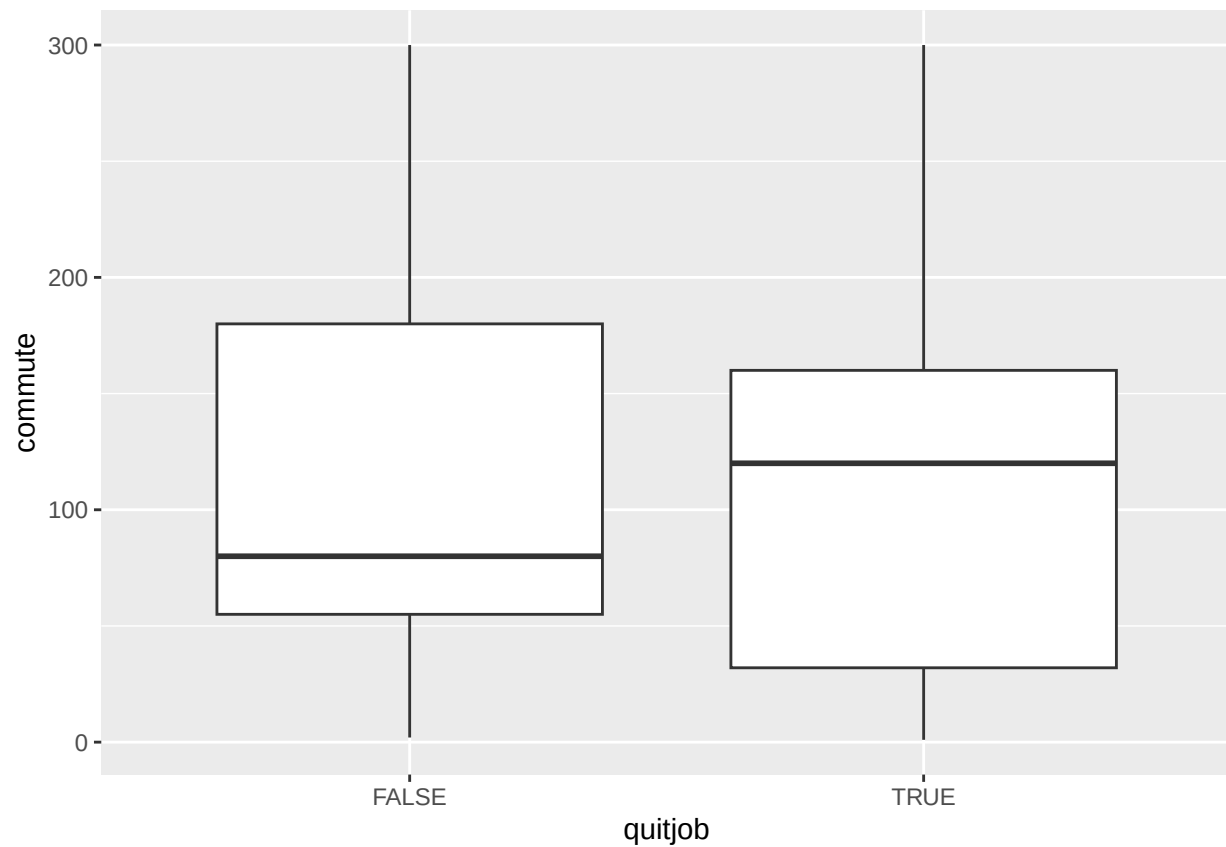


```
ggplot(df, aes(x=quitjob, y=age)) + geom_boxplot()
```



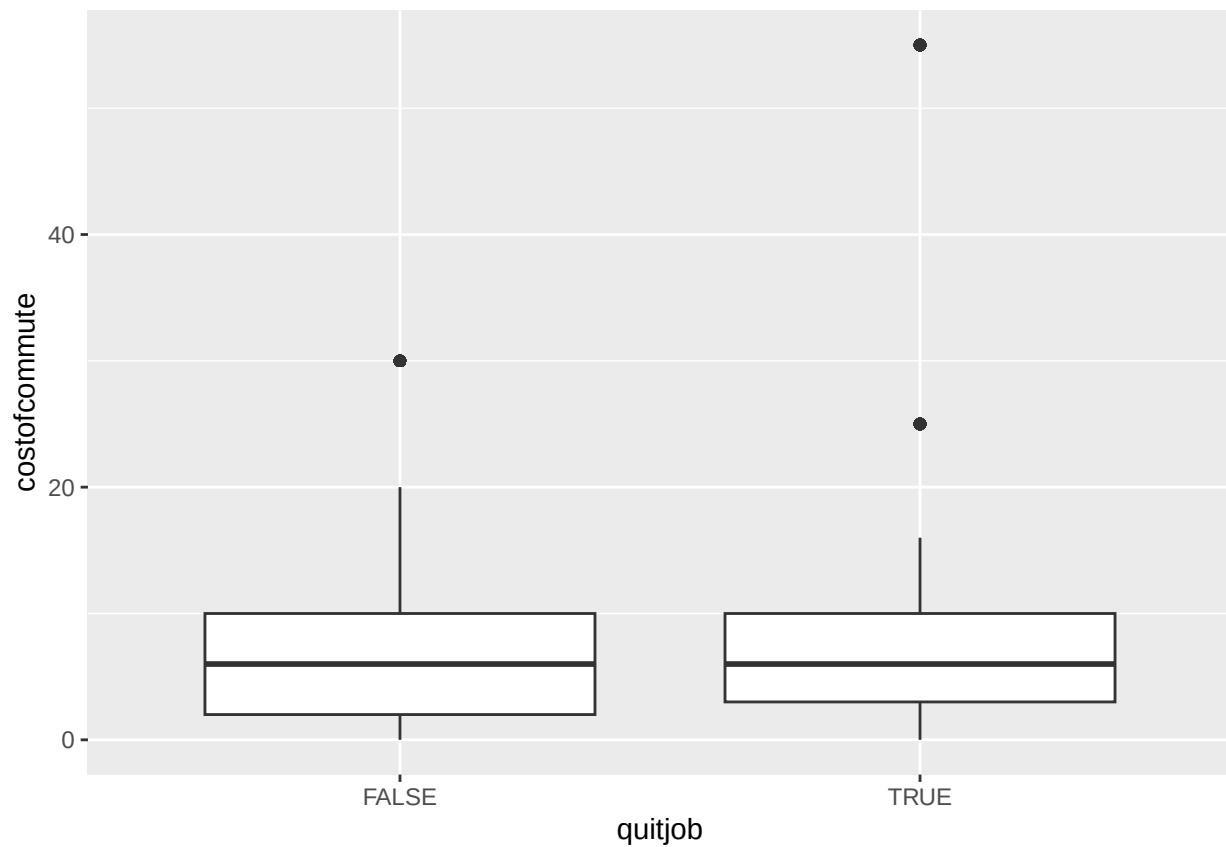
```
ggplot(df, aes(x=quitjob, y=commute)) + geom_boxplot()
```

```
## Warning: Removed 961 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



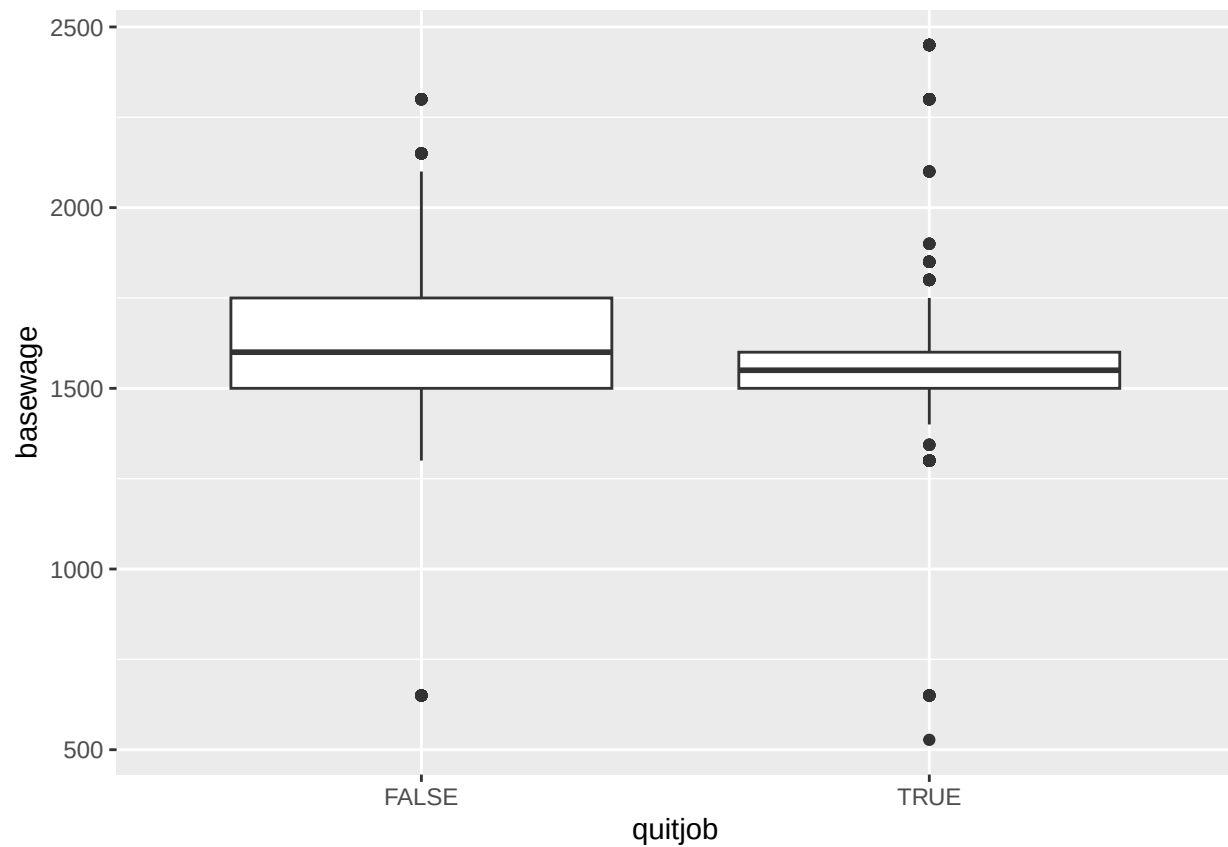
```
ggplot(df, aes(x=quitjob, y=costofcommute)) + geom_boxplot()
```

```
## Warning: Removed 591 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



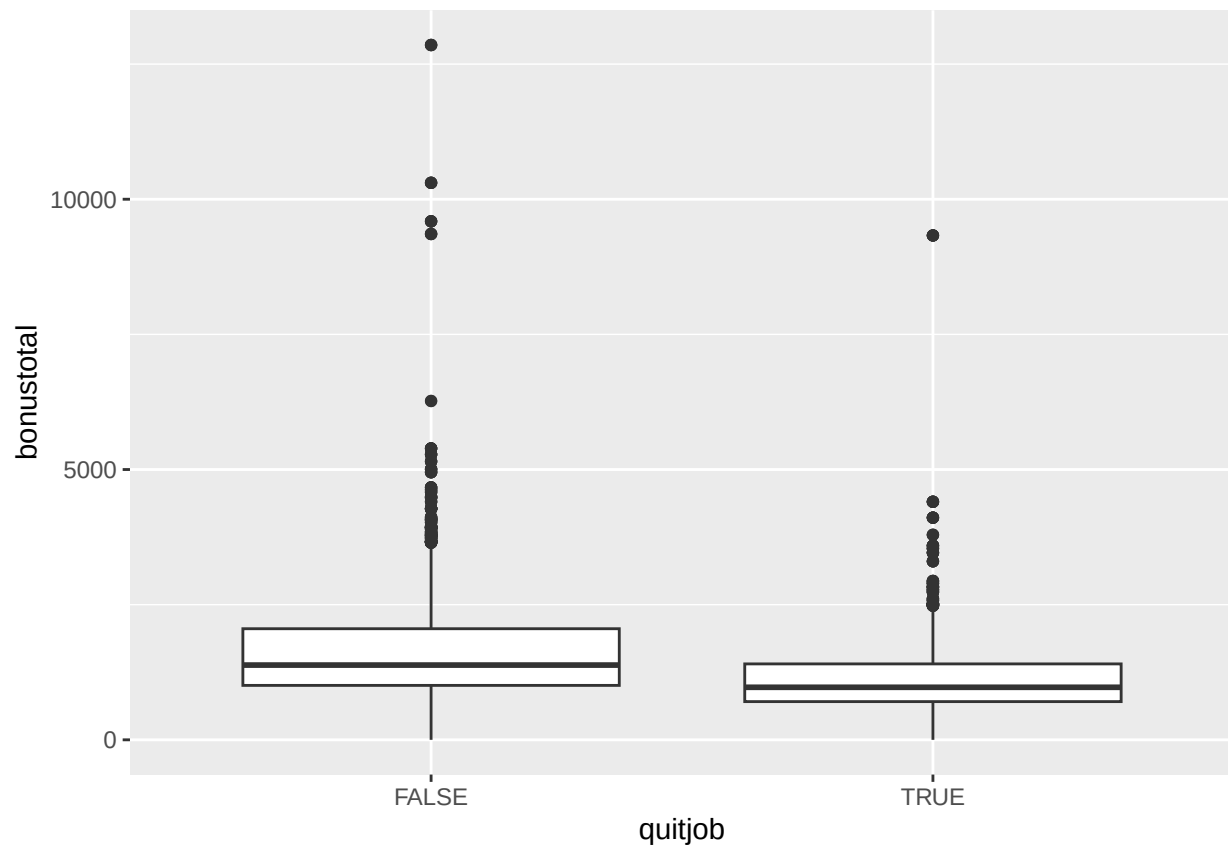
```
ggplot(df, aes(x=quitjob, y=basewage)) + geom_boxplot()
```

```
## Warning: Removed 56 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



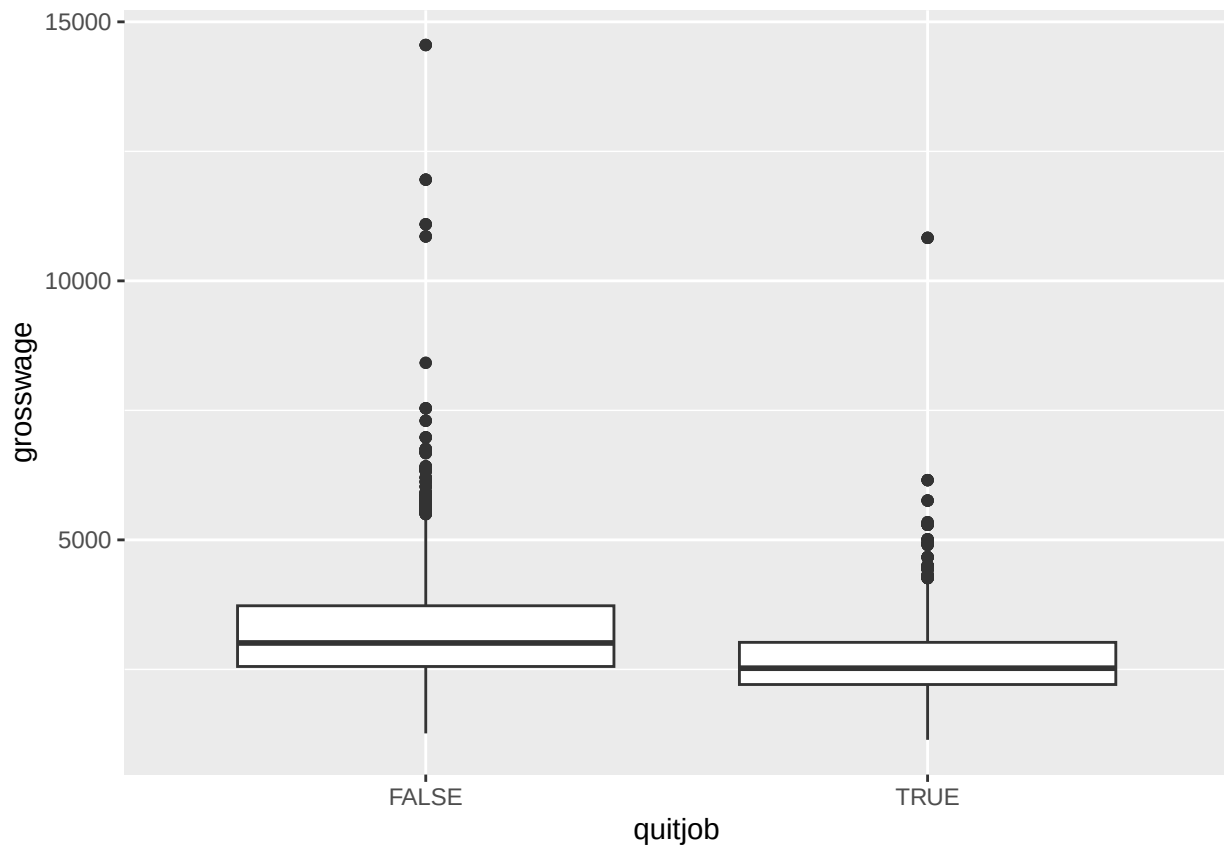
```
ggplot(df, aes(x=quitjob, y=bonustotal)) + geom_boxplot()
```

```
## Warning: Removed 56 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
ggplot(df, aes(x=quitjob, y=grosswage)) + geom_boxplot()
```

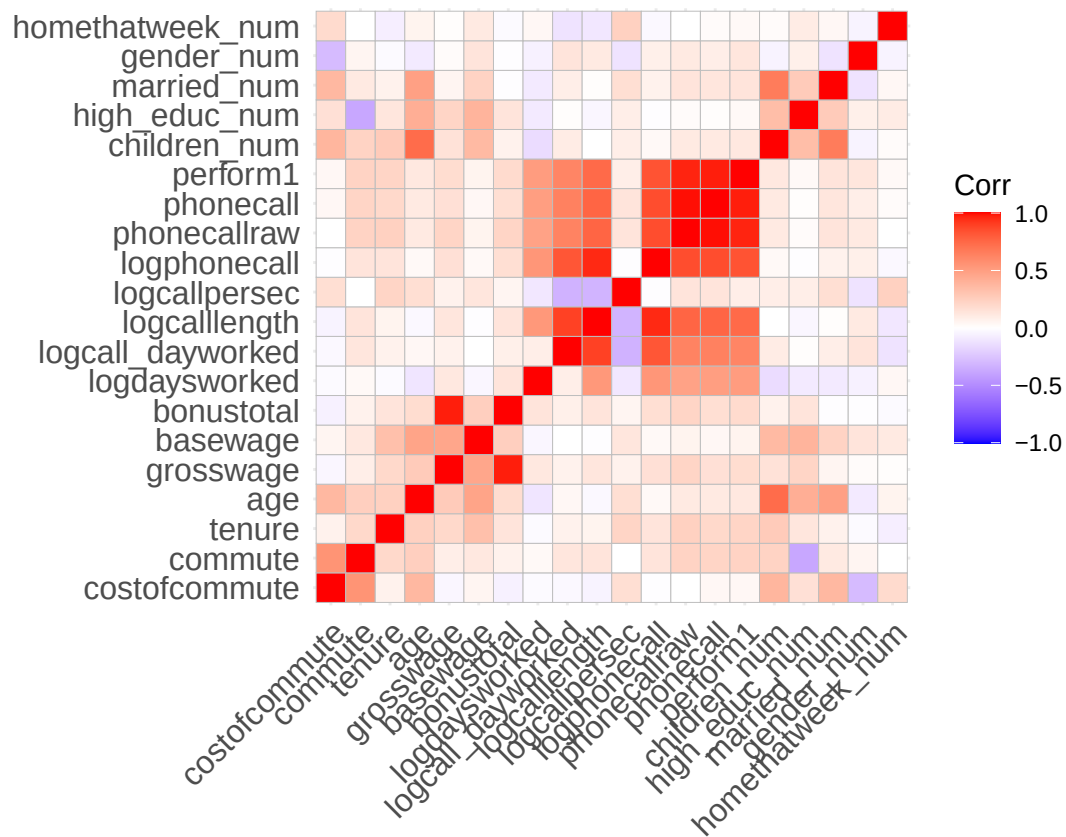
```
## Warning: Removed 160 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



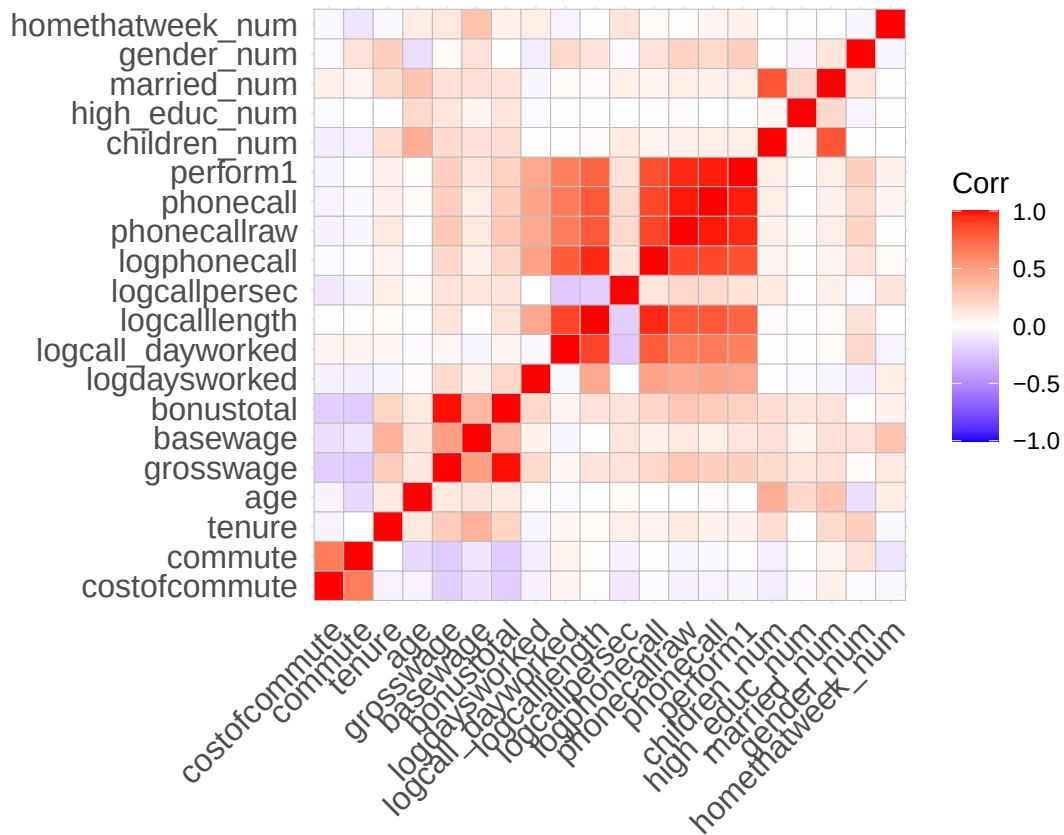
```
# Look at correlations
# subselect variable available in both groups (e.g. attitude data not available for quitters)
non_quitter_df <- non_quitter_df %>% select(costofcommute, commute, tenure, age, grosswage, basewage, bonustotal)

quitter_df <- quitter_df %>% select(costofcommute, commute, tenure, age, grosswage, basewage, bonustotal, logd)

nums <- dplyr::select(quitter_df, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
ggcorrplot(cm, lab = FALSE)
```



```
nums <- dplyr::select(non_quitter_df, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
ggcorrplot(cm, lab = FALSE)
```



```
df <- volunteer_endperiod_attitude_performance_wage
quitter_df <- df %>% filter(quitjob == TRUE)
non_quitter_df <- df %>% filter(quitjob == FALSE)
# Make some nice looking box plots for perform, commute
stats <- quitter_df %>% summarise(
  min = min(perform1, na.rm = TRUE),
  max = max(perform1, na.rm = TRUE),
  mean = mean(perform1, na.rm = TRUE),
  median = median(perform1, na.rm = TRUE),
  sd = sd(perform1, na.rm = TRUE)
)

p <- ggplot(df, aes(y = perform1, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Performance score", title = "Performance - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.3, y = 4,
  label = paste0("SD: ", round(stats$sd,2)),
  hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.6,
```

```

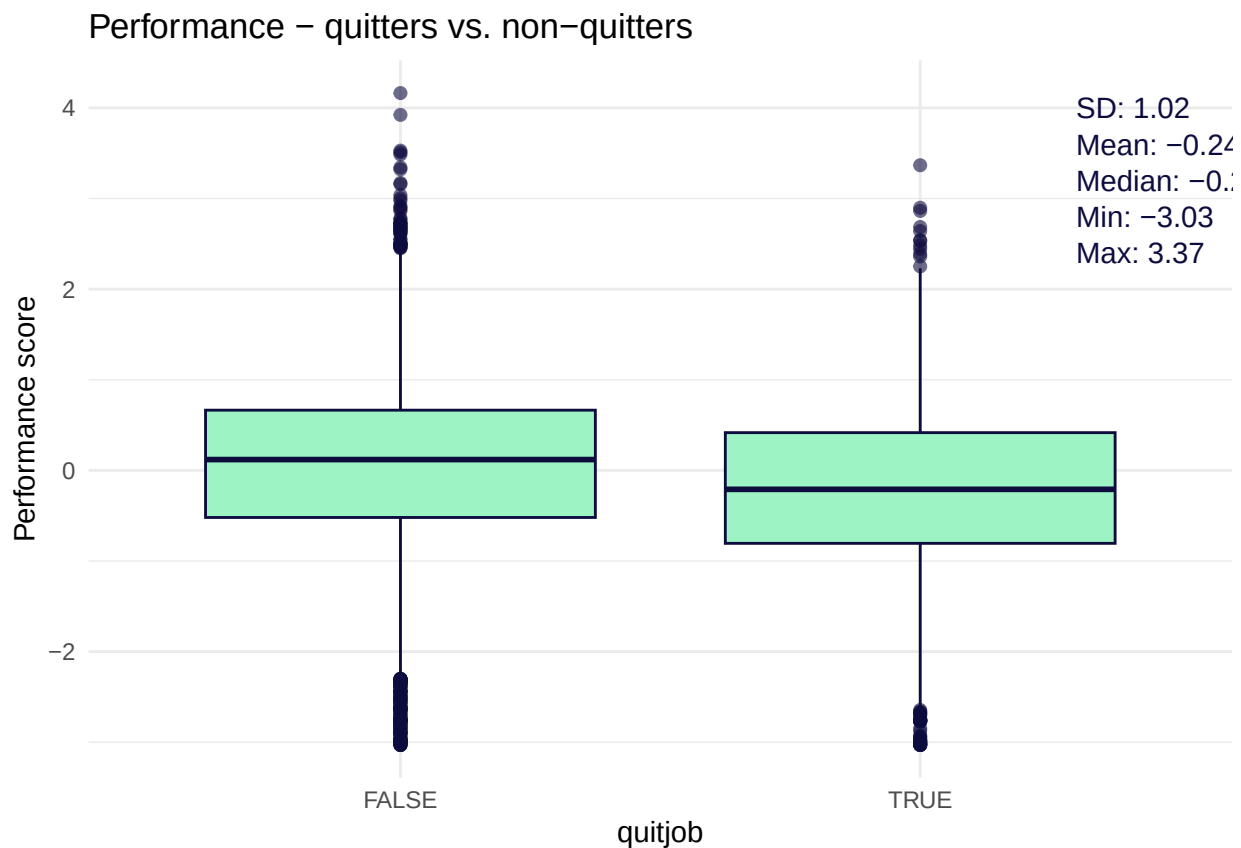
    label = paste0("Mean: ", round(stats$mean,2)),
    hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.2,
    label = paste0("Median: ", round(stats$median,2)),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.8,
    label = paste0("Min: ", round(stats$min,2)),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.4,
    label = paste0("Max: ", round(stats$max,2)),
    hjust = 0, color = "#0d0c3f")

```

```

## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



```

ggsave("performance_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)

```

```

## Warning: Removed 252 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```

```

stats <- quitter_df %>% summarise(
  min = min(commute, na.rm = TRUE),
  max = max(commute, na.rm = TRUE),
  mean = mean(commute, na.rm = TRUE),
  median = median(commute, na.rm = TRUE),
  sd = sd(commute, na.rm = TRUE)
)

```

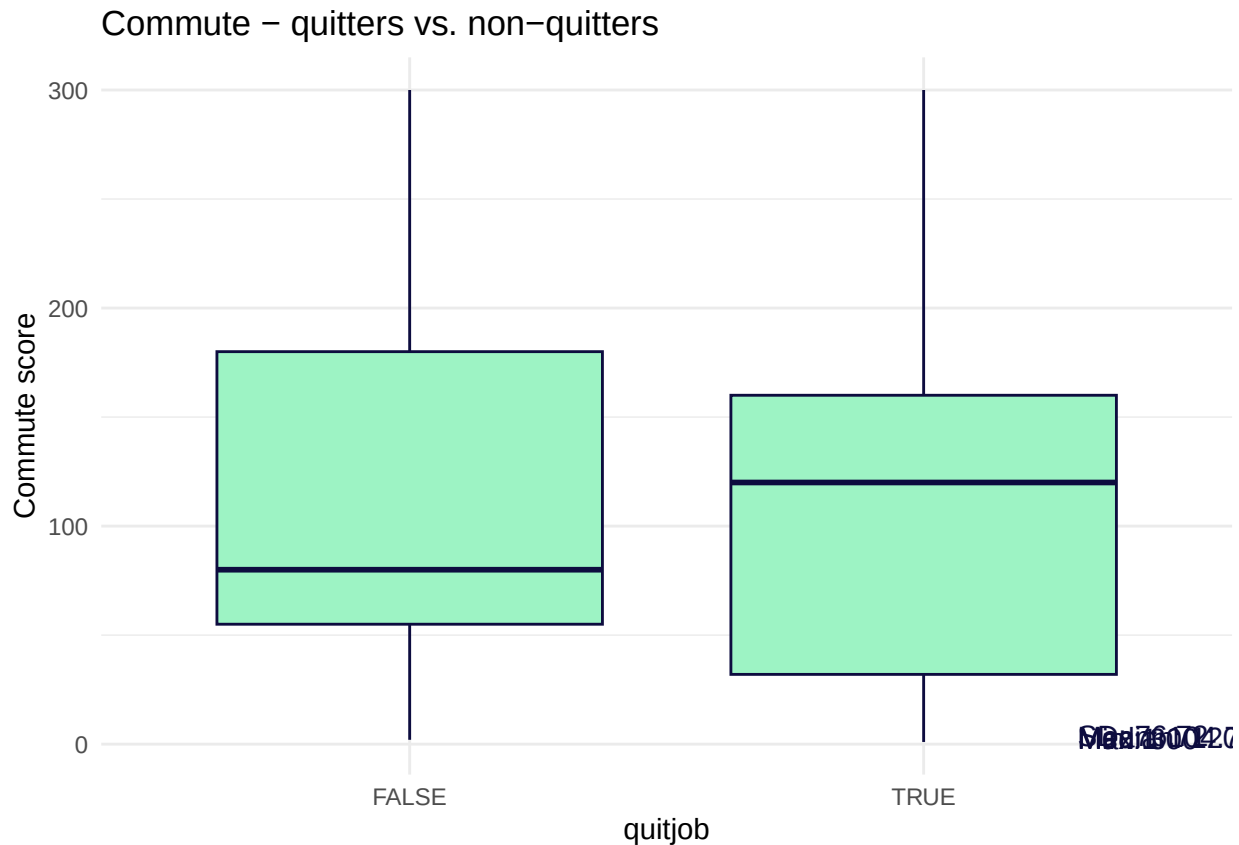
```

p <- ggplot(df, aes(y = commute, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Commute score", title = "Commute - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.3, y = 4,
  label = paste0("SD: ", round(stats$sd,2)),
  hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.6,
  label = paste0("Mean: ", round(stats$mean,2)),
  hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.3, y = 3.2,
  label = paste0("Median: ", round(stats$median,2)),
  hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.8,
  label = paste0("Min: ", round(stats$min,2)),
  hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.3, y = 2.4,
  label = paste0("Max: ", round(stats$max,2)),
  hjust = 0, color = "#0d0c3f")

## Warning: Removed 961 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



```
ggsave("commute_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 961 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
stats <- quitter_df %>% summarise(
  min = min(grosswage, na.rm = TRUE),
  max = max(grosswage, na.rm = TRUE),
  mean = mean(grosswage, na.rm = TRUE),
  median = median(grosswage, na.rm = TRUE),
  sd = sd(grosswage, na.rm = TRUE)
)

p <- ggplot(df, aes(y = grosswage, x=quitjob)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Grosswage in CNY", title = "Grosswage - quitters vs. non-quitters") +
  theme_minimal()

# add annotations
p + annotate("text", x = 2.1, y = 14000,
  label = paste0("SD: ", round(stats$sd,0), "CNY"),
  hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.1, y = 13000,
```

```

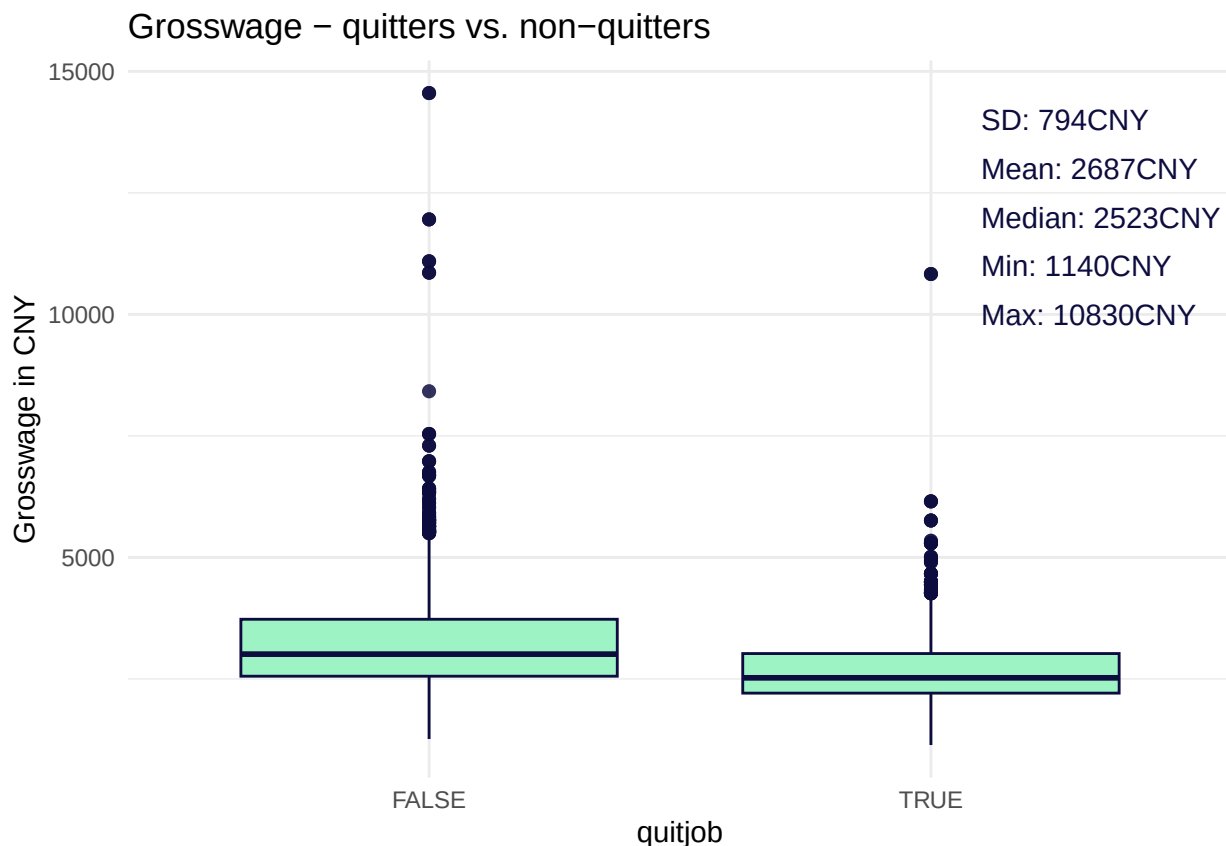
    label = paste0("Mean: ", round(stats$mean,0), "CNY"),
    hjust = 0, color = "#0d0c3f")+
  annotate("text", x = 2.1, y = 12000,
    label = paste0("Median: ", round(stats$median,0), "CNY"),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.1, y = 11000,
    label = paste0("Min: ", round(stats$min,0), "CNY"),
    hjust = 0, color = "#0d0c3f") +
  annotate("text", x = 2.1, y = 10000,
    label = paste0("Max: ", round(stats$max,0), "CNY"),
    hjust = 0, color = "#0d0c3f")

```

```

## Warning: Removed 160 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```



```

ggsave("grosswage_quitter_non_quitters.png", width = 7, height = 5, dpi = 300)

```

```

## Warning: Removed 160 rows containing non-finite outside the scale range
## (`stat_boxplot()`).

```

radar plot to compare quitter vs non quitter

```

radar_by_group(df, group_col="quitjob", include_group_indices=c(TRUE, FALSE), title="Quitter vs non-quitter")

```

```

## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "quitjob"          "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "homethatweek_num"
## [13] "perform1"         "grosswage"        "exhaustion"

```

```
## [16] "tenure"

## Warning: Using an external vector in selections was deprecated in tidysselect 1.1.0.
## i Please use `all_of()` or `any_of()` instead.
## # Was:
## data %>% select(final_selection)
##
## # Now:
## data %>% select(all_of(final_selection))
##
## See <https://tidysselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>    <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE     103.         6.59     16.7     24.2  0.513    0.163    0.966
## 2 TRUE      105.         9.14      NaN      NaN    0.450    0.239     1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1] TRUE FALSE
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>    <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE     103.         6.59     16.7     24.2  0.513    0.163    0.966
## 2 TRUE      105.         9.14      NaN      NaN    0.450    0.239     1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "FALSE" "TRUE"

## pdf
## 2
```

Generally speaking quitters perform less, work for the company shorter than average, have an average age, have slightly higher costs of commute while commuting about the same in minutes per week, earn less and work not from home in comparison to workers who stay. When begin older, quitters have a higher tendency to have children, being married and have higher education. People who were older and left at some point generally performed better than younger quitters. People who stay, tend to have lower cost of commute and lower commute times (maybe because they commit to working for the company and move closer to it).

We can already get a clue from this comparison of the differences between quitters and non quitters about the group constellation and personas.

Groups

Lets have a look on all possible combinations of categorical values and which are represented within the volunteer dataset with a threshold of at least 100 people-work-performances per group. We choose this sample size to increase our confidence. Afterwards we will do the same general analysis we did for the whole population for each group.

Important note: Because all quitters do not have attitude data, positive, negative and exhaustion appear to be “average” on the radar plots, but are actually NULL which is caused by the radar plotting function.

Available groups

```
# make it easier to process
df <- volunteer_endperiod_attitude_performance_wage
# define categorical variables
cat_cols <- c("gender", "married", "high_educ", "bedroom", "children", "promote_switch", "quitjob", "homethatweek")

# factors
df <- df %>% mutate(across(all_of(cat_cols), ~ factor(.x)))

# build grid of categories
lvls <- lapply(df[cat_cols], levels)
grid <- tidyr::expand_grid(!!!lvls)

# create summary for each category/group of people working at the company
summary_df <- df %>%
  group_by(across(all_of(cat_cols))) %>%
  summarise(
    # add number per category to filter later
    n = n(),
    across(where(is.numeric),
      list(mean = ~mean(.x, na.rm = TRUE),
           sd = ~sd(.x, na.rm = TRUE),
           se = ~sd(.x, na.rm = TRUE) / sqrt(sum(!is.na(.x))),
           med = ~median(.x, na.rm = TRUE),
           min = ~min(.x, na.rm = TRUE),
           max = ~max(.x, na.rm = TRUE)),
      .names = "{.col}_{.fn}"
    ),
    .groups = "drop"
  ) %>%
  mutate(group_index = row_number()) %>% # create an index for each group for better handling
  arrange(desc(n))

## Warning: There were 400 warnings in `summarise()`.
## The first warning was:
## i In argument: `across(...)`.
```

```
## i In group 7: `gender = female`, `married = FALSE`, `high_educ = FALSE`,
##   `bedroom = 1`, `children = FALSE`, `promote_switch = FALSE`, `quitjob =
##   TRUE`, `homethatweek = 0`.
## Caused by warning in `min()`:
## ! no non-missing arguments to min; returning Inf
## i Run `dplyr::last_dplyr_warnings()` to see the 399 remaining warnings.
```

```
# add group_index to the big d
group_characteristics <- summary_df %>% select(group_index, gender, married, bedroom, children, high_educ,
volunteer_endperiod_attitude_performance_wage <- merge(volunteer_endperiod_attitude_performance_wage, group_characteristics, by = "group_index")
# filter by number of rows per group
summary_df <- summary_df %>% filter(n>=100)
```

As a result we receive 23 distinct groups out of $8 \times 8 = 64$ possible options where each group represents a work week of an employee:

Top 6 groups by number

```
top <- summary_df %>% slice(1:6)

total_n <- summary_df %>% summarise(total_n = sum(n))
top_n <- top %>% summarise(top_n = sum(n))

top_n / total_n # about one quarter of the groups contain over 50% of the total observations

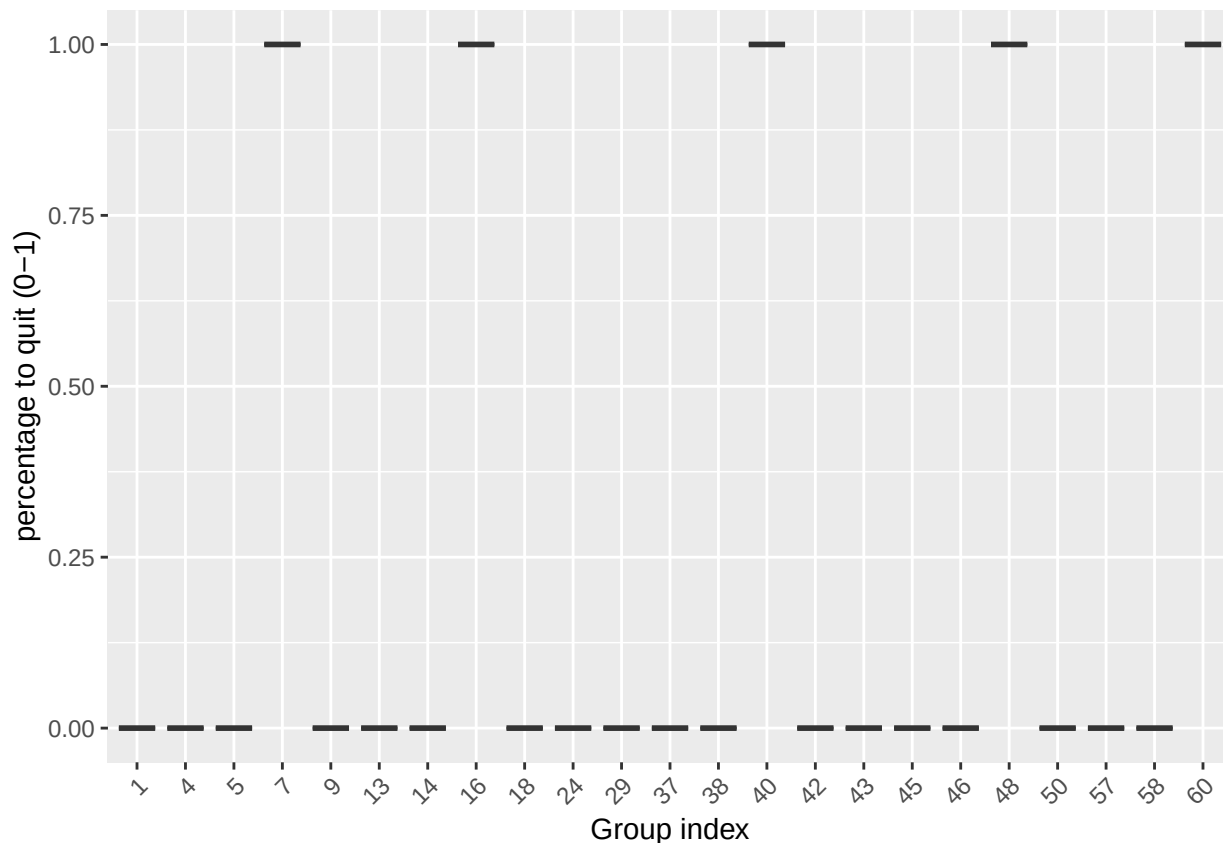
##           top_n
## 1 0.5245345

top %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)

## # A tibble: 6 x 8
##   gender married bedroom children high_educ promote_switch quitjob homethatweek
##   <fct>   <fct>   <fct>   <fct>   <fct>   <fct>       <fct>   <fct>
## 1 female FALSE     1      FALSE  FALSE  FALSE      FALSE    0
## 2 male   FALSE     1      FALSE  FALSE  FALSE      FALSE    0
## 3 male   FALSE     1      FALSE  TRUE   FALSE      FALSE    0
## 4 male   FALSE     1      FALSE  FALSE  FALSE      TRUE     0
## 5 male   FALSE     1      FALSE  FALSE  TRUE      FALSE    0
## 6 female FALSE     1      FALSE  TRUE   FALSE      FALSE    0

#

# First we look at the quitters and try to see which are more likely to quit
ggplot(summary_df, aes(x = factor(group_index), y = quitjob_num_mean)) +
  geom_boxplot() +
  labs(x = "Group index", y = "percentage to quit (0-1)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



```
df <- volunteer_endperiod_attitude_performance_wage
radar_by_group(df, group_col="group_index", include_group_indices=c(7,16,40,48,60), title="Quitter group")
```

```
## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "group_index"      "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"          "grosswage"
## [16] "exhaustion"       "tenure"
## # A tibble: 62 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>     <dbl>         <dbl>    <dbl>    <dbl>  <dbl>   <dbl>   <dbl>
## 1         1     103.          6.45    12.4     29.2      1       0       0
## 2         2     180           18     21.8     20.5      1       0       0
## 3         3     120           9      18.8     23        1       0       0
## 4         4     119.          6.71    22.0     22.4      1       0       1
## 5         5     92.0          5.12    14.5     26.6      1       0       1
## 6         6     131.          7.58    18.2     24.8      1       0       1
## 7         7     118.          4.11    NaN      NaN       1       0       1
## 8         8        20           0      NaN      NaN       1       0       1
## 9         9     112.          3.88    13.8     16.2      1       0       1
## 10        10        28           0      NaN      NaN       1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1] 7 16 40 48 60
## # A tibble: 5 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##         <int>   <dbl>         <dbl>   <dbl>   <dbl> <dbl>   <dbl>   <dbl>
## 1         7    118.         4.11     NaN     NaN     1       0       1
## 2        16    52.1         3.57     NaN     NaN     1       0       1
## 3        40    120.         8.28     NaN     NaN     0       0       1
## 4        48    33.5        11.7     NaN     NaN     0       0       1
## 5        60    101.        20.5     NaN     NaN     0       1       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7" "16" "40" "48" "60"

## pdf
## 2
```

Here we can clearly identify groups 7,16,40,48 and 60 where all people quit. Within the rest of the groups nobody quit.

Analyze quitter groups

```
df <- volunteer_endperiod_attitude_performance_wage
```

Group 7 357 observations of 6 volunteer

Persona observations:

female
not married
number of bedrooms: 1
no children
no high education
no promotion
worked for 7.6 months at the company when left (< avg)
20.5 years old (< avg)

Other average weekly observations:

not working from home
performed -0.308 (< avg)
basewage per month: 1453 (< avg)
bonus per month: 831 (« avg)
grosswage per month: 2265 (< avg)
cost of commute per month: 4 (< avg)
commute minutes per week: 103 (< avg)
logdaysworked: 1.6 (~= avg)

This groups shows a very similar correlation matrix in comparison to the overall sample, no specialities.

Interpretation

Looks like a group of female juniors (wage, performance, tenure, age), living not close by, probably the first or second job after high school(?) or even a job for a gap year. Probably left because of pursuing with school or another job for a pay increase.

Recommendation

This group is among the youngest and has a worse performance than the average. Offering working from home/remote working capabilities can be a recommended, as these group could for example continue working

during university or traveling. When performance increases a raise would be appropriate but now as first action.

```
# select group data
group7 <- df %>% filter(group_index==7)
stargazer(as.data.frame(group7), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## bedroom        357     1.000     0.000        1         1         1
## children        357     0.000     0.000        0         0         0
## high_educ       357     0.000     0.000        0         0         0
## promote_switch  357     0.000     0.000        0         0         0
## quitjob         357     1.000     0.000        1         1         1
## homethatweek    357     0.000     0.000        0         0         0
## perform1        345    -0.308     0.854     -3.031    -0.204     2.230
## phonecall       320    -0.220     0.983     -3.054    -0.157     5.828
## phonecallraw    315   411.311   137.944        22        419     1,264
## logphonecall    315     5.941     0.443        3.091     6.038     7.142
## logcallpersec   307    -5.217     0.114     -5.560    -5.198    -4.924
## logcalllength   308    11.142     0.444        8.473    11.222    11.900
## logcall_dayworked 308     9.493     0.369        7.087     9.566    10.049
## logdaysworked  357     1.603     0.325        0.000     1.609     1.946
## homethatweek_num 357     0.000     0.000        0         0         0
## basewage        357  1,451.197  160.260     527.350  1,500.000  1,750.000
## grosswage       352  2,281.242  477.662    1,140.280  2,222.000  4,487.660
## bonustotal      357   832.198   454.516        0.000     795.000  2,937.660
## age            357    20.501     1.315        19        21        22
## tenure          357     7.622     2.661         2         9         10
## commute         357   118.263    97.337        20        120        300
## gender_num      357     1.000     0.000        1         1         1
## married_num     357     0.000     0.000        0         0         0
## high_educ_num   357     0.000     0.000        0         0         0
## children_num    357     0.000     0.000        0         0         0
## costofcommute   357     4.113     4.491        0.000     4.545    10.000
## promote_switch_num 357     0.000     0.000        0         0         0
## quitjob_num     357     1.000     0.000        1         1         1
## group_index     357     7.000     0.000        7         7         7
## -----
```

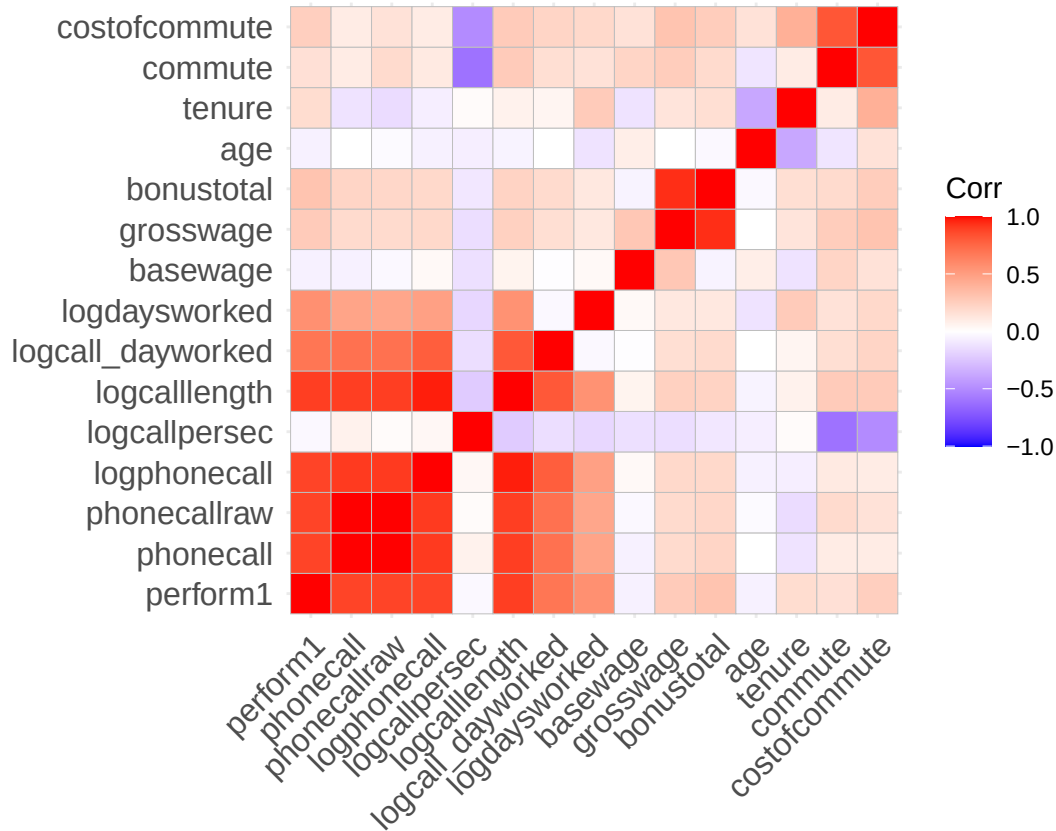
```
# group definition
#group7 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
# distinct persons
unique(group7 %>%select(personid))
```

```
##      personid
## 1      40034
## 2      38862
## 4      44256
## 5      40346
## 26     42104
## 110     39478
```

```
nums <- dplyr::select(group7, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



```
# radar plot
```

```
radar_by_group(df, group_col="group_index", include_group_indices=c(7), title="Group 7 characteristics")
```

```
## [1] "commute" "costofcommute" "negative"
## [4] "positive" "group_index" "gender_num"
## [7] "married_num" "bedroom" "children_num"
## [10] "high_educ_num" "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1" "grosswage"
## [16] "exhaustion" "tenure"
```

```
## # A tibble: 62 x 17
```

	group_index	commute	costofcommute	negative	positive	gender	married	bedroom
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
##	1	1	103.	6.45	12.4	29.2	1	0
##	2	2	180	18	21.8	20.5	1	0
##	3	3	120	9	18.8	23	1	0
##	4	4	119.	6.71	22.0	22.4	1	0
##	5	5	92.0	5.12	14.5	26.6	1	0
##	6	6	131.	7.58	18.2	24.8	1	0
##	7	7	118.	4.11	NaN	NaN	1	0
##	8	8	20	0	NaN	NaN	1	0

```
## 9          9  112.          3.88      13.8      16.2      1      0      1
## 10         10   28          0         NaN      NaN      1      0      1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 7
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>      <dbl>      <dbl>      <dbl>      <dbl> <dbl>      <dbl>      <dbl>
## 1         7  118.          4.11      NaN      NaN      1      0      1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7"

## pdf
## 2
```

Group 16 338 observations for 6 volunteers

Persona observations:

female
not married
number of bedrooms: 1
no children
high education
no promotion
worked for 23.5 months at the company when left (= avg)
23.7 years old (>= avg)

Other average weekly observations:

not working from home
performed: 0 (= avg)
basewage per month: 1598 (< avg)
bonus per month: 1362 (< avg)
grosswage per month: 2940 (<=avg)
cost of commute per month: 3.9 (< avg)
commute min per week: 52 (« avg)
logdaysworked: 1.65 (~= avg)

For this group the strong positive correlation between tenure and age stands out. For the full sample it is slightly negative. Commute and age have a much stronger negative correlation

Interpretation

Looks like a group of female juniors (wage, performance, tenure, age), who already worked for almost 2 years, living somewhat central, probably the first or second job college. Decent performance, not a top performer but average. Probably left because they did not receive a promotion and went for a higher paying job as their salary was still below average.

Recommendation

As this group has an average performance while still being young, and already has a higher education degree (so they probably won't leave for education) our client should prioritize this group and try to give employees in this group a raise and a promotion to keep them within the company. Additionally offer working from home as a benefit, though is not the greatest leverage for this group.

```
# select group data
group16 <- df %>% filter(group_index==16)
stargazer(as.data.frame(group16), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## bedroom        338      1.000      0.000        1         1         1
## children        338      0.000      0.000        0         0         0
## high_educ       338      1.000      0.000        1         1         1
## promote_switch  338      0.000      0.000        0         0         0
## quitjob         338      1.000      0.000        1         1         1
## homethatweek    338      0.000      0.000        0         0         0
## perform1        338     -0.0001      1.079     -3.031      0.049      2.863
## phonecall       320     -0.038      1.045     -3.054      0.048      2.463
## phonecallraw    307    454.397    154.865        51        457        910
## logphonecall    306      6.046      0.419      3.932      6.124      6.813
## logcallpersec   310     -5.159      0.141     -5.495     -5.151     -4.624
## logcalllength   310     11.183      0.562      5.489     11.316     11.991
## logcall_dayworked 310      9.483      0.485      3.697      9.552     10.100
## logdaysworked  337      1.645      0.395      0.000      1.792      1.946
## homethatweek_num 338      0.000      0.000        0         0         0
## basewage        338    1,598.521    93.866      1,450      1,600      1,750
## grosswage       325    2,967.382    823.790    1,655.000    2,713.000    6,153.380
## bonustotal      338    1,362.637    809.880      5.000      1,144.520    4,403.380
## age            338     23.683      1.309        22         23         26
## tenure         338     23.485      6.243        13         24         30
## commute        227     52.106     41.709        2         60        120
## gender_num      338      1.000      0.000        1         1         1
## married_num     338      0.000      0.000        0         0         0
## high_educ_num   338      1.000      0.000        1         1         1
## children_num    338      0.000      0.000        0         0         0
## costofcommute   338      3.565      2.123        0         3         6
## promote_switch_num 338      0.000      0.000        0         0         0
## quitjob_num     338      1.000      0.000        1         1         1
## group_index     338     16.000      0.000        16         16         16
## -----
```

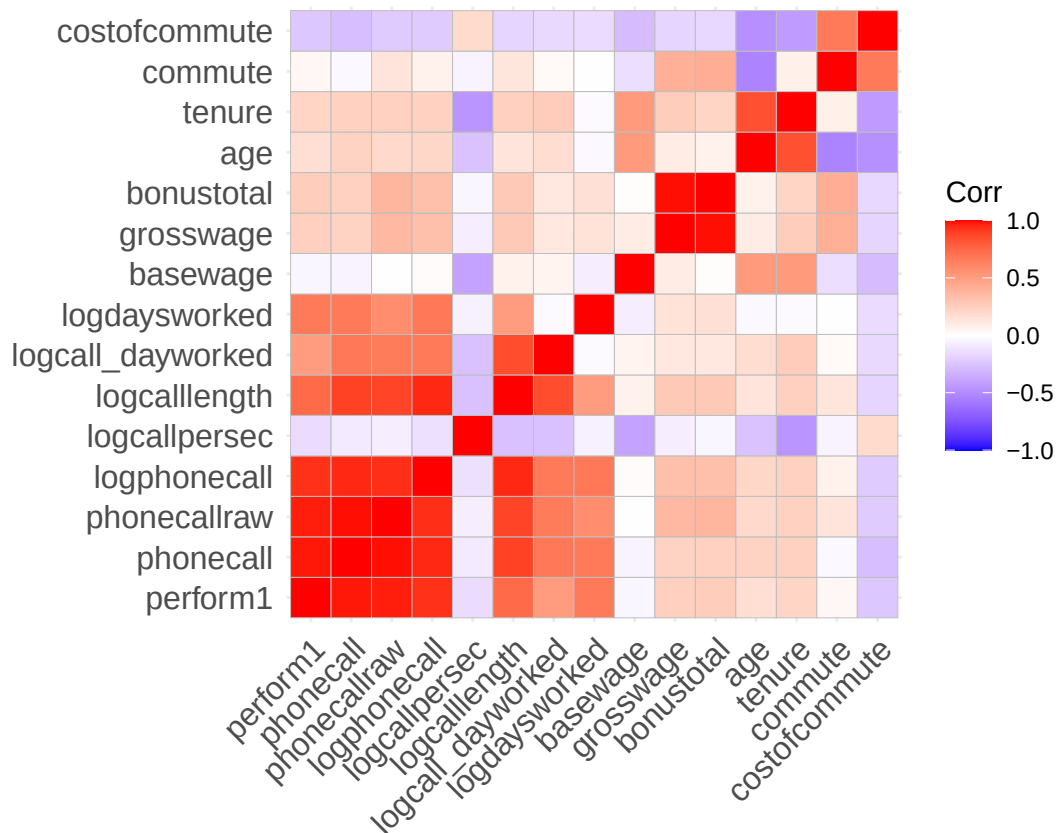
```
# group definition
#group16 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
# distinct persons
unique(group16 %>%select(personid))
```

```
##      personid
## 1      31150
## 2      37276
## 3      26634
## 5      36494
## 41     29230
## 85     26934
```

```
nums <- dplyr::select(group16, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



```
# radar plot
```

```
radar_by_group(df, group_col="group_index", include_group_indices=c(16), title="Group 16 characteristics")
```

```
## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "group_index"      "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"         "grosswage"
## [16] "exhaustion"       "tenure"
```

```
## # A tibble: 62 x 17
```

	group_index	commute	costofcommute	negative	positive	gender	married	bedroom
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	1	103.	6.45	12.4	29.2	1	0	0
## 2	2	180	18	21.8	20.5	1	0	0
## 3	3	120	9	18.8	23	1	0	0
## 4	4	119.	6.71	22.0	22.4	1	0	1
## 5	5	92.0	5.12	14.5	26.6	1	0	1
## 6	6	131.	7.58	18.2	24.8	1	0	1
## 7	7	118.	4.11	NaN	NaN	1	0	1
## 8	8	20	0	NaN	NaN	1	0	1
## 9	9	112.	3.88	13.8	16.2	1	0	1
## 10	10	28	0	NaN	NaN	1	0	1

```
## # i 52 more rows
```

```
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 16
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>    <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1      16    52.1          3.57      NaN      NaN      1      0      1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "16"

## pdf
## 2
```

Group 40 643 observations for 12 volunteers

Persona observations:

male
not married
number of bedrooms: 1
no children
no high education
no promotion
worked for 26.8 months at the company when left (= avg)
22.8 years old (<= avg)

Other average weekly observations:

not working from home
performed: -0.46 (« avg)
basewage per month: 1492 (< avg)
bonus per month: 1149 (< avg)
grosswage per month: 2604 (<=avg)
cost of commute per month: 8.3 (> avg)
commute min per week: 119 (« avg)
logdaysworked: 1.7 (~= avg)

This correlation matrix is again very similar to the full sample matrix, no outstanding differences.

Interpretation

Young male group, with no high education and family, which has never worked from home. Heavily underperformed, while working as many days as the average. Lives somewhat far away (over avg) with avg cost for commuting. Earns about 400 less than average despite the bad performance. Group7 had similar circumstances, despite commute, but a much better performance with lower salary. But the group is twice as big as this group.

Recommendation

As the performance is worse than average, trying to keep this group of workers should not be prioritized.

```
# select group data
group40 <- df %>% filter(group_index==40)
stargazer(as.data.frame(group40), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
```

## Statistic	N	Mean	St. Dev.	Min	Median	Max
## bedroom	643	1.000	0.000	1	1	1
## children	643	0.000	0.000	0	0	0
## high_educ	643	0.000	0.000	0	0	0
## promote_switch	643	0.000	0.000	0	0	0
## quitjob	643	1.000	0.000	1	1	1
## homethatweek	643	0.000	0.000	0	0	0
## perform1	642	-0.462	0.899	-3.024	-0.446	2.540
## phonecall	638	-0.392	0.900	-3.106	-0.377	2.554
## phonecallraw	631	391.097	135.837	1	385	883
## logphonecall	631	5.880	0.533	0.000	5.953	6.783
## logcallpersec	629	-5.157	0.174	-5.609	-5.162	-3.178
## logcalllength	630	11.038	0.572	3.178	11.131	11.824
## logcall_dayworked	630	9.327	0.503	2.485	9.406	10.021
## logdaysworked	642	1.697	0.265	0.000	1.792	1.946
## homethatweek_num	643	0.000	0.000	0	0	0
## basewage	638	1,491.016	141.827	650.000	1,500.000	1,900.000
## grosswage	633	2,643.891	737.108	1,360.000	2,485.170	5,011.380
## bonustotal	638	1,149.647	684.029	0.000	966.000	3,461.380
## age	643	22.879	2.293	19	23	27
## tenure	643	26.818	45.869	3	10	150
## commute	612	119.649	51.741	30	120	200
## gender_num	643	0.000	0.000	0	0	0
## married_num	643	0.000	0.000	0	0	0
## high_educ_num	643	0.000	0.000	0	0	0
## children_num	643	0.000	0.000	0	0	0
## costofcommute	643	8.283	2.400	4	10	12
## promote_switch_num	643	0.000	0.000	0	0	0
## quitjob_num	643	1.000	0.000	1	1	1
## group_index	643	40.000	0.000	40	40	40

```

# group definition
#group40 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
# distinct persons
unique(group40 %>%select(personid))

```

```

##      personid
## 1      44794
## 3      43288
## 4      38552
## 7      38842
## 19     41332
## 55     40008
## 128    43524
## 136    35822
## 177    34890
## 184    41320
## 275    39096
## 405    42096

```

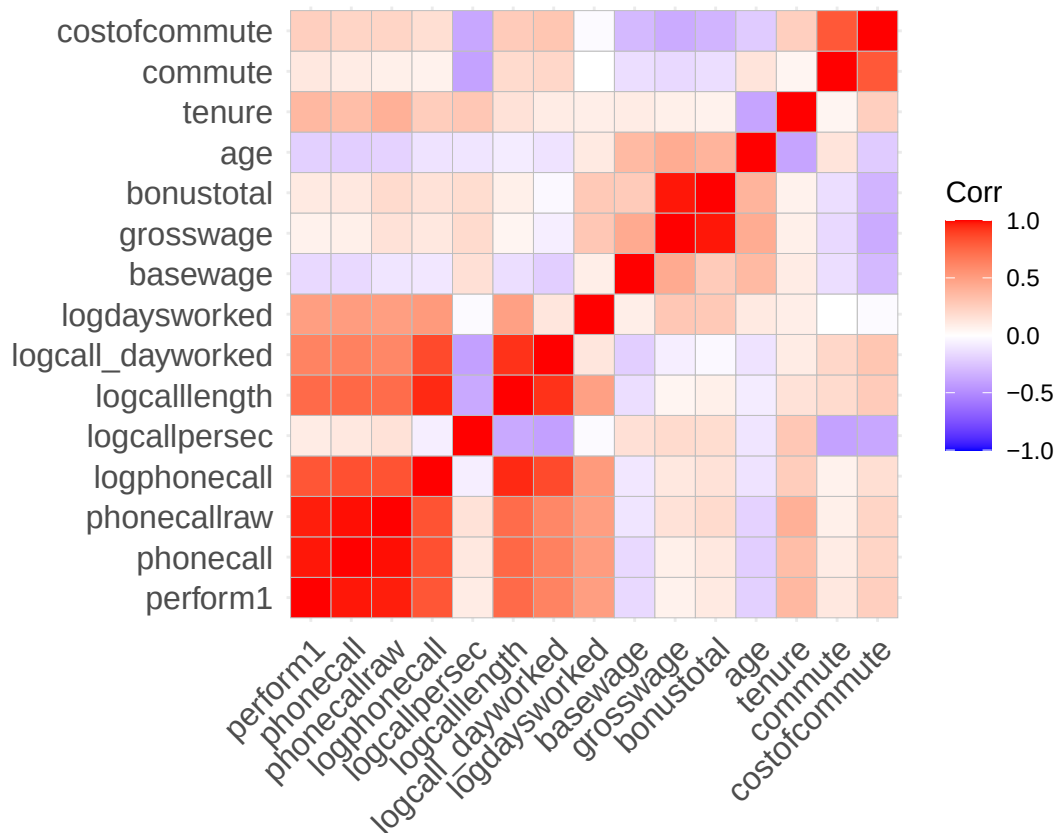
```

nums <- dplyr::select(group40, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")

```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



```
# radar plot
```

```
radar_by_group(df, group_col="group_index", include_group_indices=c(40), title="Group 40 characteristics")
```

```
## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "group_index"      "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"          "grosswage"
## [16] "exhaustion"       "tenure"
```

```
## # A tibble: 62 x 17
```

	group_index	commute	costofcommute	negative	positive	gender	married	bedroom
	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	1	103.	6.45	12.4	29.2	1	0	0
## 2	2	180	18	21.8	20.5	1	0	0
## 3	3	120	9	18.8	23	1	0	0
## 4	4	119.	6.71	22.0	22.4	1	0	1
## 5	5	92.0	5.12	14.5	26.6	1	0	1
## 6	6	131.	7.58	18.2	24.8	1	0	1
## 7	7	118.	4.11	NaN	NaN	1	0	1
## 8	8	20	0	NaN	NaN	1	0	1
## 9	9	112.	3.88	13.8	16.2	1	0	1
## 10	10	28	0	NaN	NaN	1	0	1

```
## # i 52 more rows
```

```
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 40
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>     <dbl>         <dbl>     <dbl>     <dbl> <dbl>   <dbl>   <dbl>
## 1       40    120.           8.28      NaN      NaN     0       0       1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "40"

## pdf
## 2
```

Group 48 218 observations for 4 volunteers

Persona observations:

male
 not married
 number of bedrooms: 1
 no children
 high education
 no promotion
 worked for 15.4 months at the company when left (= avg)
 21.8 years old (<= avg)

Other average weekly observations:

not working from home
 performed: -0.612 (« avg)
 basewage per month: 1566 (< avg)
 bonus per month: 1087 (< avg)
 grosswage per month: 2627 (<=avg)
 cost of commute per month: 9.5 (> avg)
 commute min per week: 33.4 (« avg)
 logdaysworked: 1.7 (~= avg)

We can see a strong correlation between costofcommute, tenure and age and a rather negative one between costofcommute, commute, tenure, age and grosswage/bonustotal. But small sample size (less confidence).

Interpretation

Young male group, with no education and no family, which has never worked from home. Heavily underperformed, even more than males with high education who quit, while working as many days as the average. Lives closer to work than the average with slightly higher than avg cost for commuting. Earns about 400 less than average despite the bad performance. Unclear why they left, probably also because they wanted more money/a promotion.

Recommendation

Because of the small sample size confidence is low. Tough, as the performance is bad, trying to keep this group of workers should not be prioritized. The company has to ask itself if it is willing to pay more money for this performance,

```
# select group data
group48 <- df %>% filter(group_index==48)
```

```
stargazer(as.data.frame(group48), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## bedroom        218     1.000     0.000        1         1         1
## children        218     0.000     0.000        0         0         0
## high_educ       218     1.000     0.000        1         1         1
## promote_switch  218     0.000     0.000        0         0         0
## quitjob         218     1.000     0.000        1         1         1
## homethatweek    218     0.000     0.000        0         0         0
## perform1        218    -0.627     0.690     -2.601    -0.614     1.202
## phonecall       216    -0.446     0.750     -2.618    -0.401     2.029
## phonecallraw    218    365.362   104.638        65      373.5     695
## logphonecall    216     5.845     0.379        4.174     5.923     6.544
## logcallpersec   217    -5.210     0.159     -5.721    -5.193    -4.581
## logcalllength   218    11.054     0.336        9.594    11.143    11.468
## logcall_dayworked 218     9.379     0.298        8.157     9.422     9.880
## logdaysworked  218     1.675     0.208        0.693     1.792     1.946
## homethatweek_num 218     0.000     0.000        0         0         0
## basewage        218  1,565.138  105.495     1,300     1,600     1,850
## grosswage       218  2,652.406  494.384   1,330.000  2,535.000  3,784.760
## bonustotal      218  1,087.269  438.508    30.000     985.340  2,134.760
## age            218    21.844     1.314        20        22        23
## tenure         218    15.486     12.637         4         4        34
## commute        218    33.482     27.988         1        30        75
## gender_num     218     0.000     0.000        0         0         0
## married_num    218     0.000     0.000        0         0         0
## high_educ_num  218     1.000     0.000        1         1         1
## children_num   218     0.000     0.000        0         0         0
## costofcommute   149    11.664     9.548         2         8        25
## promote_switch_num 218     0.000     0.000        0         0         0
## quitjob_num    218     1.000     0.000        1         1         1
## group_index     218    48.000     0.000        48        48        48
## -----
```

```
# group definition
```

```
#group48 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
```

```
# distinct persons
```

```
unique(group48 %>%select(personid))
```

```
##      personid
```

```
## 1      32804
```

```
## 2      24324
```

```
## 9      42618
```

```
## 48     42634
```

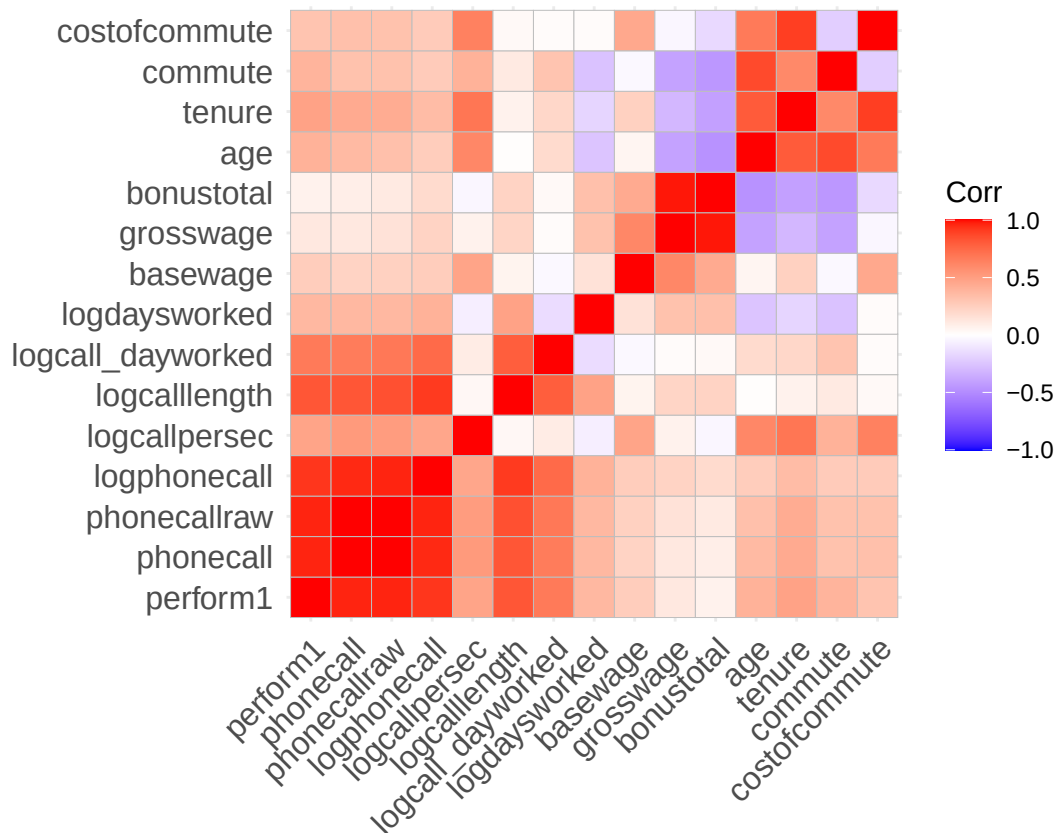
```
nums <- dplyr::select(group48, where(is.numeric))
```

```
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
```

```
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



```
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(48), title="Group 48 characteristics")

## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "group_index"      "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"         "grosswage"
## [16] "exhaustion"       "tenure"
## # A tibble: 62 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>    <dbl>         <dbl>    <dbl>    <dbl>    <dbl>    <dbl>    <dbl>
## 1         1    103.         6.45    12.4    29.2      1      0      0
## 2         2    180         18      21.8    20.5      1      0      0
## 3         3    120          9      18.8    23        1      0      0
## 4         4    119.        6.71    22.0    22.4      1      0      1
## 5         5    92.0        5.12    14.5    26.6      1      0      1
## 6         6    131.        7.58    18.2    24.8      1      0      1
## 7         7    118.        4.11    NaN     NaN       1      0      1
## 8         8     20          0      NaN     NaN       1      0      1
## 9         9    112.        3.88    13.8    16.2      1      0      1
## 10        10     28          0      NaN     NaN       1      0      1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1] 48
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>    <dbl>      <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1      48    33.5      11.7      NaN      NaN      0      0      1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "48"

## pdf
## 2
```

Group 60 202 observations for 4 volunteers

Persona observations:

male
 married
 number of bedrooms: 1
 children
 high education
 no promotion
 worked for 22 months at the company when left (> avg)
 27.8 years old (> avg)

Other average weekly observations:

not working from home
 performed: -0.22 (< avg)
 basewage per month: 1555 (< avg)
 bonus per month: 1256 (< avg)
 grosswage per month: 2784 (<=avg)
 cost of commute per month: 20.5 (» avg)
 commute min per week: 101.5 (= avg)
 logdaysworked: 1.5 (<= avg)

We can see a strong correlation age, tenure and commute. And a moderate between tenure, bonustotal, grosswage, basewage (steady salary raises(?)) But small sample size (less confidence).

Interpretation

Older than avg. male group, with high education and family, which has never worked from home. Under-performance a bit, while working as a little less days as the average. Earn about the same amount less than what they perform less. High cost of commute and average commute minutes per week. Probably left because of this, and maybe the performance also was affect by that. As the salary was steadily improved while working their, this was probably not the main reason.

Recommendation

Because of the small sample size confidence is low. Tough, as the performance is probably worse because of the high commute cost in combination with having a family. A first important action can be the offering of working from home to reduce negative aspects of commuting and increasing performance. If performance actually rises, salary should be increased as well according to that.

```
# select group data
group60 <- df %>% filter(group_index==60)
stargazer(as.data.frame(group60), type = "text", median = TRUE, header = TRUE)
```

```
##
## =====
```

## Statistic	N	Mean	St. Dev.	Min	Median	Max
## bedroom	202	1.000	0.000	1	1	1
## children	202	1.000	0.000	1	1	1
## high_educ	202	1.000	0.000	1	1	1
## promote_switch	202	0.000	0.000	0	0	0
## quitjob	202	1.000	0.000	1	1	1
## homethatweek	202	0.000	0.000	0	0	0
## perform1	201	-0.216	1.292	-2.996	-0.008	2.164
## phonecall	202	-0.182	1.343	-3.113	0.077	2.729
## phonecallraw	184	435.120	181.523	2	461	823
## logphonecall	184	5.904	0.790	0.693	6.133	6.713
## logcallpersec	184	-5.074	0.196	-5.578	-5.124	-4.533
## logcalllength	183	10.976	0.754	6.271	11.183	11.880
## logcall_dayworked	183	9.416	0.631	5.578	9.599	10.088
## logdaysworked	202	1.532	0.364	0.000	1.609	1.946
## homethatweek_num	202	0.000	0.000	0	0	0
## basewage	195	1,555.128	109.941	1,400	1,550	1,800
## grosswage	195	2,811.191	652.400	1,820.000	2,549.860	4,434.900
## bonustotal	195	1,256.063	592.845	20.000	1,054.000	2,734.900
## age	202	27.847	1.740	25	28	30
## tenure	202	22.064	16.113	8	9	47
## commute	154	101.494	99.220	30	32	240
## gender_num	202	0.000	0.000	0	0	0
## married_num	202	1.000	0.000	1	1	1
## high_educ_num	202	1.000	0.000	1	1	1
## children_num	202	1.000	0.000	1	1	1
## costofcommute	202	20.515	21.521	2	2	55
## promote_switch_num	202	0.000	0.000	0	0	0
## quitjob_num	202	1.000	0.000	1	1	1
## group_index	202	60.000	0.000	60	60	60

```

# group definition
#group60 %>% select(gender, married, bedroom, children, high_educ, promote_switch, quitjob, homethatweek)
# distinct persons
unique(group60 %>%select(personid))

```

```

## personid
## 1 40174
## 2 39942
## 6 16334
## 17 29808

```

```

nums <- dplyr::select(group60, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")

```

```

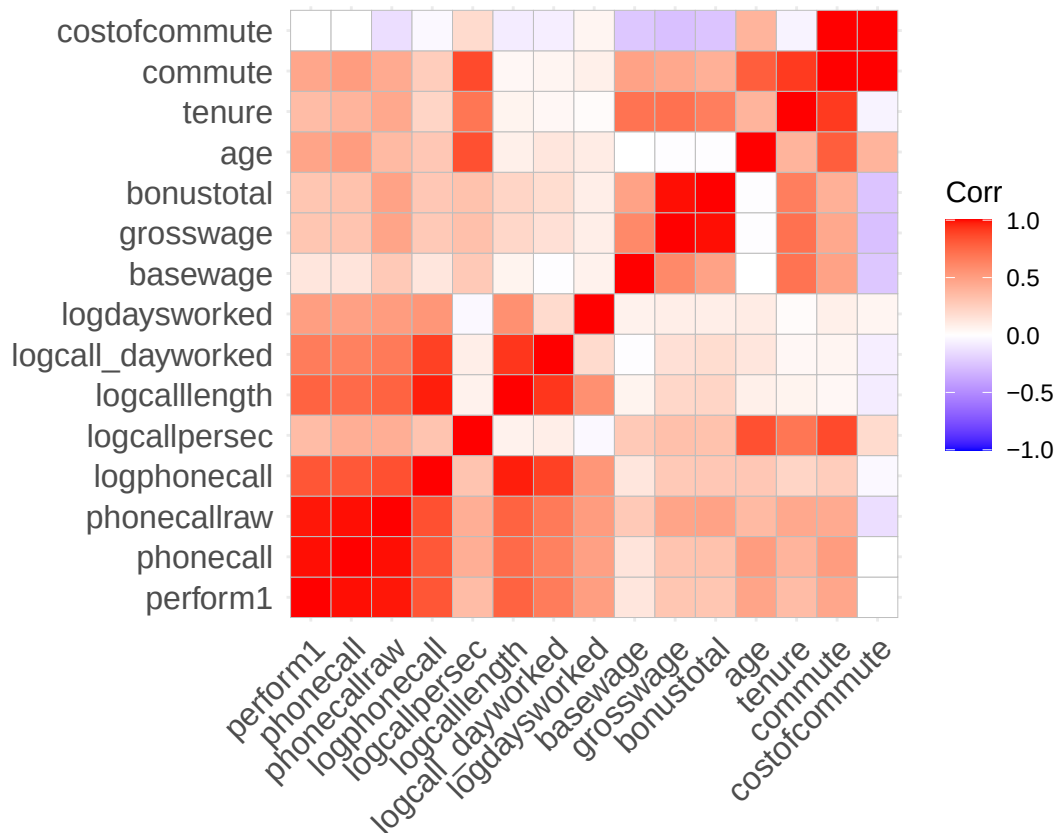
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero

```

```

ggcorrplot(cm, lab = FALSE)

```



```
# radar plot
radar_by_group(df, group_col="group_index", include_group_indices=c(60), title="Group 60 characteristics")

## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "group_index"      "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"          "grosswage"
## [16] "exhaustion"       "tenure"
## # A tibble: 62 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>     <dbl>         <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
## 1         1    103.          6.45    12.4    29.2     1       0       0
## 2         2    180           18      21.8    20.5     1       0       0
## 3         3    120           9       18.8    23       1       0       0
## 4         4    119.         6.71    22.0    22.4     1       0       1
## 5         5    92.0         5.12    14.5    26.6     1       0       1
## 6         6    131.         7.58    18.2    24.8     1       0       1
## 7         7    118.         4.11    NaN     NaN      1       0       1
## 8         8     20           0       NaN     NaN      1       0       1
## 9         9    112.         3.88    13.8    16.2     1       0       1
## 10        10     28           0       NaN     NaN      1       0       1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
```

```
## [1] 60
## # A tibble: 1 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>    <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1         60    101.         20.5     NaN     NaN      0      1      1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "60"

## pdf
## 2
```

```
stargazer( group60, type = "text", omit.stat = c("f", "adj.rsq", "ser"), no.space = TRUE, header = FALSE)
```

```
##
## =====
## Statistic          N      Mean    St. Dev.    Min      Max
## -----
## bedroom            202    1.000    0.000        1        1
## children            202    1.000    0.000        1        1
## high_educ           202    1.000    0.000        1        1
## promote_switch      202    0.000    0.000        0        0
## quitjob             202    1.000    0.000        1        1
## homethatweek        202    0.000    0.000        0        0
## perform1            201   -0.216    1.292     -2.996    2.164
## phonecall           202   -0.182    1.343     -3.113    2.729
## phonecallraw        184  435.120  181.523        2      823
## logphonecall        184    5.904    0.790     0.693    6.713
## logcallpersec       184   -5.074    0.196     -5.578   -4.533
## logcalllength       183   10.976    0.754     6.271   11.880
## logcall_dayworked   183    9.416    0.631     5.578   10.088
## logdaysworked      202    1.532    0.364     0.000    1.946
## homethatweek_num    202    0.000    0.000        0        0
## basewage            195 1,555.128 109.941    1,400    1,800
## grosswage           195 2,811.191 652.400  1,820.000 4,434.900
## bonustotal          195 1,256.063 592.845    20.000  2,734.900
## age                 202   27.847    1.740     25      30
## tenure              202   22.064    16.113      8      47
## commute             154  101.494   99.220     30     240
## gender_num          202    0.000    0.000        0        0
## married_num         202    1.000    0.000        1        1
## high_educ_num       202    1.000    0.000        1        1
## children_num        202    1.000    0.000        1        1
## costofcommute       202   20.515   21.521      2      55
## promote_switch_num  202    0.000    0.000        0        0
## quitjob_num         202    1.000    0.000        1        1
## group_index         202   60.000    0.000     60     60
## -----
```

Compare the quitter groups

```
df <- volunteer_endperiod_attitude_performance_wage

# plot characteristic radar plot for quitters (per group and summary)
```

```

quitter_group_vec<-c(7,16,40,48,60)
radar_by_group(df, group_col="group_index", include_group_indices=quitter_group_vec, title="Quitter profile")

## [1] "commute"          "costofcommute"      "negative"
## [4] "positive"         "group_index"        "gender_num"
## [7] "married_num"      "bedroom"            "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"           "grosswage"
## [16] "exhaustion"       "tenure"
## # A tibble: 62 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>     <dbl>         <dbl>     <dbl>     <dbl> <dbl>     <dbl>     <dbl>
## 1         1      103.          6.45      12.4      29.2     1         0         0
## 2         2      180          18        21.8      20.5     1         0         0
## 3         3      120           9        18.8      23       1         0         0
## 4         4      119.          6.71      22.0      22.4     1         0         1
## 5         5      92.0          5.12      14.5      26.6     1         0         1
## 6         6      131.          7.58      18.2      24.8     1         0         1
## 7         7      118.          4.11      NaN        NaN       1         0         1
## 8         8        20           0         NaN        NaN       1         0         1
## 9         9      112.          3.88      13.8      16.2     1         0         1
## 10        10       28           0         NaN        NaN       1         0         1
## # i 52 more rows
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "group_index"
## [1] 7 16 40 48 60
## # A tibble: 5 x 17
##   group_index commute costofcommute negative positive gender married bedroom
##   <int>     <dbl>         <dbl>     <dbl>     <dbl> <dbl>     <dbl>     <dbl>
## 1         7      118.          4.11      NaN        NaN       1         0         1
## 2        16      52.1          3.57      NaN        NaN       1         0         1
## 3        40      120.          8.28      NaN        NaN       0         0         1
## 4        48      33.5          11.7      NaN        NaN       0         0         1
## 5        60      101.          20.5      NaN        NaN       0         1         1
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "7" "16" "40" "48" "60"
## pdf
## 2
radar_by_group(df, group_col="quitjob", include_group_indices=c(TRUE), title="General quitter profile")

## [1] "commute"          "costofcommute"      "negative"
## [4] "positive"         "quitjob"            "gender_num"
## [7] "married_num"      "bedroom"            "children_num"
## [10] "high_educ_num"    "promote_switch_num" "homethatweek_num"
## [13] "perform1"         "grosswage"          "exhaustion"
## [16] "tenure"
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>     <dbl>     <dbl> <dbl>     <dbl>     <dbl>

```

```

## 1 FALSE      103.          6.59      16.7      24.2  0.513  0.163  0.966
## 2 TRUE       105.          9.14      NaN       NaN    0.450  0.239  1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1] TRUE
## # A tibble: 1 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>          <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 TRUE      105.          9.14      NaN      NaN    0.450  0.239    1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "TRUE"

## pdf
## 2

# plot characteristic radar plot for non quitters (per group and summary)
# get all available groups
all_groups_vec <- unique(df %>% pull(group_index))
non_quitter_group_vec <- setdiff(all_groups_vec, quitter_group_vec)

radar_by_group(df, group_col="quitjob", include_group_indices=c(FALSE), title="General non-quitter prof.

## [1] "commute"          "costofcommute"      "negative"
## [4] "positive"          "quitjob"            "gender_num"
## [7] "married_num"       "bedroom"            "children_num"
## [10] "high_educ_num"     "promote_switch_num" "homethatweek_num"
## [13] "perform1"          "grosswage"          "exhaustion"
## [16] "tenure"
## # A tibble: 2 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>          <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE      103.          6.59      16.7      24.2  0.513  0.163  0.966
## 2 TRUE       105.          9.14      NaN      NaN    0.450  0.239  1
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "quitjob"
## [1] FALSE
## # A tibble: 1 x 16
##   quitjob commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>          <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE      103.          6.59      16.7      24.2  0.513  0.163  0.966
## # i 8 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>, exhaustion <dbl>,
## #   tenure <dbl>
## [1] "FALSE"

## pdf
## 2

#radar_by_group(df)

```

Top performing 10%

Select

```
df <- volunteer_endperiod_attitude_performance_wage

#df <- na.omit(df$perform1)

# define threshold
threshold <- quantile(df$perform1, 0.9, na.rm = TRUE)

# select data
df <- df %>%
  mutate(isTop10 = if_else(perform1 >= threshold, TRUE, FALSE)) %>% filter(!is.na(isTop10))

top10 <- df %>% filter(isTop10 == TRUE)
bottom90 <- df %>% filter(isTop10 == FALSE)
```

Compare

Here we use the same approach as for the quitters:

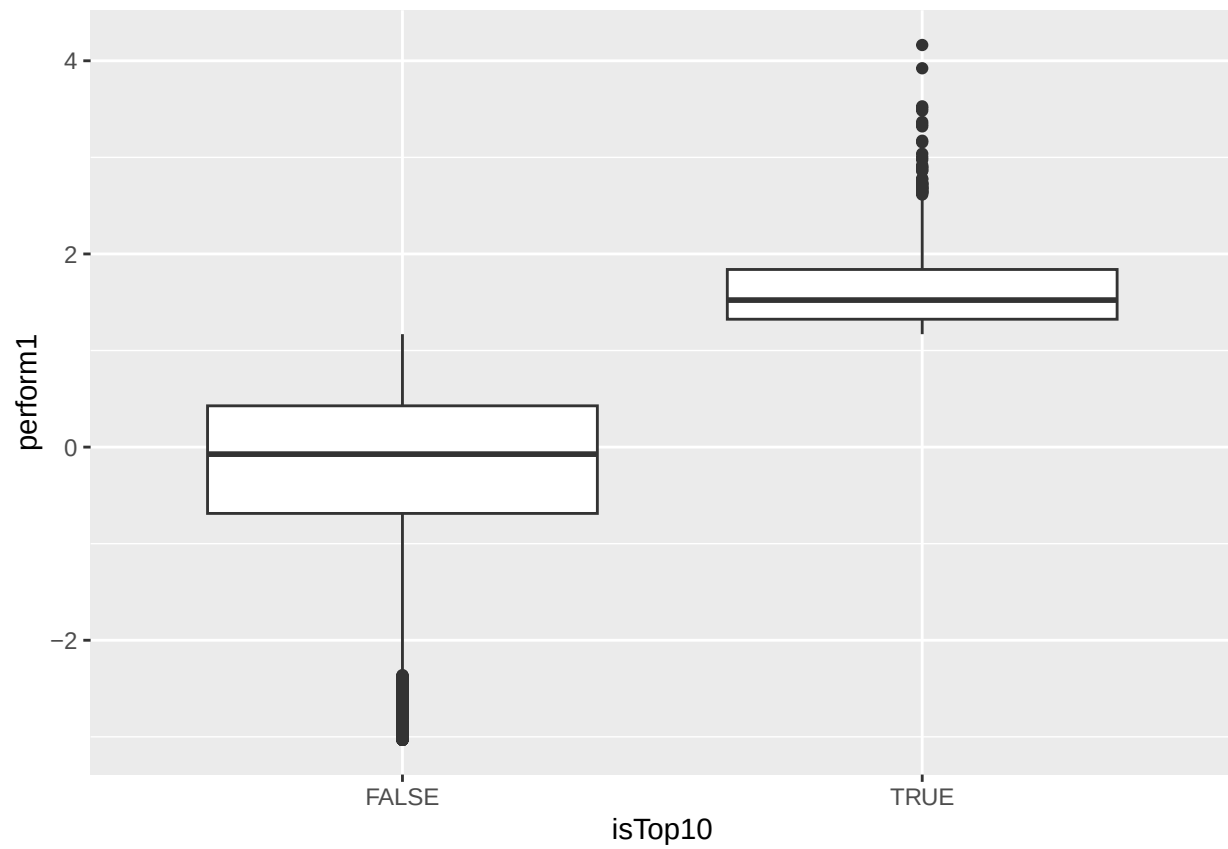
```
# compare general statistics
stargazer(as.data.frame(top10), type = "text", median = TRUE, header = TRUE)
```

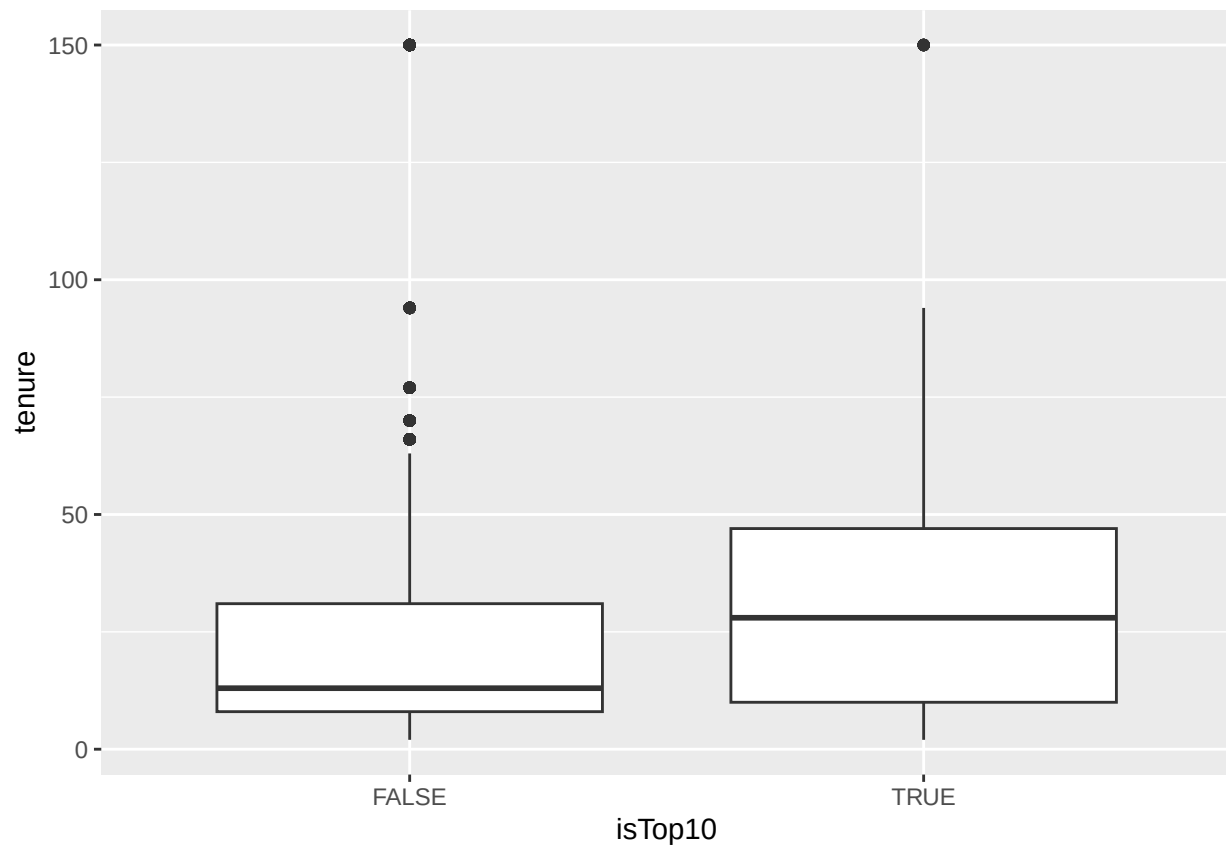
```
##
## =====
## Statistic      N      Mean    St. Dev.    Min      Median      Max
## -----
## bedroom        983    0.961    0.193      0         1         1
## children        983    0.239    0.427      0         0         1
## high_educ       983    0.481    0.500      0         0         1
## promote_switch  983    0.217    0.412      0         0         1
## quitjob         983    0.182    0.386      0         0         1
## homethatweek    983    0.254    0.436      0         0         1
## exhaustion      189    6.804    6.105      0         6        25
## negative        189   14.947    5.966      8         13       40
## positive        189   26.619    5.710      8         28       40
## perform1        983    1.643    0.430      1.169     1.522     4.163
## phonecall       982    1.480    0.488      0.039     1.415     5.828
## phonecallraw    980   665.388   93.648     426       649.5    1,264
## logphonecall    980    6.491    0.134      6.054     6.475     7.142
## logcallpersec   979   -5.125    0.140     -5.458    -5.120    -3.832
## logcalllength   980   11.615    0.160     10.231    11.610    12.116
## logcall_dayworked 981    9.818    0.162      8.285     9.827    10.359
## logdaysworked  982    1.796    0.137      1.386     1.792     1.946
## homethatweek_num 983    0.254    0.436      0         0         1
## basewage        980  1,658.520  183.865    1,300     1,600     2,300
## grosswage       969  3,750.751 1,055.672  1,375.000  3,546.670  8,418.000
## bonustotal      980  2,096.829  953.693    70.000    1,943.205  6,268.000
## age            983    23.834    5.606      1         23        34
## tenure         983    32.765    25.032      2         28       150
## commute        947   107.360   69.337      2         80       300
## gender_num      983    0.752    0.432      0         1         1
## married_num     983    0.308    0.462      0         0         1
```

```
## high_educ_num      983    0.481    0.500      0      0      1
## children_num       983    0.239    0.427      0      0      1
## costofcommute      905    6.407    6.759    0.000    4.000    55.000
## promote_switch_num 983    0.217    0.412      0      0      1
## quitjob_num        983    0.182    0.386      0      0      1
## group_index        983   23.146   17.373      1     16     62
## isTop10            983    1.000    0.000      1      1      1
## -----
stargazer(as.data.frame(bottom90), type = "text", median = TRUE, header = TRUE)
```

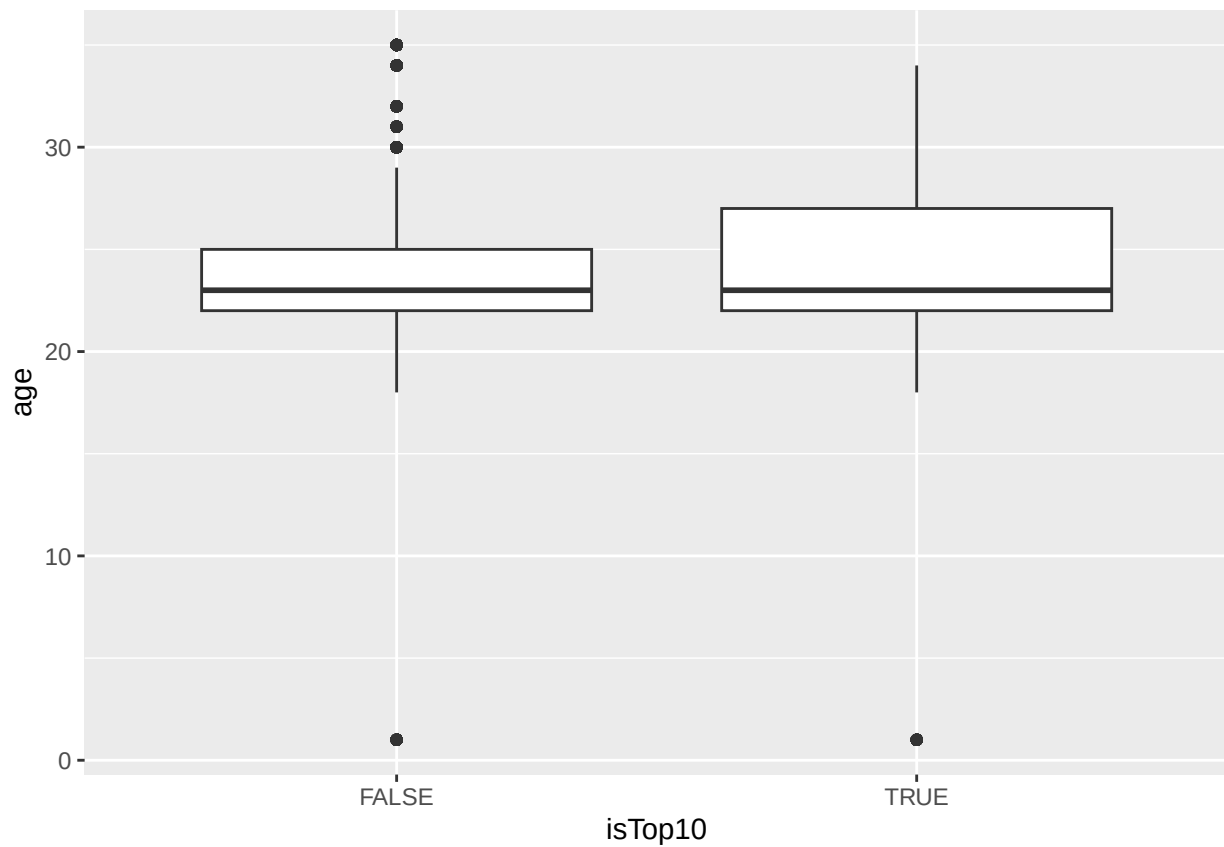
```
##
## =====
## Statistic          N      Mean    St. Dev.    Min      Median      Max
## -----
## bedroom           8,844    0.975    0.156      0         1         1
## children           8,844    0.132    0.339      0         0         1
## high_educ          8,844    0.392    0.488      0         0         1
## promote_switch     8,844    0.177    0.381      0         0         1
## quitjob            8,844    0.249    0.433      0         0         1
## homethatweek       8,844    0.190    0.392      0         0         1
## exhaustion         1,976    8.742    7.861     0.000     8.000    36.000
## negative           1,976   16.800    6.861     8.000    16.000    40.000
## positive           1,976   23.948    6.699     8.000    24.000    40.000
## perform1           8,844   -0.199    0.851    -3.031    -0.073    1.169
## phonecall          8,714   -0.172    0.854    -3.113    -0.046    3.812
## phonecallraw       8,561  414.722  123.299      1        429      977
## logphonecall       8,574    5.955    0.476     0.000     6.061    6.884
## logcallpersec      8,571   -5.175    0.160    -5.951    -5.173    -1.099
## logcalllength      8,571   11.128    0.511     2.485    11.237    11.883
## logcall_dayworked  8,569    9.438    0.448     2.485     9.514    10.186
## logdaysworked     8,830    1.674    0.288     0.000     1.792    1.946
## homethatweek_num   8,844    0.190    0.392      0         0         1
## basewage           8,794  1,590.322 170.628   527.350  1,550.000 2,450.000
## grosswage          8,702  3,009.238 942.059  1,140.280  2,808.740 14,553.000
## bonustotal         8,794  1,417.370 872.446    0.000    1,208.000 12,853.000
## age                8,844   23.444    4.026      1         23        35
## tenure             8,844   23.011   24.617      2         13       150
## commute            7,936  103.154   67.580      1         80       300
## gender_num         8,844    0.472    0.499      0         0         1
## married_num        8,844    0.170    0.375      0         0         1
## high_educ_num      8,844    0.392    0.488      0         0         1
## children_num       8,844    0.132    0.339      0         0         1
## costofcommute      8,336    7.328    7.314     0.000     6.000    55.000
## promote_switch_num 8,844    0.177    0.381      0         0         1
## quitjob_num        8,844    0.249    0.433      0         0         1
## group_index        8,844   29.117   17.646      1         37        62
## isTop10            8,844    0.000    0.000      0         0         0
## -----
```

```
# compare distributions
ggplot(df, aes(x=isTop10, y=perform1)) + geom_boxplot()
```



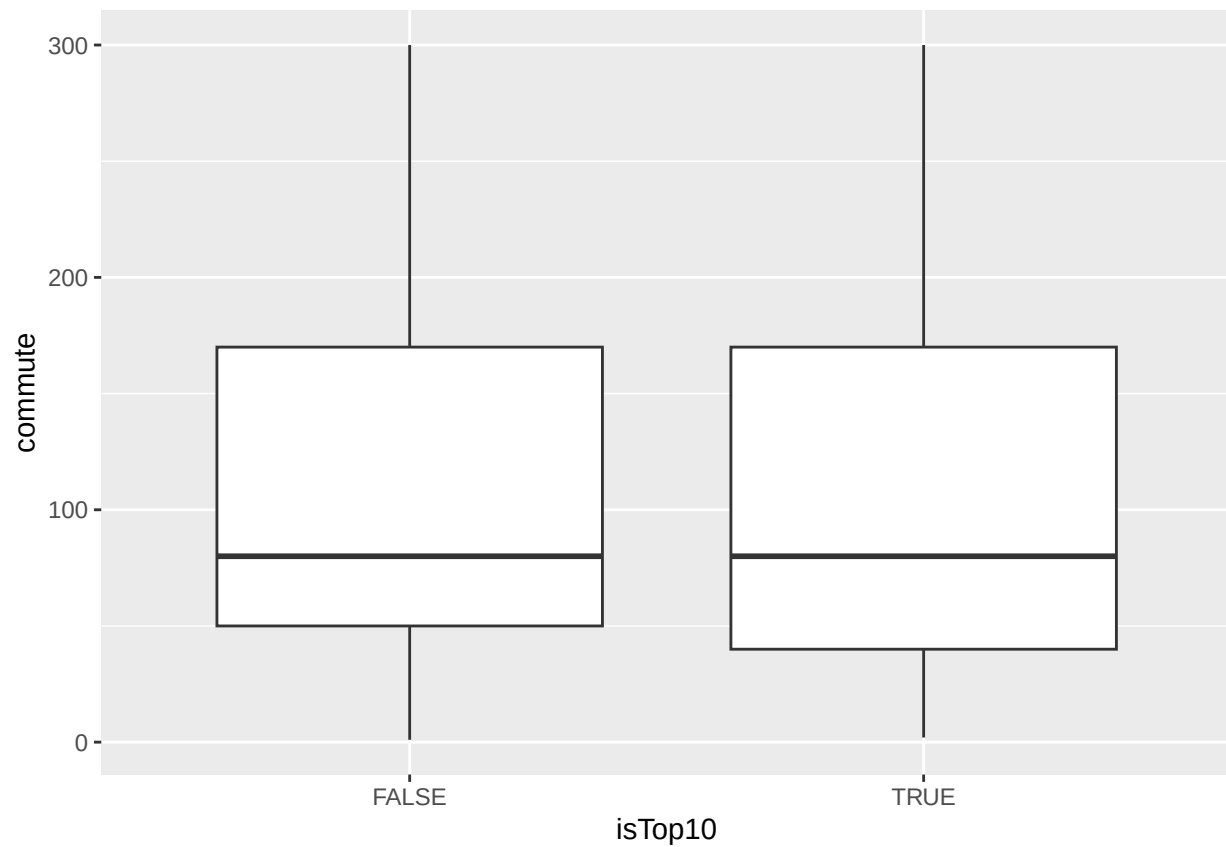


```
ggplot(df, aes(x=isTop10, y=age)) + geom_boxplot()
```



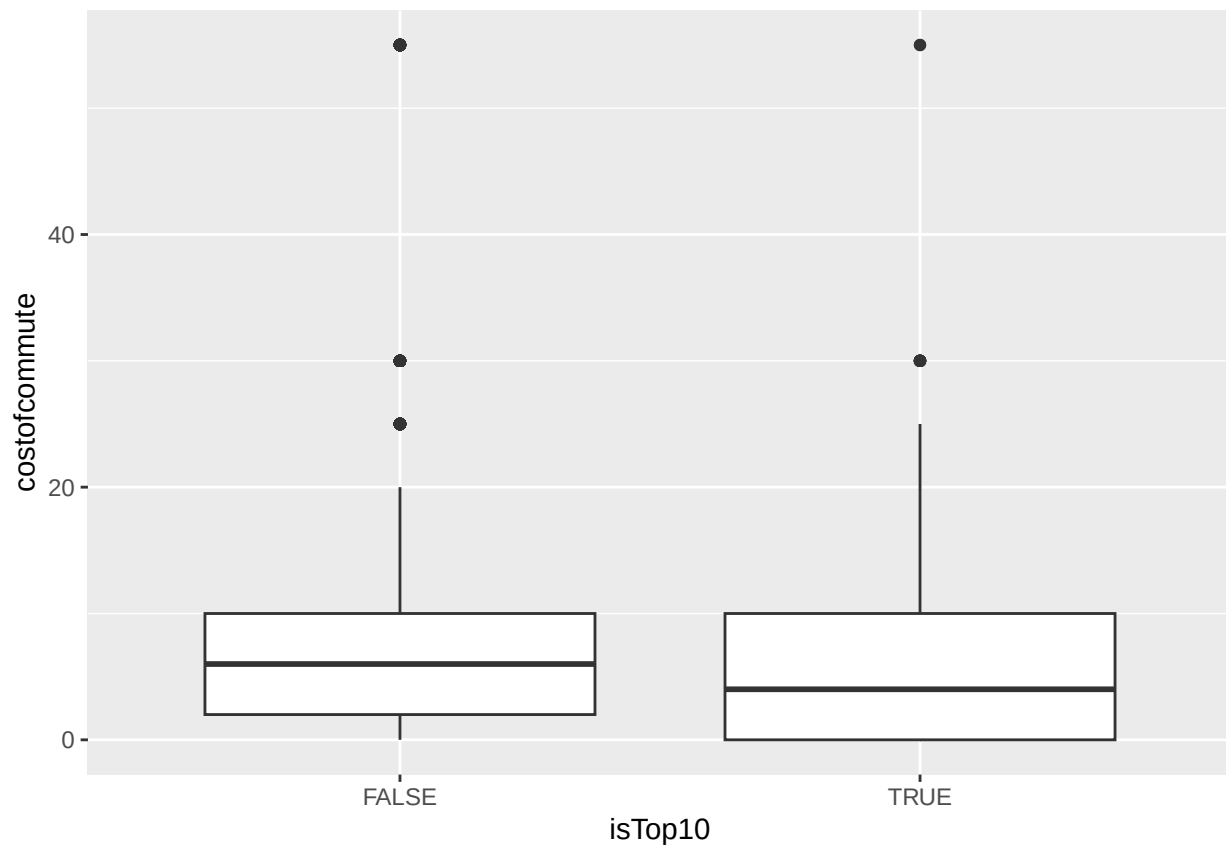
```
ggplot(df, aes(x=isTop10, y=commute)) + geom_boxplot()
```

```
## Warning: Removed 944 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



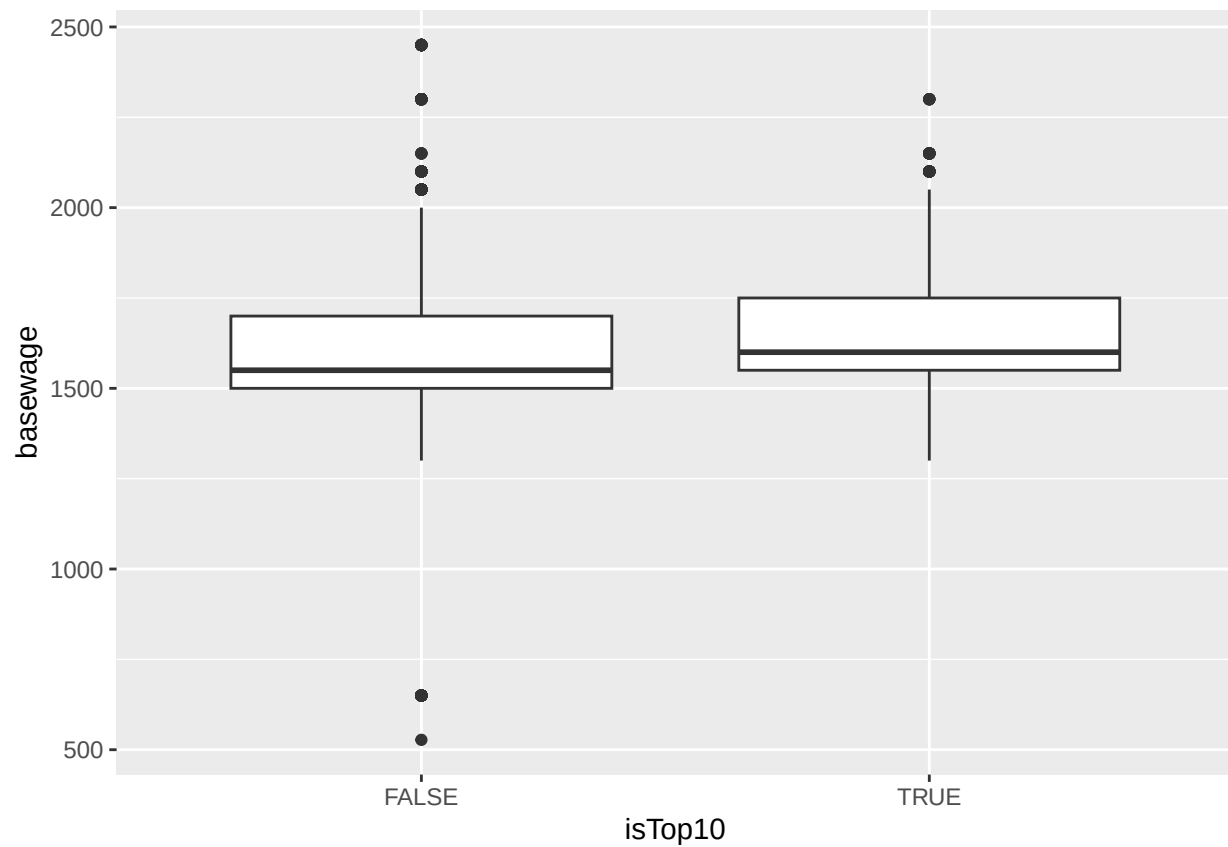
```
ggplot(df, aes(x=isTop10, y=costofcommute)) + geom_boxplot()
```

```
## Warning: Removed 586 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



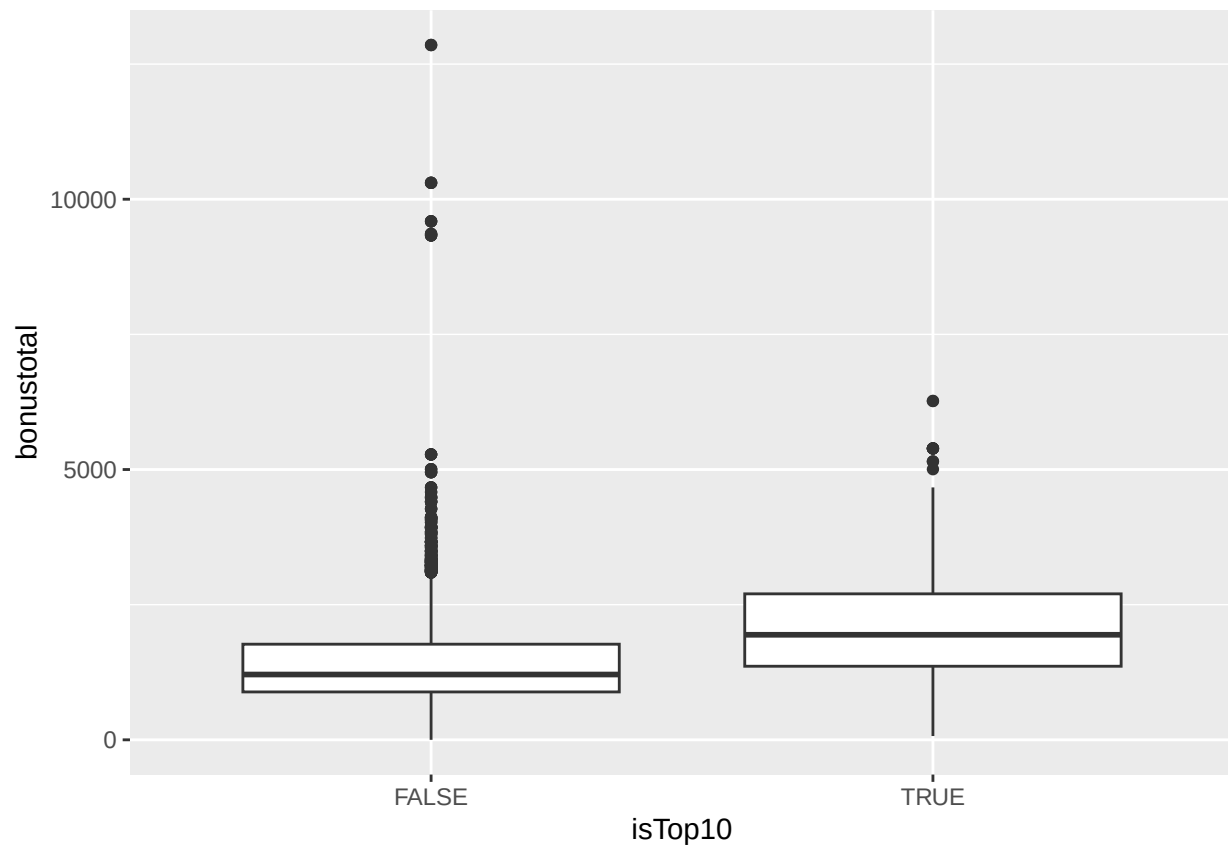
```
ggplot(df, aes(x=isTop10, y=basewage)) + geom_boxplot()
```

```
## Warning: Removed 53 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



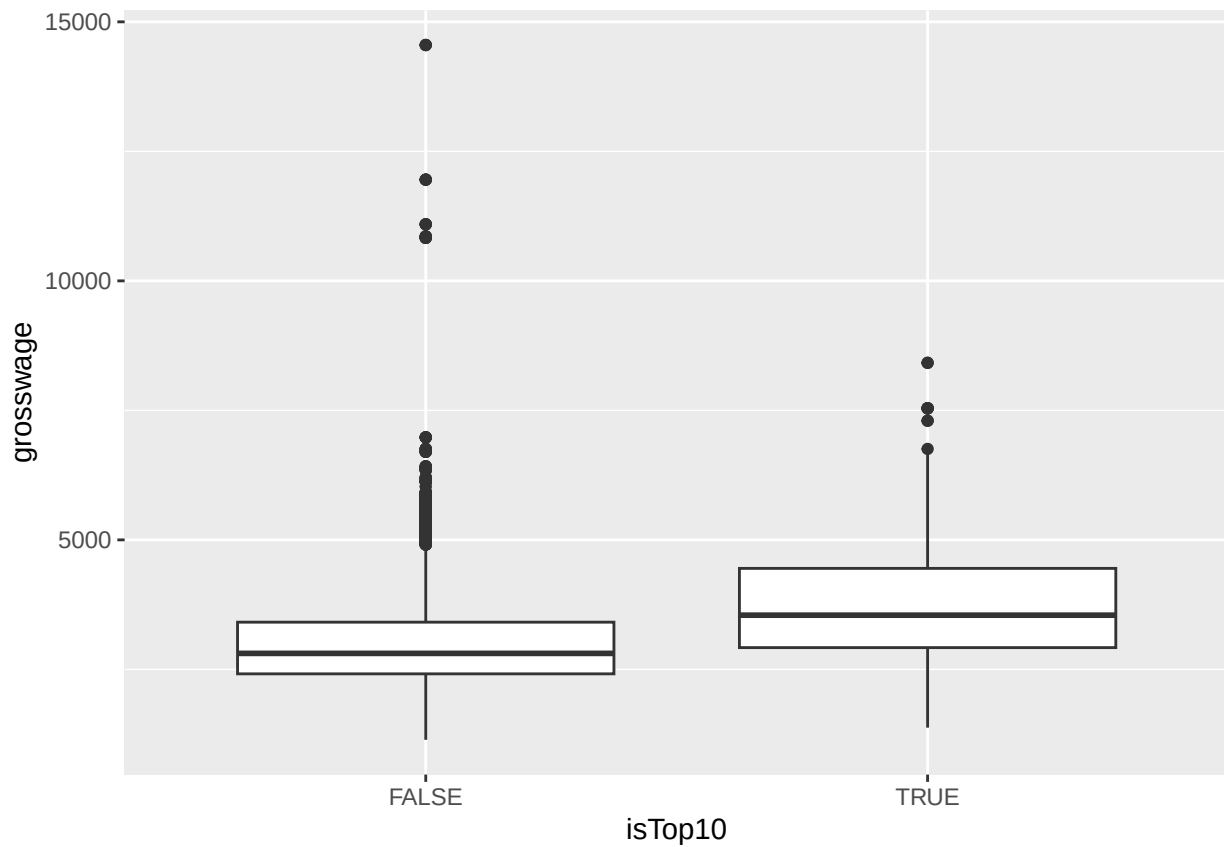
```
ggplot(df, aes(x=isTop10, y=bonustotal)) + geom_boxplot()
```

```
## Warning: Removed 53 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
ggplot(df, aes(x=isTop10, y=grosswage)) + geom_boxplot()
```

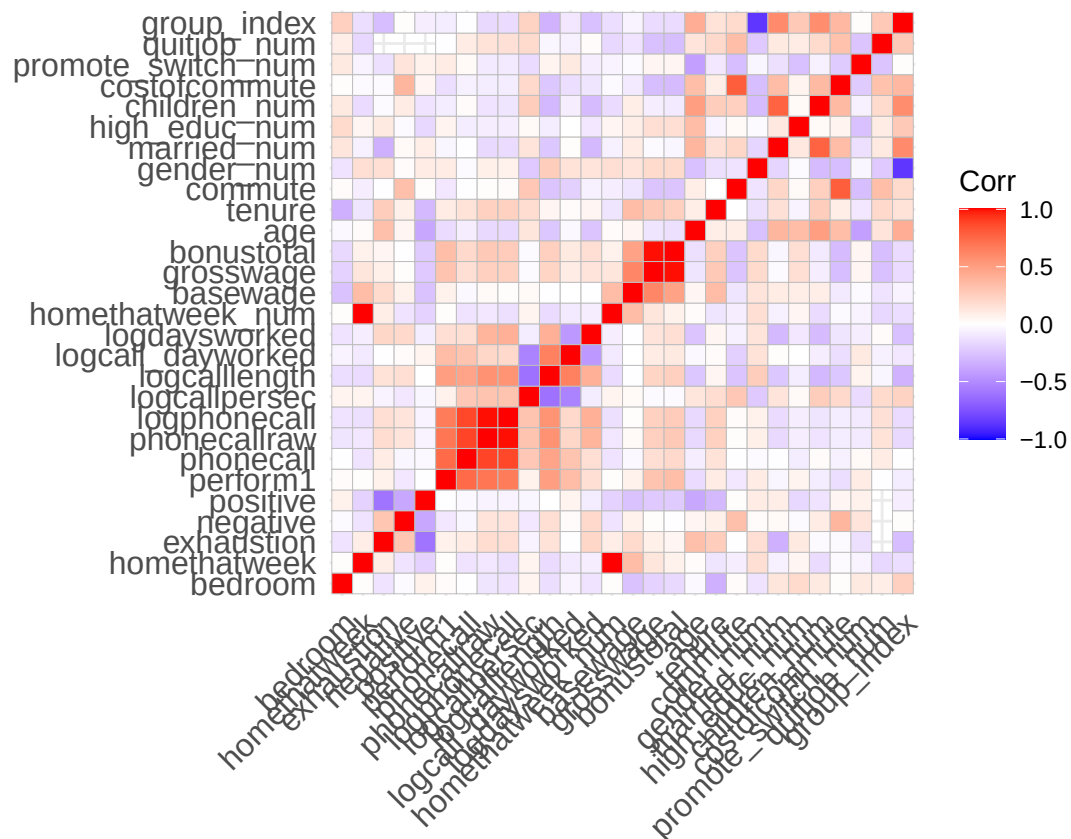
```
## Warning: Removed 156 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



```
# create correlation matrices  
nums <- dplyr::select(top10, where(is.numeric))  
cm <- cor(nums, use = "pairwise.complete.obs")
```

```
## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is  
## zero
```

```
ggcorrplot(cm, lab = FALSE)
```



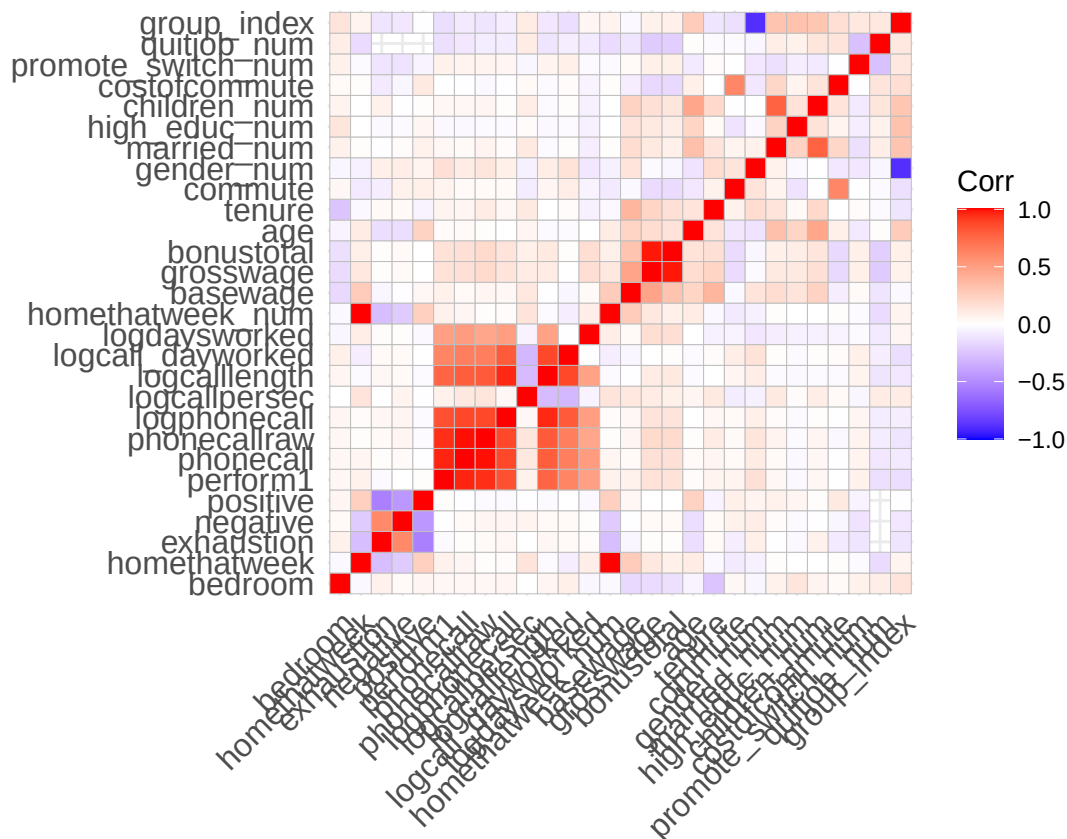
```

nums <- dplyr::select(bottom90, where(is.numeric))
cm <- cor(nums, use = "pairwise.complete.obs")

## Warning in cor(nums, use = "pairwise.complete.obs"): the standard deviation is
## zero

ggcorrplot(cm, lab = FALSE)

```



```
# radar plot
radar_by_group(df, group_col="isTop10", include_group_indices=c(TRUE, FALSE), title="Top vs low perform")

## [1] "commute"          "costofcommute"    "negative"
## [4] "positive"         "isTop10"          "gender_num"
## [7] "married_num"      "bedroom"          "children_num"
## [10] "high_educ_num"    "promote_switch_num" "quitjob_num"
## [13] "homethatweek_num" "perform1"         "grosswage"
## [16] "exhaustion"       "tenure"
## # A tibble: 2 x 17
##   isTop10 commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE     103.           7.33     16.8     23.9  0.472   0.170   0.975
## 2 TRUE      107.           6.41     14.9     26.6  0.752   0.308   0.961
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
## [1] "isTop10"
## [1] TRUE FALSE
## # A tibble: 2 x 17
##   isTop10 commute costofcommute negative positive gender married bedroom
##   <lgl>     <dbl>         <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>
## 1 FALSE     103.           7.33     16.8     23.9  0.472   0.170   0.975
## 2 TRUE      107.           6.41     14.9     26.6  0.752   0.308   0.961
## # i 9 more variables: children <dbl>, high_educ <dbl>, promote_switch <dbl>,
## #   quitjob <dbl>, homethatweek <dbl>, perform1 <dbl>, grosswage <dbl>,
## #   exhaustion <dbl>, tenure <dbl>
```

```
## [1] "FALSE" "TRUE"
```

```
## pdf
```

```
## 2
```

The average employee among the top 10% performing ones, has no children, though it is more likely than for the other 90%, works 25% of the time from home, is almost 24 years old, works for 32 months at the company and an average commute and average cost of commute. Almost 22% receive a promotion, whereas 17.7% of lower performing people receive one. They earn over the average by bonuses, but have a average base salary. Though some not high performing individuals receive enormous bonuses. Only 18% high performing people quit while 25% quit from the other 90%.

Top performer are less likely to quit when receiving a higher bonus than non performer, but more likely if they have a long commute as this is negatively correlative with their negative attitude. Working from home does not affect performance of both groups. Working from home increases positivity while lowering negativity and exhaustion for the top performer.

Recommendations

- Bonuses combined with goals/milestones to motivate people and increase performance, but only if you reach them
- check existing bonuses for not high performing people and reevaluate if that is fair, this might be a motivation killer
- offer more working from home for high performer, as it does not affect their performance and rather increases positivity

Make perform1 and grosswage poxplots look nice

```
p <- ggplot(df, aes(y = perform1, x=isTop10)) +  
  geom_boxplot(  
    outlier.alpha = 0.6,  
    outlier.size = 1.8,  
    fill = "#9df3c4",  
    color = "#0d0c3f"  
  ) +  
  labs(y = "Performance score", title = "Performance - Top-Performer vs. others") +  
  theme_minimal()  
  
# add annotations  
p
```



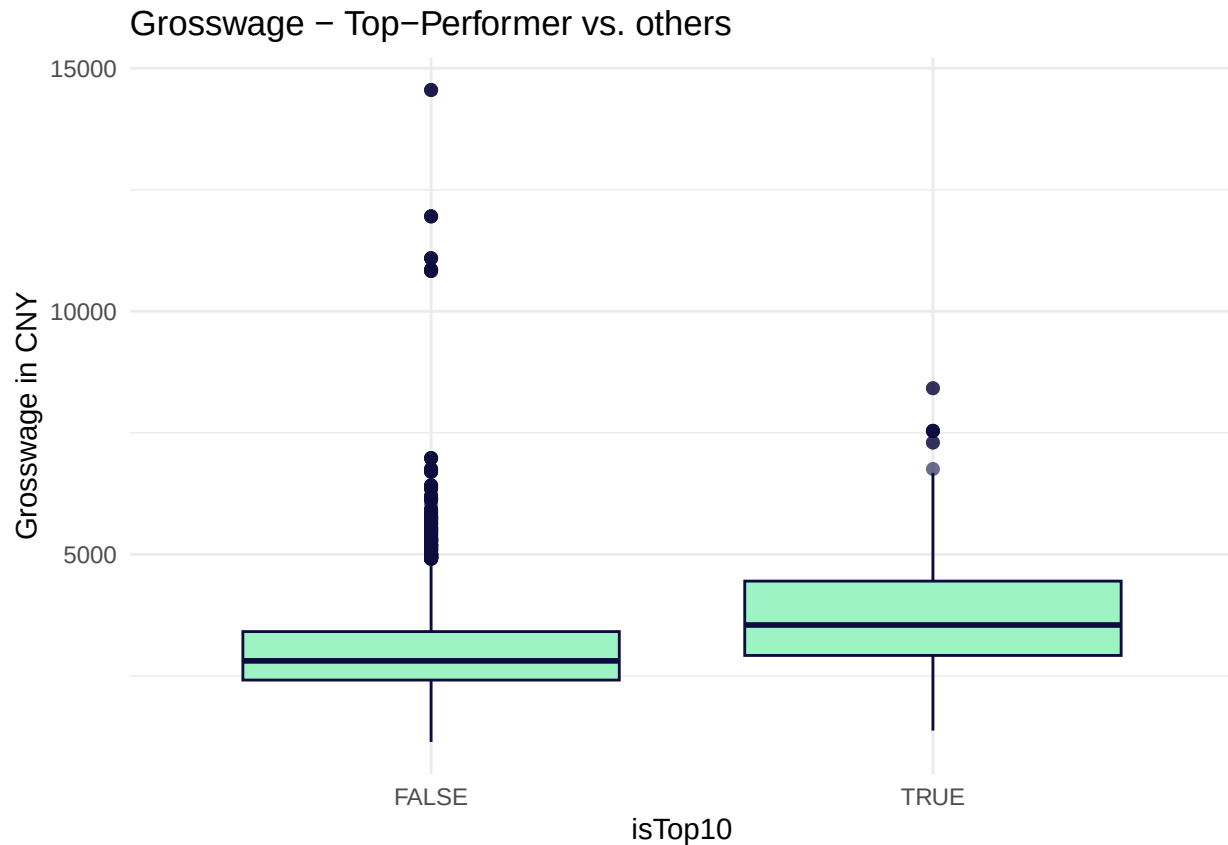
```
ggsave("performance_top_low.png", width = 7, height = 5, dpi = 300)
```

```
p <- ggplot(df, aes(y = grosswage, x=isTop10)) +
  geom_boxplot(
    outlier.alpha = 0.6,
    outlier.size = 1.8,
    fill = "#9df3c4",
    color = "#0d0c3f"
  ) +
  labs(y = "Grosswage in CNY", title = "Grosswage - Top-Performer vs. others") +
  theme_minimal()
```

```
# add annotations
```

```
p
```

```
## Warning: Removed 156 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



```
colnames(df)
```

```
## [1] "gender"      "married"      "bedroom"
## [4] "children"    "high_educ"    "promote_switch"
## [7] "quitjob"     "homethatweek" "personid"
## [10] "year_month"  "year_week"    "exhaustion"
## [13] "negative"    "positive"     "perform1"
## [16] "phonecall"   "phonecallraw" "logphonecall"
## [19] "logcallpersec" "logcalllength" "logcall_dayworked"
## [22] "logdaysworked" "date"         "homethatweek_num"
## [25] "week_date"   "basewage"     "grosswage"
## [28] "bonustotal"  "age"          "tenure"
## [31] "commute"     "gender_num"   "married_num"
## [34] "high_educ_num" "children_num" "costofcommute"
## [37] "promote_switch_num" "quitjob_num" "group_index"
## [40] "isTop10"
```

```
ggsave("grosswage_top_low.png", width = 7, height = 5, dpi = 300)
```

```
## Warning: Removed 156 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

Hypothesis testing

Quitters vs non quitters

```

# H0: There is no difference in earnings between quitters and non-quitters
# H1: There is a significant difference in earnings between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_grossw = mean(grosswage, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_grossw)) %>%
  t.test(mean_grossw ~ quitjob_num, data = ., alternative = "greater")

##
## Welch Two Sample t-test
##
## data: mean_grossw by quitjob_num
## t = 5.1008, df = 97.864, p-value = 8.285e-07
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
## 367.794 Inf
## sample estimates:
## mean in group 0 mean in group 1
## 3210.301 2664.978

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between perform1 of quitters and non-quitters
#H1: There is a significant difference between perform1 of quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_perform1 = mean(perform1, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_perform1)) %>%
  t.test(mean_perform1 ~ quitjob_num, data = ., alternative = "less")

##
## Welch Two Sample t-test
##
## data: mean_perform1 by quitjob_num
## t = 3.0921, df = 76.915, p-value = 0.9986
## alternative hypothesis: true difference in means between group 0 and group 1 is less than 0
## 95 percent confidence interval:
## -Inf 0.4601867
## sample estimates:
## mean in group 0 mean in group 1
## 0.02464293 -0.27448432

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference in tenure between quitters and non-quitters
#H1: There is a significant difference in tenure between quitters and non-quitters
volunteer_endperiod_attitude_performance_wage %>%
  group_by(personid, quitjob_num) %>%
  summarise(mean_tenure = mean(tenure, na.rm = TRUE), .groups = "drop") %>%
  filter(!is.na(mean_tenure)) %>%
  t.test(mean_tenure ~ quitjob_num, data = ., alternative = "greater")

##

```

```
## Welch Two Sample t-test
##
## data: mean_tenure by quitjob_num
## t = 0.37039, df = 64.932, p-value = 0.3561
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
## -6.401416 Inf
## sample estimates:
## mean in group 0 mean in group 1
## 23.92632 22.10000
```

we fail to reject the null hypothesis, as p-value is greater than 0.05

#H0: there is no difference in commutecost between quitters and non-quitters
#H1: there is a significant difference in commutecost between quitters and non-quitters
 volunteer_endperiod_attitude_performance_wage %>%

```
group_by(personid, quitjob_num) %>%
summarise(mean_costofcommute = mean(costofcommute, na.rm = TRUE), .groups = "drop") %>%
filter(!is.na(mean_costofcommute)) %>%
t.test(mean_costofcommute ~ quitjob_num, data = ., alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: mean_costofcommute by quitjob_num
## t = -1.3774, df = 50.59, p-value = 0.9128
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
## -5.04748 Inf
## sample estimates:
## mean in group 0 mean in group 1
## 6.608471 8.885781
```

we fail to reject the null hypothesis, as p-value is greater than 0.05

#H0: there is no difference in homeoffice_days between quitters and non-quitters
#H1: there is a significant difference in homeoffice_days between quitters and non-quitters
 volunteer_endperiod_attitude_performance_wage %>%

```
group_by(personid, quitjob_num) %>%
summarise(mean_homethatweek = mean(homethatweek_num, na.rm = TRUE), .groups = "drop") %>%
filter(!is.na(mean_homethatweek)) %>%
t.test(mean_homethatweek ~ quitjob_num, data = ., alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: mean_homethatweek by quitjob_num
## t = 4.9726, df = 113.28, p-value = 1.185e-06
## alternative hypothesis: true difference in means between group 0 and group 1 is greater than 0
## 95 percent confidence interval:
## 0.1012484 Inf
## sample estimates:
## mean in group 0 mean in group 1
```

```
##          0.22867153          0.07675795
# we reject the null hypothesis, as p-value is below 0.05

#H0: there is no correlation between age and children_num among quitters
#H1: there is a correlation between age and children_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(children_num ~ (age > median(age, na.rm = TRUE)),
    data = ., alternative = "less")

##
## Welch Two Sample t-test
##
## data:  children_num by age > median(age, na.rm = TRUE)
## t = -27.221, df = 1135.4, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -0.4341011
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      0.03795721      0.50000000
# we reject the null hypothesis, as p-value is below 0.05

#H0: there is no correlation between age and married_num among quitters
#H1: there is a correlation between age and married_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(married_num ~ (age > median(age, na.rm = TRUE)),
    data = ., alternative = "less")

##
## Welch Two Sample t-test
##
## data:  married_num by age > median(age, na.rm = TRUE)
## t = -17.35, df = 1467.5, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -0.2842208
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      0.1145618      0.4285714
# we reject the null hypothesis, as p-value is below 0.05

#H0: there is no correlation between age and high_educ_num among quitters
#H1: there is a correlation between age and high_educ_num among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(high_educ_num ~ (age > median(age, na.rm = TRUE)),
    data = ., alternative = "less")
```

```
##
## Welch Two Sample t-test
##
## data: high_educ_num by age > median(age, na.rm = TRUE)
## t = -15.705, df = 2020.9, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -0.2790255
## sample estimates:
## mean in group FALSE mean in group TRUE
##      0.3374741      0.6491597
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

```
#H0: there is no correlation between mean(perform1) and age among quitters
#H1: there is a correlation between mean(perform1) and age among quitters
volunteer_endperiod_attitude_performance_wage %>%
  filter(quitjob_num == 1) %>%
  t.test(perform1 ~ (age > median(age, na.rm = TRUE)),
        data = ., alternative = "less")
```

```
##
## Welch Two Sample t-test
##
## data: perform1 by age > median(age, na.rm = TRUE)
## t = -6.0454, df = 1700.1, p-value = 9.138e-10
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -0.1953103
## sample estimates:
## mean in group FALSE mean in group TRUE
##      -0.34598213      -0.07761384
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

Group Hypotheses

Group 7

```
#H0: There is no difference between group 7 mean(quitjob_num) compared to overall average
#H1: Group 7 has higher mean(quitjob_num) compared to overall average
t.test(group7$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "gr
```

```
##
## Welch Two Sample t-test
##
## data: group7$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##      0.7548015      Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181
```

```

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 7 mean(perform1) compared to overall average
#H1: Group 7 has lower mean(perform1) compared to overall average
t.test(group7$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group7$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -6.218, df = 377.04, p-value = 6.683e-10
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf -0.2149481
## sample estimates:
## mean of x mean of y
## -0.30751216 -0.01499374

# we reject the null hypothesis, as p-value is below 0.05

```

Group 16

```

#H0: There is no difference between group 16 mean(quitjob_num) compared to overall average
#H1: Group 16 has higher mean(quitjob_num) compared to overall average
t.test(group16$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g")

##
## Welch Two Sample t-test
##
## data: group16$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
## t = 179.52, df = 10078, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
##  0.7548015      Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 16 mean(perform1) compared to overall average
#H1: Group 16 has lower mean(perform1) compared to overall average
t.test(group16$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group16$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = 0.24946, df = 356.69, p-value = 0.5984
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf 0.113049
## sample estimates:
## mean of x mean of y
## -0.0001399055 -0.0149937416

```

```
# we fail to reject the null hypothesis, as p-value is above 0.05
```

```
#H0: There is no correlation between age and tenure
```

```
#H1: There is a correlation between age and tenure
```

```
t.test(group16$tenure ~ (group16$age > median(group16$age, na.rm = TRUE)),  
       data = group16, alternative = "less")
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group16$tenure by group16$age > median(group16$age, na.rm = TRUE)
```

```
## t = -26.446, df = 230.34, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
```

```
## 95 percent confidence interval:
```

```
##      -Inf -9.530795
```

```
## sample estimates:
```

```
## mean in group FALSE mean in group TRUE
```

```
##      18.58286      28.74847
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

Group 40

```
#H0: There is no difference between group 40 mean(quitjob_num) compared to overall average
```

```
#H1: Group 40 has higher mean(quitjob_num) compared to overall average
```

```
t.test(group40$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group40$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
```

```
## t = 179.52, df = 10078, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is greater than 0
```

```
## 95 percent confidence interval:
```

```
## 0.7548015      Inf
```

```
## sample estimates:
```

```
## mean of x mean of y
```

```
## 1.0000000 0.2382181
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

```
#H0: There is no difference between group 40 mean(perform1) compared to overall average
```

```
#H1: Group 40 has lower mean(perform1) compared to overall average
```

```
t.test(group40$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group40$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
```

```
## t = -12.126, df = 745.84, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is less than 0
```

```
## 95 percent confidence interval:
```

```
##      -Inf -0.3860627
```

```
## sample estimates:
```

```
## mean of x mean of y
```

```
## -0.46173283 -0.01499374
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

Group 48

```
#H0: There is no difference between group 48 mean(quitjob_num) compared to overall average
```

```
#H1: Group 48 has higher mean(quitjob_num) compared to overall average
```

```
t.test(group48$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group48$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
```

```
## t = 179.52, df = 10078, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is greater than 0
```

```
## 95 percent confidence interval:
```

```
## 0.7548015 Inf
```

```
## sample estimates:
```

```
## mean of x mean of y
```

```
## 1.0000000 0.2382181
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

```
#H0: There is no difference between group 48 mean(perform1) compared to overall average
```

```
#H1: Group 48 has lower mean(perform1) compared to overall average
```

```
t.test(group48$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group48$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
```

```
## t = -12.806, df = 237.14, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is less than 0
```

```
## 95 percent confidence interval:
```

```
## -Inf -0.5333124
```

```
## sample estimates:
```

```
## mean of x mean of y
```

```
## -0.62725468 -0.01499374
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

Group 60

```
#H0: There is no difference between group 60 mean(quitjob_num) compared to overall average
```

```
#H1: Group 60 has higher mean(quitjob_num) compared to overall average
```

```
t.test(group60$quitjob_num, volunteer_endperiod_attitude_performance_wage$quitjob_num, alternative = "g
```

```
##
```

```
## Welch Two Sample t-test
```

```
##
```

```
## data: group60$quitjob_num and volunteer_endperiod_attitude_performance_wage$quitjob_num
```

```
## t = 179.52, df = 10078, p-value < 2.2e-16
```

```
## alternative hypothesis: true difference in means is greater than 0
```

```
## 95 percent confidence interval:
```

```
## 0.7548015      Inf
## sample estimates:
## mean of x mean of y
## 1.0000000 0.2382181

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no difference between group 60 mean(perform1) compared to overall average
#H1: Group 60 has lower mean(perform1) compared to overall average
t.test(group60$perform1, volunteer_endperiod_attitude_performance_wage$perform1, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$perform1 and volunteer_endperiod_attitude_performance_wage$perform1
## t = -2.1893, df = 204.81, p-value = 0.01485
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
##      -Inf -0.04921906
## sample estimates:
## mean of x mean of y
## -0.21566135 -0.01499374

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between age and commute
#H1: Older Members have higher commute
t.test(group60$commute ~ (group60$age > median(group60$age, na.rm = TRUE)),
      data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$commute by group60$age > median(group60$age, na.rm = TRUE)
## t = -2116.3, df = 101, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -208.9536
## sample estimates:
## mean in group FALSE mean in group TRUE
##      30.88235      240.00000

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between age and tenure
#H1: Older Members have higher tenure
t.test(group60$tenure ~ (group60$age > median(group60$age, na.rm = TRUE)),
      data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$tenure by group60$age > median(group60$age, na.rm = TRUE)
## t = -53.988, df = 149, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
```

```
##      -Inf -32.55052
## sample estimates:
## mean in group FALSE mean in group TRUE
##      13.42      47.00

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on grosswage
#H1: Higher tenure leads to higher grosswage
t.test(group60$grosswage ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$grosswage by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -16.582, df = 120.89, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -905.1534
## sample estimates:
## mean in group FALSE mean in group TRUE
##      2321.245      3326.925

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on basewage
#H1: Higher tenure leads to higher basewage
t.test(group60$basewage ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$basewage by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -16.28, df = 188.92, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -149.6183
## sample estimates:
## mean in group FALSE mean in group TRUE
##      1474.000      1640.526

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relation between tenure on bonustotal
#H1: Higher tenure leads to higher bonustotal
t.test(group60$bonustotal ~ (group60$tenure > median(group60$tenure, na.rm = TRUE)),
       data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data: group60$bonustotal by group60$tenure > median(group60$tenure, na.rm = TRUE)
## t = -13.722, df = 122.17, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
```

```
## 95 percent confidence interval:
##      -Inf -737.7938
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      847.2447      1686.3986

# we reject the null hypothesis, as p-value is below 0.05

#H0: There is no relationship between commute and perform1
#H1: Higher commute leads to lower perform1
t.test(group60$perform1 ~ (group60$commute > median(group60$commute, na.rm = TRUE)),
       data = group60, alternative = "less")

##
## Welch Two Sample t-test
##
## data:  group60$perform1 by group60$commute > median(group60$commute, na.rm = TRUE)
## t = -6.6431, df = 113.4, p-value = 5.568e-10
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##      -Inf -0.9848502
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      -0.5102064      0.8023030

# we reject the null hypothesis, as p-value is below 0.05
```

Top 10 performers

```
#H0: The variance(basewage) among top performers is equal to the variance(basewage) among the rest
#H1: The variance(basewage) among top performers is ~ 0
var.test(top10$basewage, bottom90$basewage)

##
## F test to compare two variances
##
## data:  top10$basewage and bottom90$basewage
## F = 1.1612, num df = 979, denom df = 8793, p-value = 0.00132
## alternative hypothesis: true ratio of variances is not equal to 1
## 95 percent confidence interval:
##  1.059462 1.277145
## sample estimates:
## ratio of variances
##      1.161167

# we reject the null hypothesis, as p-value is below 0.05

#H0: The mean(grosswage) of Top 10 is equal to the mean(grosswage) of the rest
#H1: The mean(grosswage) of Top 10 is greater than the mean(grosswage) of the rest
t.test(top10$grosswage, bottom90$grosswage, alternative = "greater")

##
## Welch Two Sample t-test
##
## data:  top10$grosswage and bottom90$grosswage
## t = 20.956, df = 1146.3, p-value < 2.2e-16
```

```

## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 683.2639      Inf
## sample estimates:
## mean of x mean of y
## 3750.751 3009.238

# we reject the null hypothesis, as p-value is below 0.05

#H0 the mean(children_num) of Top 10 is equal to the mean(children_num) of the rest
#H1: The mean(children_num) of Top 10 is higher than the mean(children_num) of the rest
t.test(top10$children_num, bottom90$children_num, alternative = "greater")

##
## Welch Two Sample t-test
##
## data: top10$children_num and bottom90$children_num
## t = 7.5753, df = 1123.9, p-value = 3.729e-14
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.08347982      Inf
## sample estimates:
## mean of x mean of y
## 0.2390641 0.1324062

# we reject the null hypothesis, as p-value is below 0.05

#H0: The mean(homethatweek_num) of Top 10 is equal to the mean(homethatweek_num) of the rest
#H1: The mean(homethatweek_num) of Top 10 is = 0.25
t.test(top10$homethatweek_num, bottom90$homethatweek_num, alternative = "greater")

##
## Welch Two Sample t-test
##
## data: top10$homethatweek_num and bottom90$homethatweek_num
## t = 4.4439, df = 1165.8, p-value = 4.837e-06
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.04059306      Inf
## sample estimates:
## mean of x mean of y
## 0.2543235 0.1898462

# we reject the null hypothesis, as p-value is below 0.05
# result for top 10 is ~ 0.25 (25.4)

#H0: The mean(age) of top performs is equal to the mean(age) of the rest
#H1: The mean(age) of top performs = 24
t.test(top10$age, bottom90$age, alternative = "greater")

##
## Welch Two Sample t-test
##
## data: top10$age and bottom90$age
## t = 2.1209, df = 1097.4, p-value = 0.01708
## alternative hypothesis: true difference in means is greater than 0

```

```
## 95 percent confidence interval:
## 0.08725801      Inf
## sample estimates:
## mean of x mean of y
## 23.83418 23.44426
```

```
# we reject the null hypothesis, as p-value is below 0.05
# result for top 10 is ~ 24 (23.8)
```

```
#H0: The mean(tenure) of top performs is equal to the mean(tenure) of the rest
#H1: The mean(tenure) of top performs > mean(tenure) of the rest
t.test(top10$tenure, bottom90$tenure, alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: top10$tenure and bottom90$tenure
## t = 11.61, df = 1202.9, p-value < 2.2e-16
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 8.37142      Inf
## sample estimates:
## mean of x mean of y
## 32.76501 23.01052
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

```
#H0: the mean(promote_switch_num) of top performs is equal to the mean(promote_switch_num) of the rest
#H1: The mean(promote_switch_num) of top performs > mean(promote_switch_num) of the rest
t.test(top10$promote_switch_num, bottom90$promote_switch_num, alternative = "greater")
```

```
##
## Welch Two Sample t-test
##
## data: top10$promote_switch_num and bottom90$promote_switch_num
## t = 2.9122, df = 1176.6, p-value = 0.001828
## alternative hypothesis: true difference in means is greater than 0
## 95 percent confidence interval:
## 0.01741865      Inf
## sample estimates:
## mean of x mean of y
## 0.2166836 0.1766169
```

```
# we reject the null hypothesis, as p-value is below 0.05
```

```
#H0: The mean (quitjob_num) of top performs is equal to the mean(quitjob_num) of the rest
#H1: The mean(quitjob_num) of top performs < mean(quitjob_num) of the rest
t.test(top10$quitjob_num, bottom90$quitjob_num, alternative = "less")
```

```
##
## Welch Two Sample t-test
##
## data: top10$quitjob_num and bottom90$quitjob_num
## t = -5.1051, df = 1272.4, p-value = 1.905e-07
## alternative hypothesis: true difference in means is less than 0
## 95 percent confidence interval:
```

```
##          -Inf -0.04547338
## sample estimates:
## mean of x mean of y
## 0.1820956 0.2492085

# we reject the null hypothesis, as p-value is below 0.05

#H0: Among top vs. rest, mean(bonustotal) has no effect on quit probability
#H1: Top performers with higher mean(bonustotal) are less likely to quit than non-performers
t.test(top10$bonustotal ~ (top10$bonustotal > median(top10$bonustotal, na.rm = TRUE)),
       data = top10, alternative = "less")

##
## Welch Two Sample t-test
##
## data: top10$bonustotal by top10$bonustotal > median(top10$bonustotal, na.rm = TRUE)
## t = -40.524, df = 710.31, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group FALSE and group TRUE is less than 0
## 95 percent confidence interval:
##          -Inf -1447.914
## sample estimates:
## mean in group FALSE mean in group TRUE
##          1342.202          2851.456

# we reject the null hypothesis, as p-value is below 0.05

#H0: mean(positive) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to higher mean(positive)
t.test(top10$positive ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
       data = top10, alternative = "greater")

##
## Welch Two Sample t-test
##
## data: top10$positive by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = 3.2093, df = 186.93, p-value = 0.000783
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than 0
## 95 percent confidence interval:
##  1.104165      Inf
## sample estimates:
## mean in group FALSE mean in group TRUE
##          28.125          25.848

# we reject the null hypothesis, as p-value is below 0.05

#H0: mean(negative) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to lower mean(negative)
t.test(top10$negative ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
       data = top10, alternative = "greater")

##
## Welch Two Sample t-test
##
## data: top10$negative by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = 1.5896, df = 121.93, p-value = 0.05726
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than 0
```

```
## 95 percent confidence interval:
## -0.0628927      Inf
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      15.92188      14.44800

# we fail to reject the null hypothesis, as p-value is above 0.05

#H0: mean(exhaustion) does not differ from homethatweek_num
#H1: higher homethatweek_num leads to lower mean(exhaustion)
t.test(top10$exhaustion ~ (top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)),
      data = top10, alternative = "greater")

##
## Welch Two Sample t-test
##
## data:  top10$exhaustion by top10$homethatweek_num > median(top10$homethatweek_num, na.rm = TRUE)
## t = -1.1678, df = 126.13, p-value = 0.8775
## alternative hypothesis: true difference in means between group FALSE and group TRUE is greater than 0
## 95 percent confidence interval:
## -2.655678      Inf
## sample estimates:
## mean in group FALSE  mean in group TRUE
##      6.078125      7.176000

# we fail to reject the null hypothesis, as p-value is above 0.05
```